



BOĞAZIÇI UNIVERSITY

ME 492 PROJECT

Automated Micrometer-Adjustable Tape Casting Machine with Heated Vacuum Chuck and Cooling Fans

Group 17

Team Members:

Muhammet Ali Doğan
Buğrahan Güreşçi

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Executive Summary

This report presents the design, analysis, and manufacturing of an automated, micrometer-adjustable tape casting machine intended for the production of ceramic solid electrolyte green tapes for all-solid-state sodium-ion battery research. The primary motivation is the absence of affordable, open-architecture tape casting instruments capable of meeting the dimensional tolerances required for sub-100 μm sintered electrolyte membranes in a university laboratory setting. The project was executed exclusively during the ME 492 semester, compressing what is normally a two-semester design-and-build cycle into a single term.

The machine integrates six interdependent subsystems. The linear drive couples a NEMA 23 bipolar stepper motor to a trapezoidal lead screw through a GT2 synchronous belt at a 40:30 pulley ratio, delivering a stutter-free casting velocity in the range of 2 to 45 mm/s. The micrometer-adjustable doctor blade, 150 mm wide with a 2.0 mm metering land, provides independent 10 μm -resolution gap adjustment at each end of the blade, a resolution consistent with the usable gap window of 0.20–0.55 mm identified by the hydrodynamic design analysis. The heated vacuum chuck is a bimetallic assembly: an Al 2045 top plate perforated by 117 wire-EDM-drilled suction holes is sealed against an Al 5083 bottom plate by a Viton O-ring, and the assembly is heated by an 800 W mica plate heater regulated from ambient temperature to 150°C through a solid-state relay driven by an Inkbird ITC-100VH PID controller. Two Power Logic PL80S12M brushless DC fans, mounted in the pivoting enclosure lid, establish forced convection across the film surface and continuously sweep the solvent-laden boundary layer out of the sealed chamber. An Arduino Uno running a deterministic four-state finite-state machine governs the idle, casting, end-hold, and return phases of each cycle, with velocity locked at the instant the start command is issued to preserve a constant shear rate throughout the cast. An ATX switched-mode power supply provides stable +12 V and +5 V direct-current rails; all 220 V alternating-current loads are routed through a Siemens 16 A miniature circuit breaker, a copper distribution busbar, and an optically isolated solid-state relay.

Four analytical models were developed to validate critical design choices. First, a non-Newtonian hydrodynamic model demonstrated that a 5% casting-speed fluctuation causes unacceptable thickness variations ($> \pm 5 \mu\text{m}$), justifying the micro-stepping drive specification. Second, an enclosure drying model proved that passive evaporation stalls at 24% residual solvent, confirming that the dual-fan active ventilation is mandatory to achieve complete drying within ~ 400 seconds. Third, a 2D finite-difference thermal simulation validated the edge-mounted thermocouple placement, showing less than 1°C of surface non-uniformity. Finally, a thermal elongation analysis confirmed that the shear mismatch between the bimetallic chuck's aluminum alloys remains within safe limits, ensuring the long-term integrity of the Viton seal.

Manufacturing was carried out with substantial industrial support. Precision CNC milling and wire-EDM drilling were sponsored by voluntary manufacturers. The total resource value of the prototype is approximately TL 34,392, of which the project team expended TL 3,240 directly, with the remainder covered by the university, sponsoring firms, and in-kind donations.

Functional validation was performed by casting a polymer electrolyte test slurry on aluminium foil substrates at blade gaps of 100 μm and 250 μm . Light-microscopy cross-section imaging confirmed that the 250 μm setting yielded a dried tape of 80 μm thickness, demonstrating that the machine operates successfully within the geometrically relevant range for NASICON electrolyte research. Principal recommendations for future development include upgrading the stepper driver supply to 24–36 V to extend the accessible speed range, precision lapping the vacuum chuck surface to within 5 μm flatness, and performing rheological characterisation of the actual NASICON slurry to validate the theoretical casting model experimentally.

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1 Introduction

1.1 Problem Statement

Thin layers and uniform films are mainly used in high-tech components where space, precision, and high surface area are crucial. The large-scale demand for cost-effective high-performance thin-film materials manufacturing that are mostly used in many applications from solid-state battery electrolytes and piezoelectric multilayered devices [1, 2] to gas diffusion layers with gradient pore sizes for SPE electrolyzers [3] has increased rapidly. Tape casting usage has also expanded to consumer electronics where specialized film sheets according to user needs are utilized in the manufacturing of electronic product cases [4].

However, accomplishing the strict dimensional tolerances and defect-free microstructures depends on controllable fluid dynamics of the slurry, the dispersion of particles, and the thermodynamic kinetics of the drying phase [5, 6]. Tape casting is still performed manually or with rudimentary applicators in many laboratory environments. Manual drawing causes slurry to have unavoidable fluctuations in the casting velocity, which destabilizes the shear rate applied to the fluid [6]. Since the complex slurries mostly exhibit non-Newtonian, shear-thinning behavior, velocity variations dramatically changes the fluid's actual viscosity during the cast, which lead to uneven particle distribution and localized thickness gradients [6, 7].

Furthermore, improper preparation of slurry and ambient drying conditions lead to severe structural flaws. Insufficient multi-stage milling and de-airing can lead to defects such as “zipper-bag” formations, where agglomerated ceramic particles become trapped within the binder, or “displacement effects”, which appear as long streaks extending across the film [8].

Additionally, there is a growing health mandate for environmental and occupational to replace highly toxic organic solvents such as benzene, toluene, and xylene, with less toxic alternatives like ethanol or butanol. This requires even more precise thermal controls over drying to avoid warping the green body [4]. There are different engineering perspectives need for an open-architecture, highly precise tape casting machine that is capable to maintain constant shear rates, controlled bottom-up thermal gradients, and environmentally friendly slurry compositions. This project addresses this need specifically in the context of production of $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ (NASICON) solid electrolyte green tapes for all-solid-state sodium-ion batteries, an application where dimensional control at the micrometre scale is critical to the electrochemical performance of the final cell [1].

1.2 Literature and Patent Review

1.2.1 Historical context and the shift from art to science

The origins of the tape casting process is going back to the mid-1940s, pioneered by Glenn N. Howatt during the Second World War. It is first appeared to manufacture thin piezoelectric materials (US Patent 2,486,410) [2]. Historically, ceramic manufacturing was considered as an “art” which relies strongly on operator intuition and trial-and-error [9]. Today, tape casting machines have become a highly scalable, scientifically rigorous process capable of creating large-area colloidal self-assemblies and advanced composite nanocomposites [10].

1.2.2 Sodium-ion batteries and NASICON solid electrolytes

The increasing demand for stationary energy storage devices has deepened the search for battery technologies that are both cost-effective and resource-efficient. Lithium-ion batteries (LIBs) currently dominate portable electronics and electric vehicle markets. However, the scarcity of lithium in the Earth's crust (approximately 20 ppm) and the geopolitical concentration of its reserves have

raised serious long-term supply security concerns [11]. Sodium, on the other hand, is the fourth most abundant element in the Earth’s crust (approximately 23 600 ppm) and is available as unlimited in ocean reserves, making sodium-ion batteries (SIBs) a fundamentally more sustainable alternative for large-scale stationary energy storage applications [11].

Replacing flammable organic liquid electrolytes with solid-state electrolytes represents a significant advancement toward inherently safer battery designs. In an all-solid-state sodium-ion battery components (ASS-SIB), the ceramic solid electrolyte simultaneously performs the roles of ion-conducting medium and mechanical separator between electrodes [12]. The risk of electrolyte decomposition, thermal runaway, and electrolyte leakage that are inherent to liquid-electrolyte systems are eliminated with this architecture, making ASS-SIBs particularly attractive for grid-scale storage installations where reliability and non-flammability are paramount requirements [12].

Among the inorganic solid electrolytes explored for ASS-SIBs, NASICON-type oxides have emerged as the most promising candidates for room-temperature applications. The acronym NASICON stands for NA Super Ionic CONductor, and the composition $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ (NZSP) represents the benchmark member of the family [1]. Its open, three-dimensional framework of corner-sharing ZrO_6 octahedra and SiO_4/PO_4 tetrahedra forms interconnected ion-conduction channels that support rapid Na^+ migration, with room-temperature ionic conductivities in the range of 10^{-3} to $5 \times 10^{-3} \text{ S cm}^{-1}$ achievable through controlled dopant substitution [1]. This three-dimensional conduction network is especially beneficial for polycrystalline electrolyte sheets, as it prevents anisotropically oriented crystallites from forming blocking grain boundaries, unlike in strictly two-dimensional conductors such as $\beta''\text{-Al}_2\text{O}_3$ [1].

The practical distribution of NASICON electrolytes in solid-state cells is contingent on producing geometrically precise, dense ceramic membranes with tightly controlled thicknesses. As established in the literature [1], the electrolyte layer must be thinner than approximately $100 \mu\text{m}$ to preserve competitive energy density relative to conventional polymer-separator cells. At the same time, the membrane must be free of through-porosity to prevent electronic short circuits and to achieve ionic conductivity approaching single-crystal benchmarks. These strict geometric and microstructural requirements constitute the primary manufacturing challenge addressed in this project.

1.2.3 Tape casting as the preferred fabrication route for NASICON solid electrolytes

Several approaches have been investigated for producing thin NASICON ceramic sheets, including axial die-pressing followed by calendering, cold isostatic pressing, and screen printing. Each of these methods presents fundamental limitations when the target geometry is a large-area, uniform membrane with a sintered thickness below $100 \mu\text{m}$. Die-pressing inherently produces thick pellets and cannot scale to large planar geometries without prohibitive tooling costs. Screen printing can deposit thin layers but is constrained to relatively low-viscosity inks and produces surface textures that degrade ionic conductivity at grain boundaries [1].

Tape casting is uniquely capable of addressing these constraints simultaneously. The process begins with the preparation of a homogeneous slurry composed of NASICON ceramic powder, an organic solvent, a dispersant, a polymeric binder, and a plasticizer. The slurry is degassed and its rheology is carefully adjusted to achieve the shear-thinning behavior required for stable flow beneath the Doctor Blade [6]. The slurry is then cast at a controlled velocity onto a flexible polymer carrier film, leaving a wet green film of precisely defined wet thickness h as set by the Doctor Blade gap. Figure 1 illustrates the fundamental geometry of this process, showing the slurry reservoir, the Doctor Blade gap h , and the relative motion v_0 between the blade and the substrate that governs the film-forming hydrodynamics. Following a temperature-controlled drying phase, the free-standing green tape is peeled from the carrier substrate, optionally laminated into multilayer stacks, and

ultimately subjected to organic binder burnout followed by high-temperature sintering to densify the ceramic [1, 13].

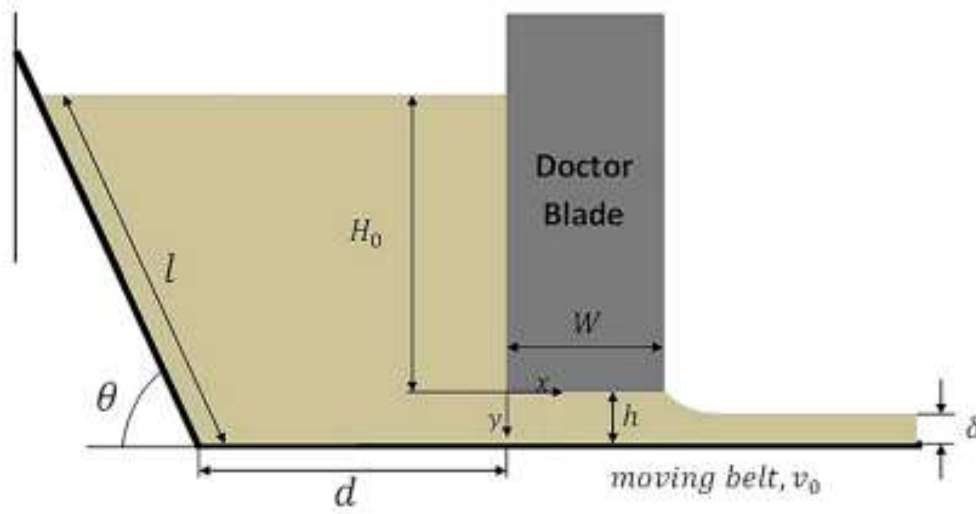


Figure 1: Schematic of the tape casting process: Doctor Blade gap h and traverse velocity v_0 govern slurry flow and green tape thickness δ . Source: [6]

A seminal demonstration of this route for NASICON was reported by Okubo et al. [13], who produced a dense NZSP sheet with a sintered thickness of $35\ \mu\text{m}$ by combining tape casting with a glass-phase sintering aid and a reduced densification temperature of $1100\ ^\circ\text{C}$. The resulting sheet achieved a room-temperature ionic conductivity of $4.4 \times 10^{-4}\ \text{S cm}^{-1}$, confirming that tape casting can deliver the sub- $100\ \mu\text{m}$ membrane geometry demanded by practical battery cells without sacrificing electrolyte quality [13]. An important challenge specific to NASICON is solvent selection. Early recipes employed toluene, which provides excellent dispersibility for NZSP powder; however, its acute toxicity has motivated a shift toward ethanol-based systems [13]. Ethanol-compatible formulations require more careful thermal management during drying, because the higher vapor pressure of ethanol accelerates the evaporation front, making the avoidance of premature surface skin formation a critical constraint. This constraint directly motivates the integration of a bottom-up heated vacuum chuck in the machine described in this report.

1.2.4 Fluid dynamics, Doctor Blade mechanics, and automation

The fundamental mechanics of tape casting rely on shearing a fluid between a stationary Doctor Blade and a moving substrate. Analytical models demonstrate that the overall fluid flow is governed by a combination of pressure-driven flow and shear-driven flow [6]. Recent numerical and experimental studies utilizing smoothed particle hydrodynamics demonstrate that the specific geometric shape of the Doctor Blade directly influences the flow inside the casting unit, ultimately dictating particle orientation and final tape thickness [14].

Maintaining optimal contact stress and structural rigidity in the Doctor Blade mechanism is critical for achieving a consistent layer [15]. To combat human-induced velocity fluctuations, recent advancements have shown that integrating computer numerical control (CNC) mechanisms with Doctor Blade technology successfully eliminates manual error, allowing for highly homogeneous deposition of small-molecule thin films over large areas [16]. Furthermore, state-of-the-art quality control now explores real-time machine vision and deep learning (e.g., YOLOv5 algorithms) to continuously measure coating thickness and identify microscopic defects in film-coated products [17].

The analytical treatment of the flow in the Doctor Blade region provides quantitative insight into the physical parameters that govern tape thickness. Under steady-state, one-dimensional flow conditions with the geometry shown in Figure 1, the pressure gradient at the entrance of the blade channel is determined by the hydrostatic head H_0 of the slurry reservoir [6]:

$$\frac{dp}{dx} = -\frac{\rho g H_0}{W} \quad (1)$$

where ρ is the slurry density, g is gravitational acceleration, and W is the Doctor Blade width. The total volumetric flow rate per unit width through the gap of height h is the superposition of the substrate drag-driven (Couette) and pressure-driven (Poiseuille) contributions; applying mass conservation then yields the wet tape thickness δ immediately downstream of the blade:

$$\delta = \frac{1}{v_0} \int_0^h u(y) dy \quad (2)$$

where $u(y)$ is the velocity profile across the gap and v_0 is the substrate traverse velocity [6]. For a Newtonian fluid this integral yields a result in which δ decreases hyperbolically with v_0 : as the substrate accelerates, the drag-flow component stretches the slurry progressively thinner, while a reduction in H_0 as the reservoir empties simultaneously reduces the pressure contribution and further thins the tape [6]. For a target sintered thickness of 35–100 μm in NASICON electrolyte sheets, the corresponding wet green tape must be cast at thicknesses of approximately 100–300 μm ; a 5 % fluctuation in casting speed at moderate velocities therefore introduces thickness gradients of 10–15 μm per cast length, which is incompatible with the dense, pinhole-free electrolyte microstructure required for solid-state battery operation [13].

The analysis becomes considerably more complex when the non-Newtonian rheology of ceramic slurries is accounted for. NASICON and other high-solid-loading ceramic suspensions exhibit shear-thinning power-law behaviour, in which the apparent viscosity decreases with increasing shear rate according to:

$$\mu_a = k \dot{\gamma}^{n-1}, \quad n < 1 \quad (3)$$

where k is the consistency index and n is the power-law exponent [6]. Beneath the Doctor Blade, the dominant shear rate is approximated by $\dot{\gamma} \approx v_0/h$, so that any fluctuation in substrate velocity simultaneously alters both the drag force driving the tape and the apparent viscosity of the slurry itself. This non-linear coupling amplifies the sensitivity of δ to velocity perturbations well beyond what Newtonian theory predicts: a sudden acceleration not only mechanically stretches the tape thinner, but also transiently reduces the slurry viscosity, further destabilising the flow regime beneath the blade and producing localised thickness gradients that propagate several blade-widths downstream before damping out [14]. It is precisely this non-linear amplification in shear-thinning suspensions that motivates the use of a precision micro-stepping motor drive in the present machine, rather than a conventional DC motor, to maintain a deterministic and stutter-free casting velocity throughout the entire stroke.

1.2.5 Thermodynamics of drying kinetics

Following the physical shaping of the tape, the wet cast must undergo a highly controlled drying phase. The drying of cast slip takes place in two distinct stages: a fluid stage controlled by rapid surface evaporation, and a solid stage controlled by much slower solid-state diffusion through the polymeric binder [5].

A major limiting factor in high-quality film casting is managing the evaporation rate during the initial fluid stage. If heat or excessive airflow is applied from the top down, rapid evaporation

causes a polymeric skin to form over the surface of the tape, trapping vapor and retarding further drying [18]. Industrial patents, such as those filed by Steinmueller (US Patent 6,682,598), highlight the necessity of complex apparatuses featuring drying tunnels and carrier gas controls to manage solvent evaporation without disrupting the tape structure [18]. For compact, laboratory-scale applicators, a highly uniform bottom-up heating mechanism is required to drive the thermal gradient from the substrate upward, preventing premature skinning.

The two-stage drying model can be characterised in terms of the competition between two transport processes operating in succession. During the first, constant-rate stage, the evaporation front resides at the free surface of the tape and the drying rate is governed by the external mass-transfer coefficient and the partial pressure difference of the solvent between the film surface and the surrounding atmosphere [5]. For ethanol-based NASICON slurries, the higher saturation vapour pressure of ethanol (≈ 5.8 kPa at 20°C) compared to toluene (≈ 3.8 kPa) means that the constant-rate period is shorter, the evaporation front retreats more rapidly into the bulk of the tape, and the probability of forming a surface skin before the tape has adequately consolidated is elevated [18]. Once the surface is no longer saturated with solvent vapour, the process enters the falling-rate stage, in which the drying rate is governed by solid-state diffusion of residual solvent through the consolidating binder matrix; this stage is substantially slower and determines the total drying time of the green tape.

A critical design implication of this two-stage model concerns the direction of the thermal driving force during the drying process. When heat is applied conductively from the substrate upward, the highest temperature exists at the substrate-film interface, driving solvent vapour upward through the still-open pore network toward the free surface. This configuration maintains the evaporation front at the top surface throughout the constant-rate period, preventing the formation of a blocking skin and ensuring that the binder consolidates gradually and uniformly across the full tape thickness [18]. The significance of precise temperature regulation is further underscored by the thermal sensitivity of the polymeric binder system: for polyvinyl butyral (PVB), the standard binder family used in NASICON tape casting, the onset of binder softening occurs within a narrow temperature window near the glass transition temperature, and uncontrolled heating above this threshold can produce localised densification of the binder phase that traps residual porosity in the final sintered electrolyte [13]. In the machine described in this report, a 800 W mica plate heater positioned beneath the vacuum chuck provides this bottom-up conductive heating, regulated by a proportional-integral-derivative (PID) controller to a set-point stability of $\pm 0.1^\circ\text{C}$ across the full casting surface.

1.2.6 Advanced materials and precursor tape casting

Innovations in slurry formulations have expanded the capabilities of tape casting. For instance, the “precursor tape casting” method utilizes chemical precursors instead of final powder materials, allowing for low-temperature annealing (rather than high-temperature sintering) to form nanostructured materials with grain sizes between 1 and 100 nanometers [19]. Additionally, the process is increasingly used for advanced layered configurations, such as sequential casting and vacuum plasma spraying to create gradient porous structures for SPE electrolyzers [3], and processing Si_3N_4 with TiN composites for high-toughness micro-laminates [8].

1.2.7 Commercial tape casting equipment: state of the art and limitations

The commercial market for tape casting equipment encompasses a wide range of systems, from hand-operated film applicators to large-scale, continuous roll-to-roll industrial machines. A critical review of these offerings reveals a persistent gap between what is commercially available at

accessible cost and what is technically required for laboratory-scale NASICON solid electrolyte research.

At the high end of the market, manufacturers such as Hohsen Corporation (Japan) and Tape Casting Warehouse (United States) offer fully integrated automated casters featuring heated drying chambers, controlled-atmosphere enclosures, and programmable traverse stages. Their capital costs, however, consistently exceed \$50,000 and can approach \$200,000 for fully equipped configurations, placing them beyond the budget of most university research groups. Furthermore, these architectures are optimized for established industrial ceramics (multilayer capacitor dielectrics, solid oxide fuel cell electrolytes) rather than emerging NASICON formulations where blade geometry, substrate temperature, and casting speed parameters remain subjects of active optimization [1, 14].

Mid-range motorized Doctor Blade coaters (e.g., MTI MSK-AFA-MR200) offer a more accessible entry point with motorized traverse drives and optionally heated platens. However, critical deficiencies persist for NASICON applications. The substrate hold-down mechanism in these systems relies on static adhesion to a flat glass or aluminum platen rather than an active porous-vacuum matrix, leaving the flexible carrier substrate susceptible to edge lift-off and planarity loss when loaded with high-viscosity ceramic slurries. Furthermore, the absence of integrated vacuum fixation means that thermal expansion of the carrier film during heating introduces variable curvature across the casting surface, producing gradients in the effective blade gap h that translate directly into unacceptable green-tape thickness non-uniformity [7]. The proprietary, closed architecture of these systems further prevents the researcher from exchanging blade assemblies or modifying slurry reservoir geometry for process optimization.

At the entry level, research groups frequently employ hand-operated bar coaters or simple motorized film applicators. While affordable, these instruments provide no automated velocity control, no vacuum substrate fixation, and no thermal management. The casting speed must be set manually, and operator-induced fluctuations in traverse velocity are translated directly into variable shear rates $\dot{\gamma} = v_0/h$ across the slurry, which in turn produce spatially non-uniform apparent viscosity profiles in shear-thinning NASICON suspensions [6, 14]. A clear gap therefore exists between affordable but imprecise manual applicators and prohibitively expensive industrial casters, and it is precisely this engineering gap that the machine designed in this project is intended to fill.

1.3 Significance of the Project

The primary objective of this project is to bridge the gap between error-prone manual casting methods and prohibitively expensive commercial automated systems. The proposed Automated Micrometer-Adjustable Tape Casting Machine introduces a highly controlled, open-architecture mechatronic solution tailored for precise laboratory-scale materials engineering.

The engineering significance of this design is rooted in the targeted mitigation of the physical and thermodynamic challenges identified in the literature:

1. **Shear-Rate Stabilization via Automation:** By utilizing a high-torque NEMA 23 stepper motor paired with a robust linear guide, the system functions similarly to advanced CNC mechanisms [16], eliminating human-induced velocity variations. This ensures a constant shear rate, which is mathematically required to maintain uniform apparent viscosity in non-Newtonian slurries [6].
2. **Micrometer Precision and Blade Geometry:** Acknowledging the critical impact of Doctor Blade shape and contact mechanics [14, 15], the integration of precision micrometer heads allows the blade gap to be adjusted with micron-level accuracy, directly controlling the shear-driven flow volume.

3. **Bottom-Up Thermal Management:** The integration of a custom-engineered heated vacuum chuck directly addresses the thermodynamic failure points of film drying. By applying controlled conductive heat through the bottom of the substrate, the machine prevents surface skinning and allows volatile, less-toxic solvents to diffuse naturally, eliminating blistering and mud-cracking defects [18, 4].
4. **NASICON Solid Electrolyte Compatibility:** The machine is specifically dimensioned to produce NZSP green tapes suitable for all-solid-state sodium-ion battery research. The 200×200 mm heated vacuum chuck effective area accommodates the substrate areas required for meaningful electrochemical testing, while the micrometer-adjustable blade gap enables casting of wet films in the 80–200 μm range that yield, after sintering densification, the sub-100 μm solid electrolyte membranes reported as optimal in the literature [1, 13].

The theoretical framework established in the preceding sections also provides direct justification for a key mechanical feature of the present design: the active porous vacuum hold-down system. During bottom-up heating, the flexible polymer carrier undergoes thermal expansion while in contact with the heated chuck surface; without active fixation, differential expansion causes the carrier to buckle and partially lift off the surface, introducing a spatially varying thermal contact resistance across the casting area. The resulting substrate temperature non-uniformity translates directly into spatial variation in the solvent evaporation rate during the critical constant-rate drying stage, producing green-tape thickness gradients that are indistinguishable from blade-gap non-uniformity during process inspection. The porous vacuum chuck designed for this machine — featuring 117 micro-perforated holes distributed across an Al 2045 top plate and producing a clamping force to address this requirement by maintaining intimate thermal contact between the carrier and the heated surface throughout the entire casting and drying cycle.

Ultimately, this project provides a scalable blueprint for an industrial-grade automated caster. It empowers researchers to achieve exceptional repeatability and defect-free thin films without the financial barrier or closed-architecture limitations of proprietary commercial units.

2 Design Process

2.1 Design Criteria and Product Design Specifications

The primary objective of the Automated Micrometer-Adjustable Tape Casting Machine is to eliminate the severe thickness variations and particle dispersion defects associated with manual film casting. As established in the theoretical framework, the behavior of non-Newtonian ceramic slurries and battery pastes is governed by the shear rate equation:

$$\dot{\gamma} = \frac{v}{h} \quad (4)$$

where $\dot{\gamma}$ is the shear rate, v is the traverse velocity of the blade, and h is the gap height. In manual casting, inconsistent human pulling forces cause v to fluctuate, which drastically alters the fluid's apparent viscosity and leads to defects. To prevent this, the engineering design criteria were explicitly derived from these process variables, alongside strict manufacturing and safety constraints.

2.1.1 Establishment of design criteria

Based on the physical, thermodynamic, and operational requirements of laboratory-scale tape casting, fourteen specific design criteria were established and defined:

1. **Precision Gap Control:** The ability to rigidly maintain and adjust the Doctor Blade in exact micrometer increments to dictate wet film thickness (h).
2. **Velocity Stability:** The ability of the linear drive to maintain a perfectly constant, stutter-free traverse speed (v) under viscous fluid loads.
3. **Thermal Uniformity:** The ability of the substrate bed to provide even, bottom-up heating without localized hot-spots to prevent surface skinning during solvent evaporation.
4. **Vibration Damping:** The capacity of the chassis to absorb micro-stepping vibrations from the motor before they propagate into the wet cast film.
5. **Vacuum Hold-Down Strength:** The ability to secure flexible polymer carrier substrates completely flat against the bed under negative pressure.
6. **Manufacturability:** The feasibility of fabricating the custom components using accessible university workshop tools (e.g., laser cutting, basic tapping).
7. **Ease of Assembly:** The simplicity of putting the machine together using standard fasteners rather than complex permanent welds.
8. **Modularity:** The open-architecture capability to swap, upgrade, or repair subsystems (heaters, motors) independently.
9. **Cost Effectiveness:** The ability to procure and manufacture the system strictly within the allocated undergraduate project budget.
10. **Material Availability:** Reliance on commercial off-the-shelf (COTS) mechatronic components rather than specialized, long-lead-time industrial parts.
11. **Safety:** The strict isolation of 220V electrical paths, EMI filtering, circuit breakers, and thermal insulation to protect the operator.
12. **Maintenance:** The ease with which the Doctor Blade and bed can be cleaned of hardened ceramic slurries and corrosive solvents.
13. **Ergonomics:** The accessibility and intuitive placement of the user interface, including the KCD power switch and micrometer adjustment knobs.
14. **Weight:** The overall mass of the machine. (Note: Portability is a non-issue for stationary lab equipment; higher weight is generally preferred for stability).

2.1.2 Binary dominance matrix

To objectively prioritize these fourteen criteria and guide the detailed design phase, a Binary Dominance Matrix was utilized (Figure 1). Each criterion was evaluated against the others, assigning a '1' to the dominant feature and a '0' to the subordinate feature.

The matrix reveals that safety is the absolute highest priority, dictating all electrical integration. Following Safety, the core physics parameters—velocity Stability, precision gap control, and thermal uniformity—dominate the matrix. Conversely, weight scored a zero; because a heavier machine actively improves vibration damping, reducing mass was deemed entirely unnecessary for a stationary laboratory instrument.

BINARY DOMINANCE MATRIX: CRITERIA EVALUATION																
Criteria	Precision Gap Control	Velocity Stability	Thermal Uniformity	Vibration Dampigng	Vacuum Down Strength	Manufacturability	Ease of Assembly	Modularity	Cost Effectiveness	Material Availability	Safety	Maintenance	Ergonomics	Weight	Total	Weighting
Precision Gap Control	#	1	1	1	1	0	1	1	1	1	0	1	1	1	11	0,121
Velocity Stability	0	#	1	1	1	0	1	1	1	1	0	1	1	1	10	0,11
Thermal Uniformity	0	0	#	0	0	0	1	1	0	0	0	1	1	1	5	0,055
Vibration Dampigng	0	0	1	#	0	0	1	1	1	1	0	1	1	1	8	0,088
Vacuum Hold-Down Strength	0	0	1	1	#	0	1	1	0	0	0	1	1	1	7	0,077
Manufacturability	1	1	1	1	1	#	1	1	0	0	0	1	1	1	10	0,11
Ease of Assembly	0	0	0	0	0	0	#	0	0	0	0	1	1	1	3	0,033
Modularity	0	0	0	0	0	0	1	#	0	0	0	1	1	1	4	0,044
Cost Effectiveness	0	0	1	0	1	1	1	1	#	1	0	1	1	1	9	0,099
Material Availability	0	0	1	0	1	1	1	1	0	#	0	1	1	1	8	0,088
Safety	1	1	1	1	1	1	1	1	1	1	#	1	1	1	13	0,143
Maintenance	0	0	0	0	0	0	0	0	0	0	0	#	0	1	1	0,011
Ergonomics	0	0	0	0	0	0	0	0	0	0	0	1	#	1	2	0,022
Weight	0	0	0	0	0	0	0	0	0	0	0	0	0	#	0	0
															91	1,001

Figure 1: Binary dominance matrix

2.1.3 Product design specifications (PDS)

Having prioritized the design criteria, the top-ranking requirements were translated into quantifiable Product Design Specifications (PDS) to serve as strict engineering boundaries:

- **Electrical Safety Rating:** The system must utilize a 16A miniature circuit breaker, EMI-filtered power entry, and mechanical KCD phase-breaking switches to safely isolate 220V AC lines from the +12 V / +5 V DC logic loops.
- **Traverse Velocity Range:** The NEMA 23 linear actuator must be capable of delivering constant, stutter-free casting speeds ranging from 2 mm/s to 45 mm/s .
- **Gap Adjustment Resolution:** The Doctor Blade must utilize precision micrometer heads capable of adjusting the casting gap h in increments of 10 μm , with a total operational range of 0 to 5000 μm .
- **Thermal Operating Range:** The 200 $\times$ 200 mm mica heating bed must reach a maximum steady-state temperature of at least 120°C with uniform planar heat distribution to evaporate common solvents.
- **Structural Mass:** The base must utilize a high-density, vibration-absorbing material (such as MDF paired with heavy aluminum extrusion) to ensure micro-stepping vibrations do not exceed an amplitude that would disturb the wet film.

2.2 Overview of Possible Solutions

During the conceptualization phase, an extensive market and literature review was conducted to evaluate existing commercial solutions against the specific thermodynamic, kinematic, and eco-

nomic requirements of the project. Six distinct coating concepts were identified and benchmarked:

- **Concept A: Commercial Bar Coater (e.g., Ossila) [20].** This system utilizes a motorized linear drive to pull a Meyer bar (wire-wound rod) across a substrate. While it provides baseline kinematic stability, it lacks micrometer adjustment, an integrated heated bed, and a vacuum chuck, making it unsuitable for volatile, non-Newtonian ceramic slurries.
- **Concept B: Commercial Heated Tape Caster (e.g., MTI MSK-AFA-MR200) [21].** A high-end, benchtop Doctor Blade coater featuring an integrated heated bed and motorized drive. While it meets the thermal and kinematic requirements, it utilizes a closed, proprietary architecture, lacks a true porous vacuum-hold-down matrix (relying instead on flat glass or metal plates), and is prohibitively expensive for standard laboratory budgets.



Figure 2: Concept A solution



Figure 3: Concept B solution

- **Concept C: Compact Entry-Level Caster (e.g., MTI MSK-AFA-EC150) [22].** A smaller, entry-level motorized film applicator. It provides an automated traverse velocity but entirely lacks the bottom-up thermal management and vacuum fixation required to prevent the "skinning" and mud-cracking defects in advanced ceramics.
- **Concept D: Ultrasonic Spray Pyrolysis Unit (e.g., MTI MSK-USP-02) [23].** This concept utilizes ultrasonic atomization to deposit films onto a heated plate. While excellent for nano-scale films, spray systems are fundamentally incompatible with high-viscosity, high-solid-loading ceramic slurries, which rapidly clog the ultrasonic nozzles.



Figure 4: Concept C solution



Figure 5: Concept D solution

- **Concept E: Proposed Open-Architecture Tape Caster (Our Design).** A custom-engineered system integrating a high-torque NEMA 23 linear drive, micrometer-adjustable Doctor Blade, a custom 200×200 mm heated vacuum chuck, and cooling cover. This concept is designed to offer industrial-grade precision and thermal control using accessible, modular mechatronics at a fraction of commercial costs.
- **Concept F: Automated Spray Coater (e.g., Ossila Spray Coater) [24].** A programmable, low-temperature spray deposition system. Similar to Concept D, it lacks the shear-driven mechanics required to properly align and distribute high-viscosity ceramic pastes and solid-state battery electrolytes.



Figure 6: Concept E solution



Figure 7: Concept F solution

2.2.1 Decision matrix and concept selection

To select the optimal path forward, a rigorous decision matrix (Figure 8) was constructed. Fourteen specific evaluation criteria were established using the Binary Dominance Matrix, resulting in weightings ranging from 0.143 (highest priority) down to 0.

Each concept was evaluated on a scale from 0 to 100 (in 10-point increments). The raw score was then multiplied by the criterion's weight to determine the weighted score, presented in the table as raw (weighted).

2.2.2 Justification of selection

As demonstrated by the Decision Matrix, concept E which is our project, achieved the highest overall score, decisively outperforming the closest commercial competitor, the MTI MR200 (Concept B).

The evaluation reveals that the highest-weighted criteria are safety, precision gap control, and velocity stability, which quickly eliminated the spray-based architectures (concepts D and F) due to their inability to process thick slurries. While the high-end commercial MTI MR200 matched the proposed project in the high-level kinematic and thermal capabilities, it lost significant ground in the specialized mid-weighted criteria.

Specifically, concept E provides a vastly superior vacuum substrate hold mechanism tailored for flexible polymer carrier substrates, exceptional component modularity due to its open-architecture design, and unparalleled cost-effectiveness for a university setting. Because concept E is the only

Criteria	Weighting	Concept A	Concept B	Concept C	Concept D	Concept E	Concept F
Safety	0.143	90	100	80	80	100	80
Precision Gap Control	0.121	60	100	90	80	80	20
Velocity Stability	0.11	70	90	80	90	90	80
Manufacturability	0.11	60	70	60	50	90	60
Cost Effectiveness	0.099	40	30	20	30	100	30
Vibration Damping	0.088	80	70	70	70	80	90
Material Availability	0.088	10	20	30	30	100	10
Vacuum Hold-Down Strength	0.077	60	70	70	80	100	60
Thermal Uniformity	0.055	80	20	60	60	80	40
Modularity	0.044	80	70	60	50	90	80
Ease of Assembly	0.033	40	30	30	30	100	40
Ergonomics	0.022	80	100	70	80	90	80
Maintenance	0.011	60	60	60	70	70	70
Weight	0	60	40	60	40	60	40
Percentage of Satisfaction		62.59	68.31	63.36	64.35	91.63	55.22

Figure 8: Decision matrix of possible concepts

solution that satisfies both the complex fluid physics and the economic constraints of the project without compromise, it was officially selected for detailed design and manufacturing.

2.3 Detailed Design and Analysis

All technical drawings derived from the analyses conducted in this section can be found in the Appendix B (6).

2.3.1 Structural chassis and enclosure subsystem

The structural foundation of the Automated Tape Casting Machine was engineered to satisfy three primary design criteria: maximum geometric rigidity to support the linear kinematics, superior vibration isolation, and high chemical resistivity against volatile coating solvents. To achieve this, a hybrid-material architecture was developed, utilizing Medium-Density Fiberboard (MDF), aluminum extrusion profiles, and stainless steel sheet metal.

Base Plate and Vibration Isolation The fundamental requirement of the tape casting process is that the wet film must not be disturbed during the drying phase. Stepper motors, by their electromechanical nature, generate high-frequency micro-vibrations during operation. If the entire chassis were constructed from rigid metals (such as a solid steel or aluminum plate), these harmonic vibrations would propagate directly into the casting bed, causing microscopic ripples (chatter marks) in the non-Newtonian ceramic slurry.

To mitigate this, the main base plate was constructed from heavy-duty Medium-Density Fiberboard (MDF). MDF possesses a significantly higher vibration damping coefficient and acoustic absorption rate than isotropic metals, effectively acting as a mechanical low-pass filter. It absorbs and dissipates the resonant frequencies generated by the NEMA 23 motor before they can reach the aluminum vacuum chuck.

Internal Skeleton and Kinematic Foundation While MDF provides excellent damping, it lacks the precise geometric linearity required to mount linear guide rails. Therefore, the internal load-bearing skeleton was constructed from modular aluminum extrusion profiles securely bolted to the MDF base using threaded insert nuts. The aluminum profiles provide an exceptionally straight, torsion-resistant framework. This ensures that the dual linear guide rails remain perfectly parallel across the entire casting stroke, preventing mechanical binding and ensuring a constant traverse velocity (v). Furthermore, the modular nature of the extrusions aligns with the open-architecture design criterion, allowing for future modifications or sensor attachments.



Figure 9: Base MDF plate

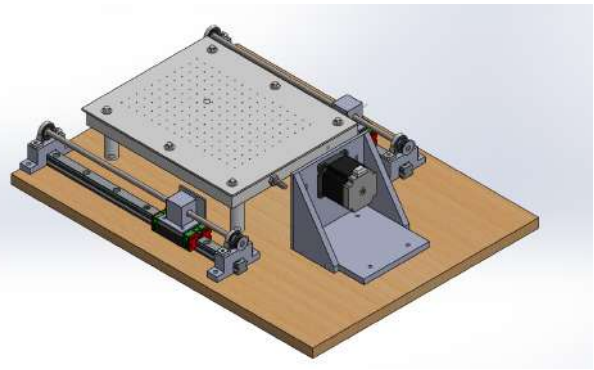


Figure 10: Assembly on base MDF plate

Outer Enclosure and Chemical Resistivity The outer enclosure serves a dual purpose: protecting the high-voltage electrical components (PID, SSR, and ATX PSU) from accidental fluid spills, and providing a safe, cleanable work surface. The enclosure panels were fabricated from precision laser-cut stainless steel.

The selection of stainless steel over standard mild steel or acrylic was driven by the thermodynamic and chemical realities of advanced materials processing. Tape casting slurries frequently utilize aggressive, volatile organic solvents (such as ethanol, toluene, or butanol) which act as severe plasticizers or corrosive agents to standard materials. Stainless steel ensures complete chemical compatibility, allowing the operator to easily wipe away spilled ceramic binders or solvents without degrading the machine's structural integrity.

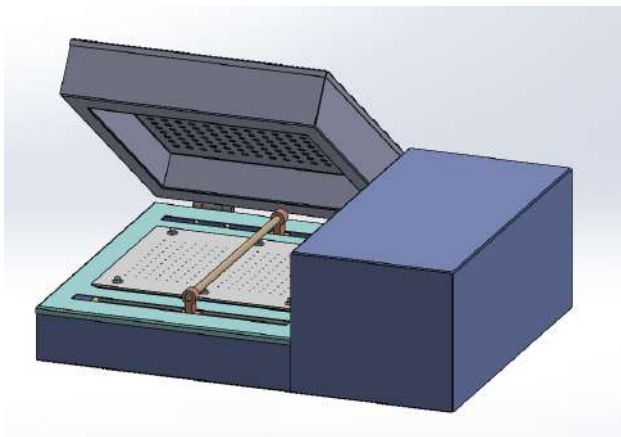


Figure 11: Enclosure front view

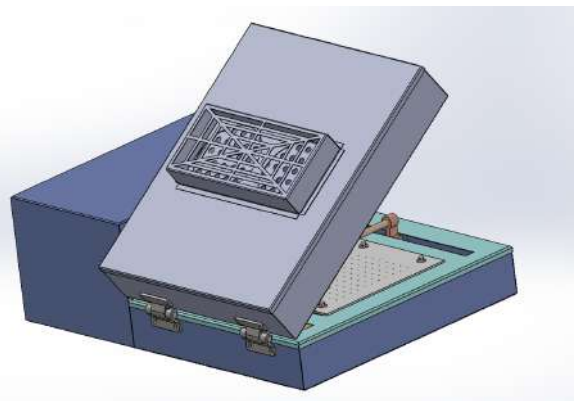


Figure 12: Enclosure behind view

Manufacturing and Fastening Adaptations During the manufacturing phase, the extreme hardness and high yield strength of the stainless steel sheet metal presented an assembly challenge. Initially, M3 threaded rivet nuts were specified to fasten the enclosure panels to the internal aluminum frame. However, the manual mechanical deformation required to set the rivets in the rigid stainless steel resulted in stripped threads.

To solve this without compromising the aesthetic or structural integrity of the enclosure, an alternative fastening methodology was engineered. For edge-mounted connections, sliding mechanical "U-nut" clips were utilized. For blind mounting points located centrally on the panels, a "chemical welding" process was adopted using industrial steel-reinforced epoxy. This allowed standard M3 nuts to be permanently bonded to the roughened interior surface of the stainless steel panels, resulting in a clean, flush exterior capable of withstanding the necessary clamping forces.

2.3.2 Kinematics and linear drive subsystem

The traverse motion of the Doctor Blade is produced by a mechanical chain that converts discrete rotary steps into precise linear displacement. A NEMA 23 bipolar stepper motor was selected as the prime mover, owing to its high holding torque, typically between 1.2 and 3.0 Nm, and its inherent ability to translate switched electrical excitations into fixed angular increments without position feedback [25]. The motor advances in nominal full steps of 1.8° , equivalent to 200 steps per revolution. This open-loop behaviour suits the application, because the dominant load is a predictable viscous drag rather than an erratic disturbance.

The motor is commanded by a TB6600 driver, which serves as the power interface between the low-current logic of the Arduino and the high-current windings of the motor. The driver accepts three logic-level signals, namely pulse (PUL), direction (DIR), and enable (EN), and it converts each incoming pulse into one increment of coil excitation [25]. Phase current is bounded between 0.5 and 4.0 A through onboard DIP switches, while a large aluminium heat-sink and internal over-current, over-temperature, and under-voltage protection safeguard the device during continuous operation.

A central design decision concerns the supply voltage of the driver. Although the TB6600 accepts any voltage between 9 and 40 V, it is powered exclusively from the +12 V rail of the ATX supply, since no higher-voltage source exists within the selected power architecture. The supply voltage strongly governs the torque available at speed, because the winding inductance resists the rapid current reversal demanded at each step, and a low driving voltage cannot force the current to rise quickly enough [25]. Consequently, the developed torque collapses beyond a threshold speed, and the motor stalls while emitting an audible tone. This electromechanical limit is the physical reason the casting speed was capped at 45 mm/s, a window across which the 12 V rail sustains stable torque.

Vibration is suppressed primarily through micro-stepping. Coarse full-stepping excites mechanical resonance at low speeds and produces ripples that would propagate into the wet film. The driver was therefore configured for 1600 pulses per revolution, an eight-fold subdivision of the native step, which smooths the current waveform and yields essentially continuous motion. This configuration directly satisfies the velocity-stability and vibration-damping criteria established previously.

Motion Transmission and Velocity Resolution

Rotary motion is transferred to the blade carriage through a GT2 synchronous belt and a trapezoidal lead screw. A 40:30 pulley pair imposes a speed-increasing ratio (GR) of 1.33:1, so the lead screw rotates faster than the motor shaft. With a screw lead of 8 mm, a single motor revolution therefore advances the carriage by

$$d_{rev} = \text{Lead} \times \text{GR} = 8 \text{ mm} \times 1.33 = 10.66 \text{ mm}. \quad (5)$$

The firmware regulates speed by inserting an equal delay between successive pulse edges, since each step pulse comprises one HIGH interval and one LOW interval of identical length. The relation between the commanded linear velocity v (mm/s) and the inter-pulse delay t_d (μs) follows from the requirement that the pulse rate equal the product of the revolution rate and the micro-steps per revolution:

$$t_d = \frac{10^6 \cdot (\text{Lead} \cdot \text{GR})}{2 \cdot v \cdot N_\mu}, \quad (6)$$

where N_μ is the micro-step resolution in pulses per revolution. Substituting $d_{rev} = 10.66 \text{ mm}$ and

$N_\mu = 1600$ collapses the expression into a single firmware constant:

$$t_d \approx \frac{3333}{v} \quad [\mu s]. \quad (7)$$

For the maximum casting speed of 45 mm/s this yields a delay of roughly 74 μs , whereas a representative slow cast of 6 mm/s requires about 555 μs . These values confirm that the resolution is more than adequate for micron-scale film control. The governing drive parameters are summarised in Table 1, and the physical arrangement of the linear drive subsystem is shown in Figure 13.

Table 1: Linear drive subsystem parameters

Parameter	Value
Motor type	NEMA 23 bipolar stepper (1.8°/step, 200 steps/rev)
Driver	TB6600, set to 1600 pulses/rev (1/8 micro-step)
Driver supply voltage	+12 V (from ATX PSU)
Belt ratio (GR)	1.33:1 (40:30 GT2 pulleys)
Lead screw lead	8 mm (trapezoidal)
Linear travel per motor revolution	10.66 mm
Step-delay constant	$t_d = 3333/v \mu s$
Forward (casting) speed range	$\approx 2\text{--}45$ mm/s
Return speed	45 mm/s (fixed)

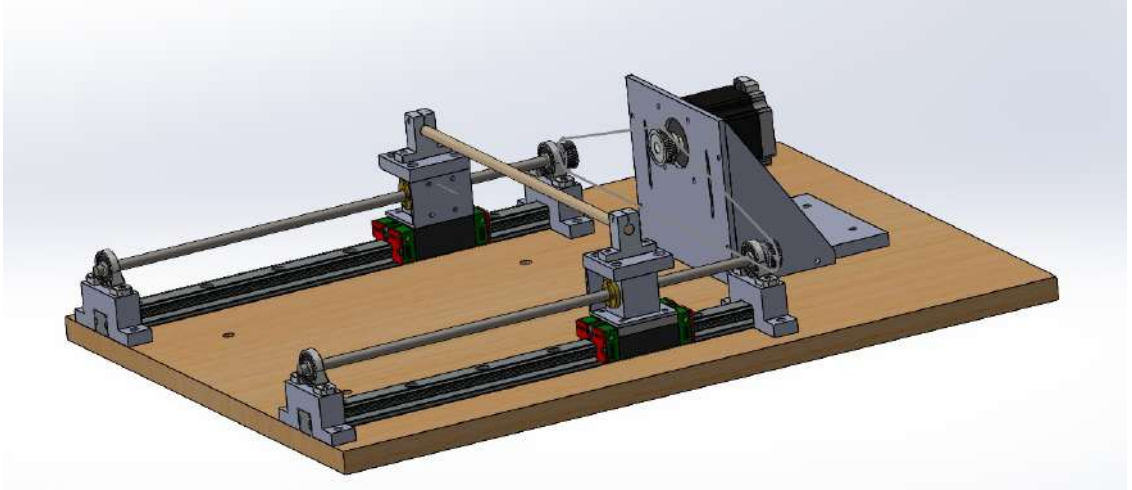


Figure 13: Layout of the linear drive subsystem (NEMA 23 motor, GT2 belt, lead screw, and carriage on the linear rails).

2.3.3 Fluid dynamics and Doctor Blade assembly

The Doctor Blade assembly is the component in which the casting physics described in the literature review is converted into a quantitative design specification. Whereas Section 1.2.4 established the governing flow relations in general terms, this subsection applies those relations to the specific geometry and operating envelope of the present machine in order to fix three design quantities:

the usable blade-gap range, the casting-speed range, and the substrate temperature window. A reduced-order hydrodynamic model is used for this purpose, because the objective is not to resolve the full velocity field but to bound the parameter combinations that yield a defect-free green tape. The related matlab code is in Appendix C (6).

Reduced-order casting model. Under the steady, one-dimensional approximation introduced previously, the slurry beneath the land of the blade is sheared between the moving carrier substrate and the stationary blade. The flow is the superposition of a drag-driven (Couette) component, produced by the substrate moving at velocity v_0 , and a pressure-driven (Poiseuille) component, fed by the hydrostatic head H_0 of the slurry pool retained behind the blade. Integrating the resulting velocity profile across a gap of height h and imposing mass conservation gives the wet tape thickness immediately downstream of the blade,

$$\delta = \frac{h}{2} + \frac{\rho g H_0 h^3}{12 \mu v_0 L_b}, \quad (8)$$

where ρ is the slurry density, g the gravitational acceleration, L_b the blade land length, and μ the slurry viscosity [6]. The first term is the drag-flow limit, in which the tape is simply the half-gap film carried forward by the substrate; the second term is the pressure-driven excess that thickens the tape at low speed or large gap.

Ceramic casting slurries are not Newtonian. As noted in Section 1.2.4, NASICON suspensions are strongly shear-thinning and obey the power-law relation $\mu_a = k \dot{\gamma}^{n-1}$ with $n < 1$. Because the dominant shear rate beneath the land is $\dot{\gamma} \approx v_0/h$, the apparent viscosity is itself a function of the operating point. Substituting $\mu = k(v_0/h)^{n-1}$ into Eq. (8) yields the design expression used throughout this analysis,

$$\delta(h, v_0) = \frac{h}{2} + \frac{\rho g H_0 h^{2+n}}{12 k v_0^n L_b}. \quad (9)$$

The consistency index k is temperature dependent, and this dependence is represented by an Arrhenius law referenced to the lower bound of the casting window,

$$k(T) = k_{\text{ref}} \exp \left[\frac{E_a}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{ref}}} \right) \right], \quad (10)$$

where E_a is the activation energy of the consistency, R the universal gas constant, and T_{ref} the reference temperature. The rheological and geometric inputs used in the evaluation of Eqs. (9) and (10) are summarised in Table 2. These values were not measured on the project slurry; they are representative figures for ethanol-based, high-solid-loading ceramic tape-casting suspensions reported in the literature [6, 7, 13], adopted here to make the parametric trends concrete and to size the mechanism.

Operating window. Evaluating Eq. (9) over the full gap and speed ranges at a representative casting temperature of 60 °C produces the thickness map shown in Figure 14. The map confirms that the blade gap is the primary control variable: because the drag term contributes $h/2$ to the wet thickness, the contours of constant δ run almost horizontally, and the casting speed acts only as a secondary trim through the pressure term. For a target wet thickness of 100–300 μm , which sinters down to the sub-100 μm membranes required for NASICON electrolytes [13], the usable gap lies between approximately 0.20 and 0.55 mm. This result directly justifies the integration of precision micrometer heads on the blade carriage: the working gap is a small fraction of the 5 mm mechanical range, so a 10 μm setting resolution corresponds to only a few percent of the target

Table 2: Assumed slurry and geometric parameters for the casting analysis

Parameter	Symbol	Value
Slurry density	ρ	1700 kg m ⁻³
Power-law index	n	0.45
Consistency index (at 40 °C)	k_{ref}	12 Pa s ^{n}
Activation energy of consistency	E_a	30 kJ mol ⁻¹
Reservoir head	H_0	10 mm (range 5–15)
Blade land length	L_b	2.0 mm
Blade width	W	150 mm
Blade-gap range	h	0–5 mm
Casting-speed range	v_0	2–45 mm s ⁻¹

thickness, whereas a coarse hand-set blade would be unusable in this regime.

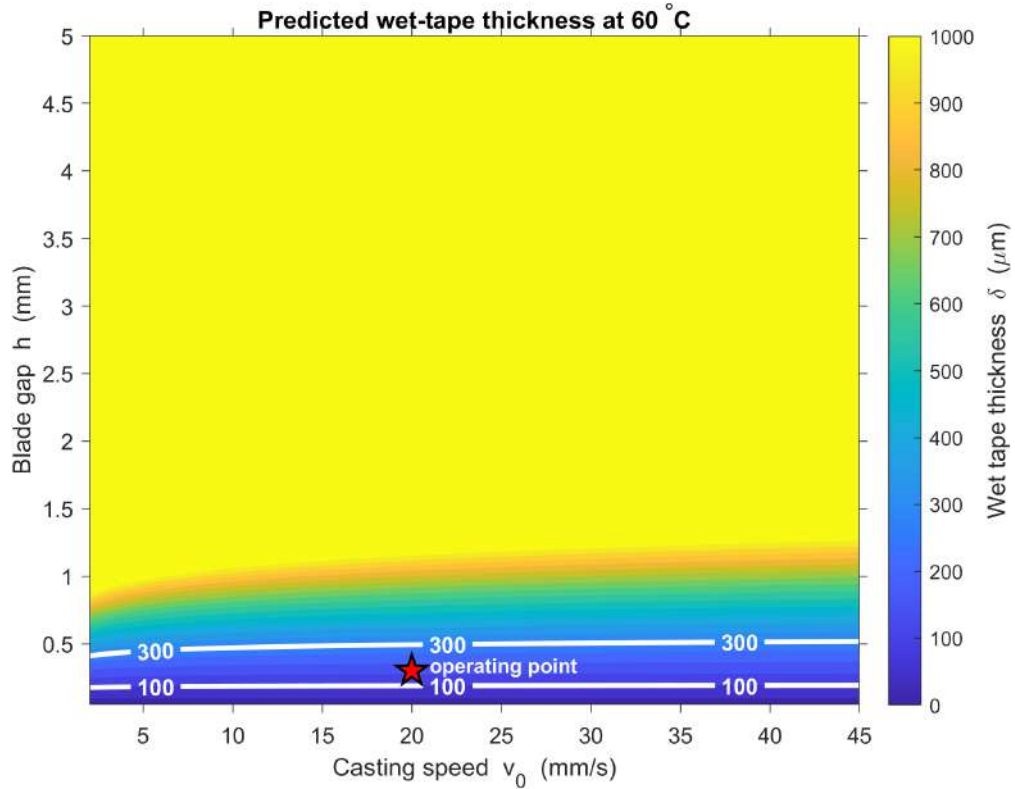


Figure 14: Predicted wet-tape thickness as a function of blade gap and casting speed at 60 °C. The white contours bound the 100–300 μm target window, and the marker indicates a representative operating point.

Influence of substrate temperature. The strong temperature dependence of the slurry rheology is shown in Figure 15, which plots the apparent viscosity against shear rate for substrate temperatures between 40 and 80 °C. Two features are relevant to the design. First, across the shaded band of shear rates actually generated by the machine, the suspension is never Newtonian: the apparent viscosity falls by more than an order of magnitude as the operating point shifts from a

slow, wide-gap cast to a fast, narrow-gap cast. Second, a temperature increase of 40 °C reduces the consistency by roughly a factor of four, so even a few degrees of spatial non-uniformity across the casting bed would translate into a measurable local change in viscosity and, through Eq. (9), into a local thickness gradient that is indistinguishable from a blade-gap error during inspection. This sensitivity is the design driver for the temperature-uniform heated vacuum chuck analysed in Section 2.3.5.

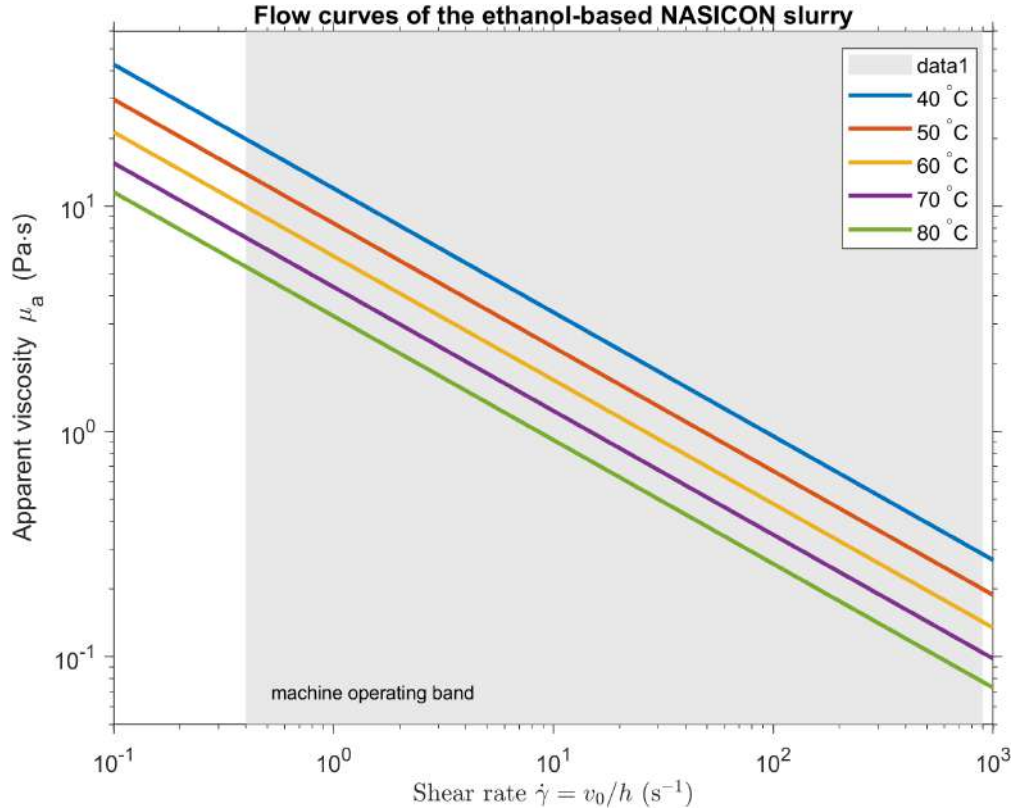


Figure 15: Flow curves of the assumed ethanol-based NASICON slurry at several casting temperatures. The shaded band marks the shear-rate range $\dot{\gamma} = v_0/h$ accessible to the machine.

Velocity sensitivity. Only the pressure term of Eq. (9) depends on casting speed, so the sensitivity of the wet thickness to a velocity fluctuation is

$$\left| \frac{\partial \delta}{\partial v_0} \right| = \frac{n \rho g H_0 h^{2+n}}{12 k L_b} v_0^{-(n+1)}. \quad (11)$$

Figure 16 evaluates the resulting thickness error for a 5 % velocity fluctuation as a function of casting speed for three representative gaps. The error grows sharply at low speed, where the pressure-driven contribution is largest, and at the slowest casts it exceeds a $\pm 5 \mu\text{m}$ tolerance that is significant for a sub-100 μm membrane. This behaviour is the quantitative justification for the micro-stepping linear drive specified in Section 2.3.2: a conventional drive subject to stick-slip or speed ripple would imprint these fluctuations onto the film, whereas the deterministic, stutter-free traverse holds v_0 constant and suppresses the dominant source of thickness scatter.

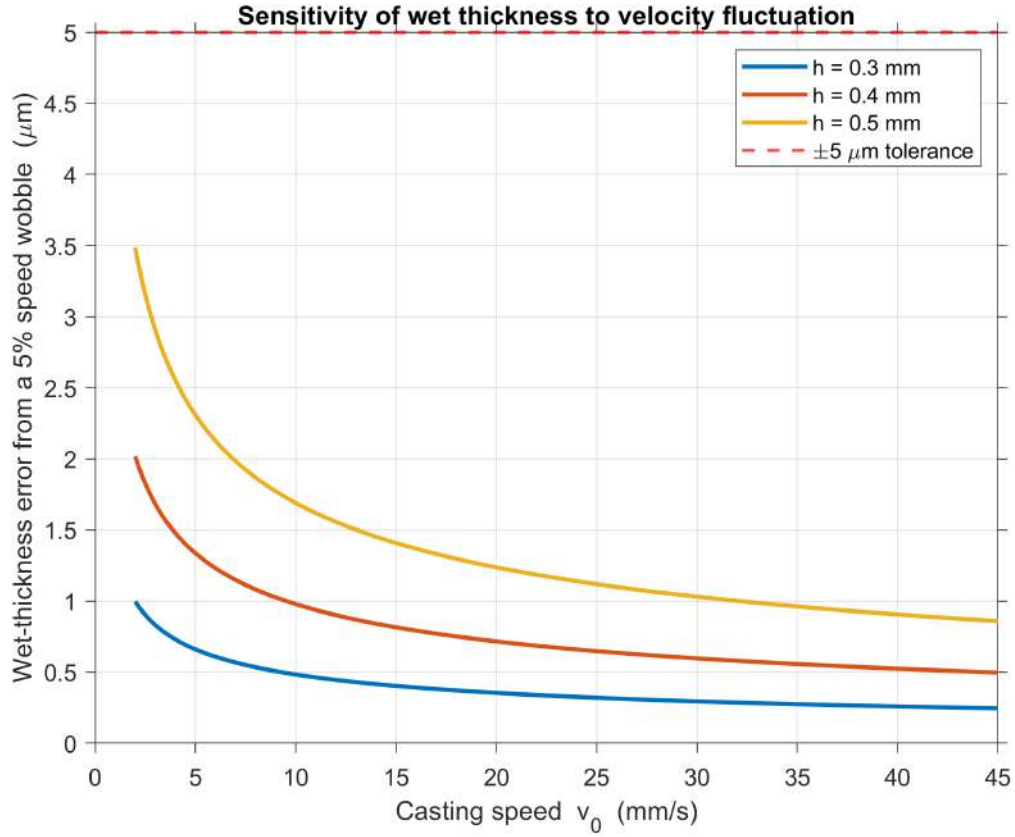


Figure 16: Wet-thickness error produced by a 5 % casting-speed fluctuation, as a function of speed for three blade gaps. The dashed line marks a $\pm 5 \mu\text{m}$ reference tolerance.

Mechanical realisation of the blade assembly. The foregoing analysis is embodied in a rigid, micrometer-adjustable Doctor Blade assembly. The blade is held in an aluminium U-frame that spans the 150 mm casting width and provides the stiff datum required to keep the gap h constant across the full blade length; any bending of the blade under the hydrostatic load would appear directly as a lateral thickness gradient. Two precision micrometer heads set the gap independently at each end, allowing both the absolute gap and a deliberate cross-cast taper to be dialled in with $10 \mu\text{m}$ resolution, consistent with the operating window identified in Figure 14. A stainless-steel blade was selected for its wear resistance and chemical compatibility with the volatile ethanol-based slurry, and its tip is chamfered to define a short, well-controlled land length L_b ; a longer land would increase the pressure-driven term in Eq. (9) and amplify the velocity sensitivity of Eq. (11). Light preload springs maintain the blade seating without over-constraining the frame, so that the gap set by the micrometers is preserved during casting. The detailed manufacturing of these parts is described in Section 3.2.2.

2.3.4 Forced-convection drying and ventilation subsystem

After deposition, the cast slurry film must release its volatile solvent in a controlled manner while still resting on the heated bed, so that the binder network consolidates uniformly without forming a surface skin that would trap solvent beneath it. The ventilation subsystem governs this stage. Two Power Logic PL80S12M brushless DC fans ($80 \times 80 \times 25$ mm, 12 V, 0.14 A per unit) are mounted in the pivoting lid of the casting chamber. When the lid is closed and the fans are energised, they establish a forced-convection flow across the film surface, continuously sweeping the solvent-laden boundary layer out of the enclosure. The purpose of this analysis is to quantify the solvent-removal benefit of forced ventilation relative to the unventilated baseline in which the chamber is closed

but the fans remain off.

Convective mass-transfer model. The combined volumetric airflow of the two fans is $Q = 2 \times 25 \text{ CFM} \approx 0.024 \text{ m}^3 \text{ s}^{-1}$. Distributed across the chamber cross-section, this yields a mean velocity over the film of

$$u = \frac{Q}{L_{\text{ch}} W_{\text{ch}}} = \frac{0.024}{0.40 \times 0.35} \approx 0.169 \text{ m s}^{-1}, \quad (12)$$

where L_{ch} and W_{ch} are the internal plan dimensions of the closed chamber. The corresponding Reynolds number over the film length $L_f = 0.20 \text{ m}$ is $Re_L = uL_f/\nu \approx 1.9 \times 10^3$, which lies well within the laminar regime ($Re_L \ll 5 \times 10^5$). The Sherwood number then follows from the classical laminar flat-plate correlation [30],

$$Sh = 0.664 Re_L^{0.5} Sc^{1/3}, \quad (13)$$

in which the Schmidt number $Sc = \nu/D_{\text{EtOH-air}} \approx 1.31$ uses the temperature-corrected diffusivity of ethanol vapour in air ($D \approx 1.37 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$). Substituting gives $Sh \approx 31.5$, from which the forced-convection mass-transfer coefficient is $k_m^{\text{forced}} = ShD/L_f \approx 2.15 \times 10^{-3} \text{ m s}^{-1}$. With the fans off, the quiescent air supports only natural convection, for which an order-of-magnitude estimate gives $k_m^{\text{natural}} \approx 2.0 \times 10^{-4} \text{ m s}^{-1}$, roughly an order of magnitude lower.

Coupled evaporation and enclosure model. The solvent surface concentration at the bed temperature is set by the ethanol saturation pressure, evaluated from the Antoine equation ($\log_{10}(P^*/\text{mmHg}) = 8.204 - 1642.9/(230.3 + T_C)$). At a representative bed temperature of 48°C this gives $P^* \approx 26.7 \text{ kPa}$ and a saturation mass concentration $C^* \approx 0.46 \text{ kg m}^{-3}$. The drying dynamics are then described by a well-mixed enclosure model that couples the solvent mass remaining in the film, m , to the vapour concentration in the chamber air, C_{ch} :

$$\frac{dm}{dt} = -k_m A_f (xC^* - C_{\text{ch}}), \quad (14)$$

$$\frac{dC_{\text{ch}}}{dt} = \frac{k_m A_f (xC^* - C_{\text{ch}}) - QC_{\text{ch}}}{V_{\text{ch}}}, \quad (15)$$

where $x = m/m_0$ is the instantaneous solvent depletion fraction, A_f the exposed film area, and V_{ch} the chamber volume. The evaporation driving force is the difference between the surface concentration xC^* and the bulk chamber concentration C_{ch} . In the unventilated case ($Q = 0$) the evaporated solvent has nowhere to escape, so C_{ch} rises and progressively suppresses the driving force; in the ventilated case the fans continuously replace the chamber air, holding C_{ch} near zero. Equations (14) and (15) were integrated numerically using the MATLAB script listed in Appendix D, with film parameters consistent with Section 2.3.3 (wet thickness $250 \mu\text{m}$, solvent volume fraction 0.45, giving an initial ethanol mass of approximately 2.66 g in the 18.2-litre chamber).

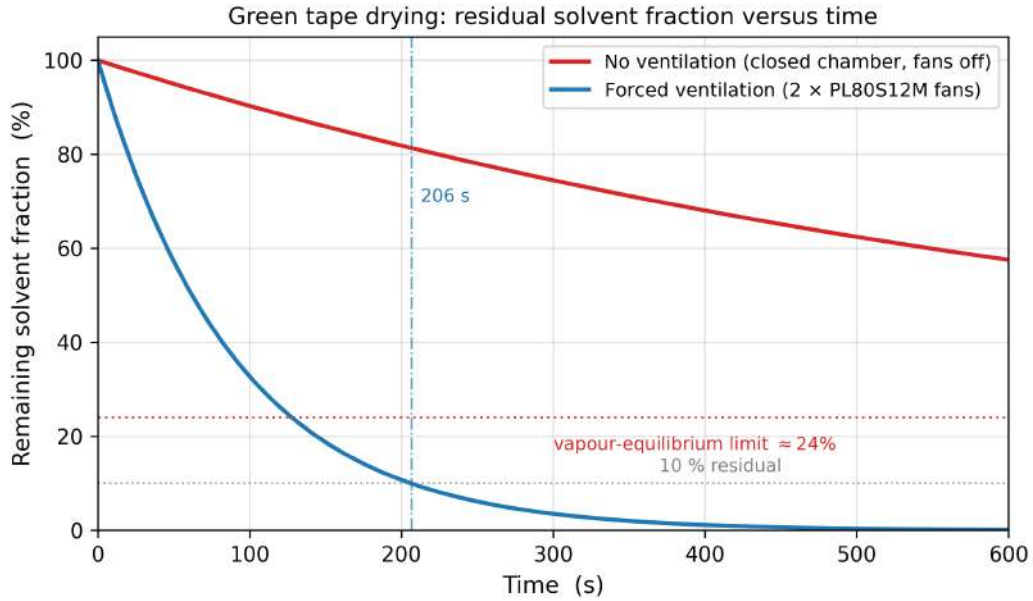


Figure 17: Residual solvent fraction in the cast film as a function of time for the unventilated and forced-ventilation configurations. With two fans, the film reaches 10 % residual solvent in approximately 206 s and dries essentially completely within about 400 s. Without ventilation, the closed chamber approaches vapour equilibrium and drying stalls at a residual fraction of roughly 24 %, leaving a substantial quantity of solvent trapped in the film.

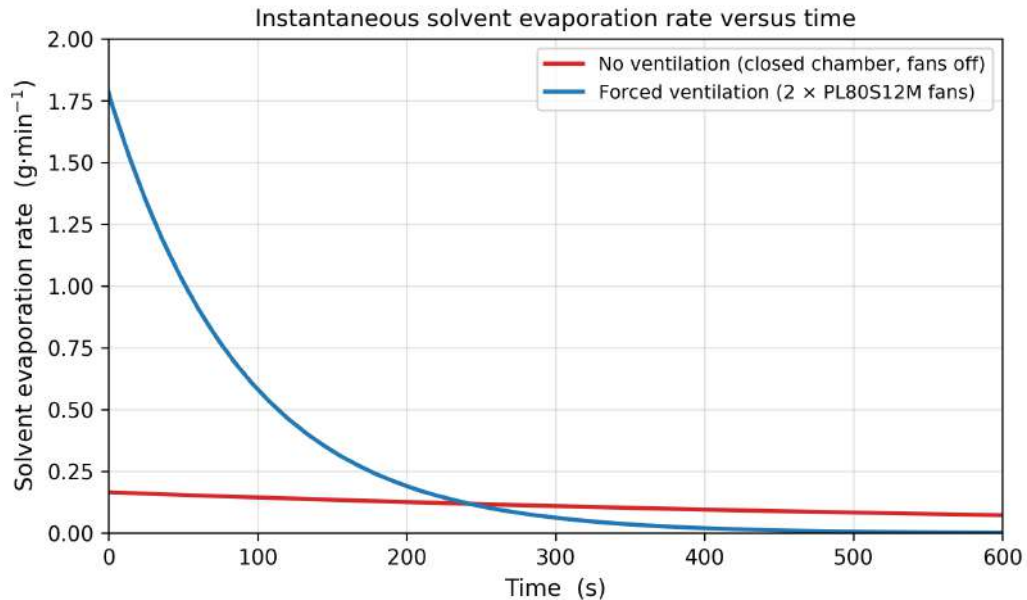


Figure 18: Instantaneous solvent evaporation rate versus time. Forced ventilation produces an initial evaporation rate of about 1.8 g min^{-1} , roughly an order of magnitude above the unventilated rate, and sustains it until the film is nearly depleted. The unventilated rate remains low throughout because the accumulating chamber vapour suppresses the evaporation driving force.

The results in Figures 17 and 18, see Appendix D (6) for the matlab code, confirm that forced ventilation is a functional requirement of the process rather than an auxiliary feature. The two fans raise the mass-transfer coefficient by an order of magnitude and, more critically, prevent the

vapour saturation that would otherwise arrest drying in the sealed chamber. Without them, the film cannot be dried below approximately 24 % residual solvent regardless of holding time, which would compromise the structural integrity of the green tape. The analysis therefore validates the selection of the two PL80S12M fans and the closed-chamber ventilation strategy adopted in the final design.

2.3.5 2D heat transfer and PID control simulation

A 2D finite difference method (FDM) simulation was conducted in MATLAB to analyze the steady-state thermal gradients across the heating bed and result are in Figure 19. The model evaluates the 20 mm Al 5083 bottom plate and the 5 mm Al 2045 top plate, incorporating an 800 W centrally located heater and a simulated PID controller. The thermocouple is positioned at the edge of the bottom plate. The simulation determines the thermal lag and steady-state temperature distribution, verifying the thermal homogeneity of the casting surface. The MATLAB script is provided in Appendix E (6).

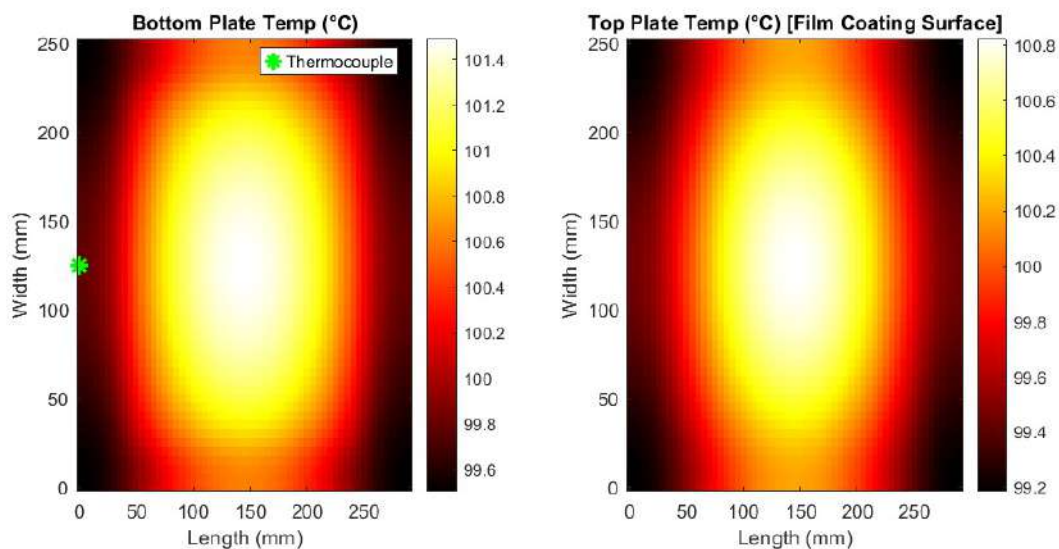


Figure 19: Steady-state 2D temperature distribution across the bottom and top plates under PID control.

2.3.6 Thermal elongation and shear mismatch analysis

To assess the thermomechanical stability of the vacuum chuck, the thermal elongation of the top and bottom plates was simulated and the results are in Figure 20. The analysis compares the dimensional changes of the Al 5083 bottom plate and the Al 2045 top plate from ambient temperature (22°C) to 120°C. The model quantifies the planar expansion along the X and Y axes and calculates the maximum shear mismatch between the layers to ensure the structural integrity of the stacked assembly. The MATLAB script is detailed in Appendix F 6).

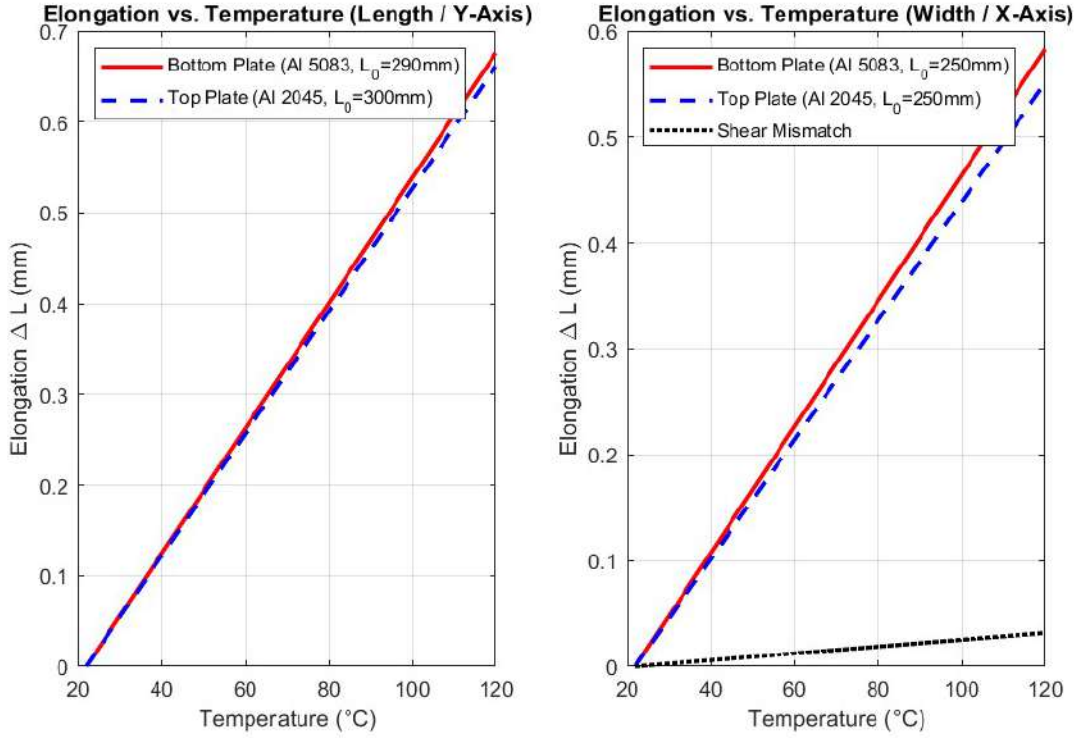


Figure 20: Thermal elongation profiles and resulting shear mismatch between the bottom and top plates.

2.3.7 Electrical control and safety architecture

The electrical system was architected around the principle established by the Binary Dominance Matrix, in which operator safety dominates every other criterion. The design therefore enforces a strict separation between the 220 V alternating-current domain and the low-voltage direct-current logic domain, and it routes mains power through a cascade of protective stages before any energy reaches a load. The complete wiring schematic is provided in Appendix H (6).

Mains Entry and Protective Cascade

Mains power enters through an IEC C14 inlet that integrates an EMI/RFI line filter. This filter attenuates conducted high-frequency noise in both directions: it shields the Arduino, the LCD, and the PID controller from supply transients and harmonics that could otherwise cause logic lock-ups or erroneous temperature readings, and it prevents the switching noise produced by the motor driver and the solid-state relay from being injected back into the laboratory mains [26]. The filtered phase conductor passes through a Siemens 16 A miniature circuit breaker, which provides over-current and short-circuit protection for the entire assembly. From the breaker the phase reaches an illuminated KCD rocker switch, whose terminals were soldered rather than crimped to guarantee a vibration-proof joint. Routing the neutral conductor through the second pole of the same switch completes the circuit for its integrated indicator lamp. To ensure comprehensive electrical safety and equipotential bonding, a dedicated protective earth (PE) network was established. Two distinct grounding lines—one extracted from the perforated metal plate supporting the ATX PSU, the motor driver, and the Arduino, and another from the external machine casing—were routed to a centralized earth node. This configuration guarantees that any fault current is immediately shunted to the ground, preventing the metallic chassis from becoming energized.

Power Distribution Bus

Downstream of the main switch, both the phase and the neutral conductors terminate on an

isolated four-rail copper busbar. Dedicating one rail to the common phase and a second rail to the common neutral creates a clean, standards-compliant distribution node from which every subsystem is tapped. This arrangement replaces fragile point-to-point wiring with a single mechanically robust junction and simplifies fault isolation during maintenance.

Low-Voltage Conversion and Common Ground

Direct-current power is produced by a repurposed ATX switched-mode power supply, which rectifies and regulates the 220 V input into stable +12 V and +5 V rails [27]. Because such supplies expect a motherboard start command, the unit is forced into operation by permanently bonding its PS_ON line to ground, which simulates the power-on handshake. The supply also offers intrinsic short-circuit and over-current protection, entering a latched shutdown state when a fault is detected and thereby protecting the downstream electronics. The +12 V rail feeds both the TB6600 driver and the Arduino regulator input (V_{in}), whereas the regulated +5 V rail establishes the common positive line that powers the LCD and the potentiometer wiper. All return paths are consolidated onto a single common-ground node, which is essential for coherent logic-level signalling between the controller and the driver.

Thermal Regulation Loop

Substrate temperature is governed by an Inkbird ITC-100VH controller, which implements a fuzzy-logic PID algorithm with an auto-tuning function [28]. The controller reads temperature through a K-type thermocouple and, instead of switching the load mechanically, issues a 12 V direct-current pulse train, denoted by the "VH" suffix in its model designation. This pulse drives the control side of a solid-state relay (SSR), whose input stage energises an internal opto-coupler; the opto-coupler triggers a light-sensitive semiconductor on the output stage, which switches the 220 V supply to the 800 W mica heater [29]. The decisive advantage of this approach is complete optical isolation, because no electrical connection exists between the low-voltage control circuit and the high-voltage load, so a mains-side fault cannot propagate into the PID controller or the Arduino [29]. The relay is mounted on a heat-sink with thermal compound to dissipate its conduction losses. As a final safeguard, a dedicated manual switch was inserted in the heater phase, since the PID controller energises immediately at power-up and would otherwise drive the heater whenever the stored set-value exceeded the measured value before the operator was ready.

2.3.8 Control firmware and motion sequencing

The supervisory logic runs on an Arduino Uno, which coordinates user input, motion generation, and status display. The complete commented source listing is provided in Appendix G (6), and the pin allocation is summarised in Table 3. Buttons and limit switches are configured with internal pull-up resistors, so each input requires only a signal wire and a ground wire, eliminating external biasing components.

Finite-State Machine

Program flow is organised as a four-state finite-state machine, which guarantees deterministic and repeatable behaviour throughout a casting cycle. In the idle state the controller continuously samples the speed potentiometer, refreshes the display, and holds the driver disabled, so the carriage can be repositioned by hand. Pressing the start button transitions the machine into the coating state, in which the direction line is set forward, the driver is enabled, and pulses are issued at the locked delay until the end limit switch is reached. The machine then enters a hold state and waits for a reset command, after which the return state drives the carriage backward at a fixed 45 mm/s until the home limit switch restores the idle state. Figure 21 depicts these transitions.

Velocity Locking and Signal Conditioning

A defining feature of the firmware is that the commanded velocity is frozen at the instant the start button is pressed. Capturing the potentiometer value once, rather than reading it continuously,

prevents accidental mid-cast adjustments and thereby preserves the constant shear rate required for uniform film thickness. The analog reading is further conditioned by a hysteresis filter that ignores fluctuations smaller than three ADC counts, which suppresses electrical noise and prevents jitter in both the displayed and the executed speed.

Pulse Generation and Display Strategy

Step pulses are produced by a helper routine that toggles the pulse pin HIGH and LOW with an identical delay on each edge, reproducing the timing derived in the previous subsection. A custom delay function preserves microsecond accuracy across the full speed range by splitting long intervals into millisecond and microsecond components. Crucially, the LCD is never refreshed while the motor is moving, because the blocking nature of display writes would disturb the precise pulse timing and reintroduce the stutter that the micro-stepping configuration was designed to remove.

Table 3: Arduino Uno pin allocation (as-built configuration)

Function	Arduino Pin
Step pulse (PUL)	D9
Direction (DIR)	D13
Driver enable (EN)	D7
Speed potentiometer (analog)	A0
Home limit switch	D3
End limit switch	D2
Start button (forward)	D4
Reset button (return)	D5
LCD register-select / enable (RS / E)	D12 / D11
LCD data lines (D4–D7)	D10, D6, A1, A2

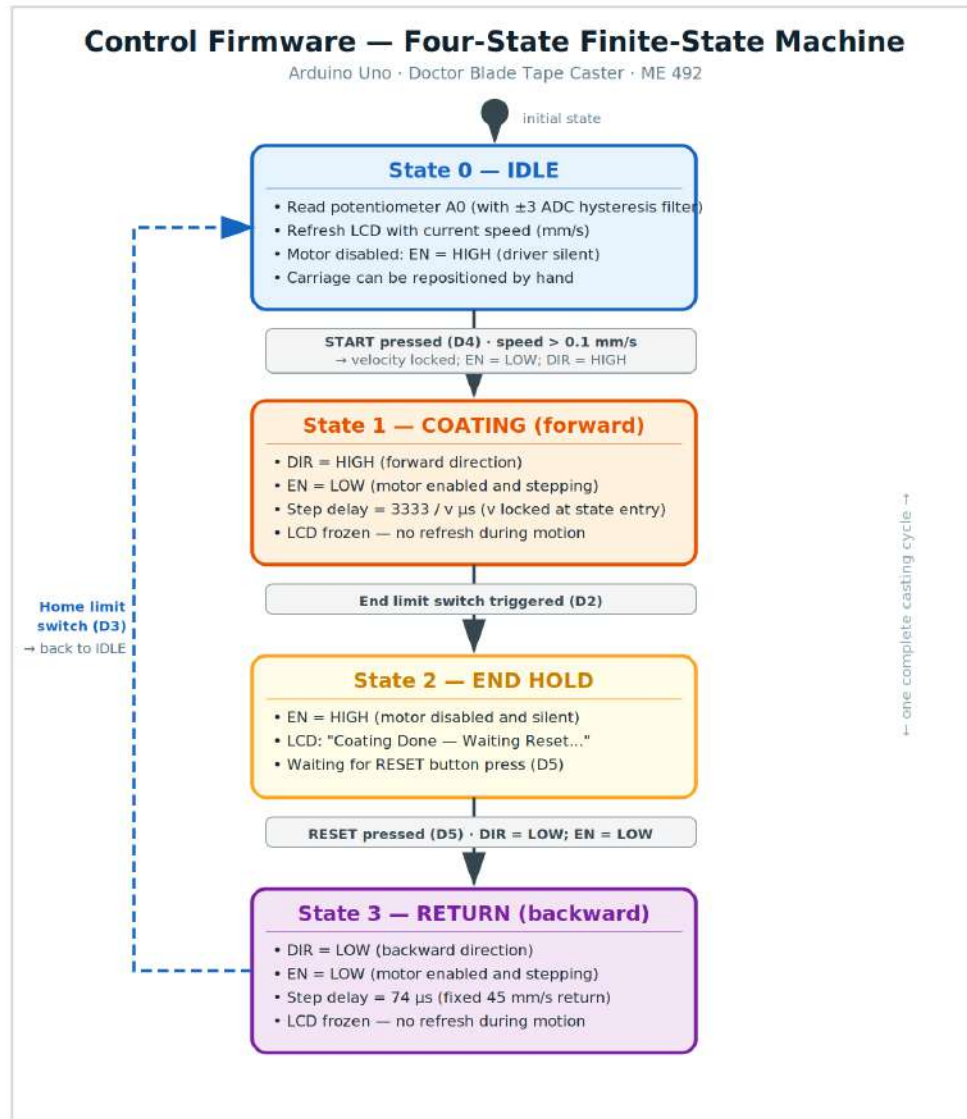


Figure 21: Four-state finite-state machine governing the casting cycle (idle, coating, end hold, and return).

3 Manufacturing Process

To facilitate systematic development and parallel workstreams, the manufacturing phase was organised around seven functionally distinct modules: the vacuum chuck, the Doctor Blade, the linear driving system, the heating system, the cooling system, the enclosure, and the user interface. This modular decomposition allowed each subsystem to be analysed, fabricated, and verified independently before integration onto the common chassis. Material properties referenced throughout this section are tabulated in Appendix A (6).

3.1 Selection of Materials

- **Main Chassis:** A Medium-Density Fiberboard (MDF) base plate was selected as the main chassis, measuring 450 mm × 600 mm × 18 mm. Two principal factors drove this decision: the material's inherently high damping coefficient, which attenuates stepper-motor micro-vibrations before they reach the casting bed, and its ease of processing, since a large number

of through-holes and threaded inserts were required for subsystem mounting.

- **Vacuum Chuck Metals:** Al 2045 aluminium alloy was selected for the top plate of the vacuum chuck on account of its ready availability, ease of machining, and compatibility with the required surface finish. As the top plate is the thinner of the two mating components, its thermal elongation under operating conditions is more pronounced. Al 5083 alloy was therefore specified for the bottom plate; this material pairing minimises the differential in-plane expansion between the two plates during heating and reduces the risk of thermally induced distortion of the assembly.
- **Thermal Seal:** Because the vacuum chuck operates at temperatures exceeding 150 °C, thermal isolation between the heated assembly and the MDF base plate is essential to protect both the structural chassis and the underlying electronic components. Solid PTFE (Teflon) standoffs were selected as the primary thermal break on account of their low thermal conductivity and excellent resistance to sustained elevated temperatures. Mineral wool insulation was additionally applied to seal thermally sensitive interface surfaces.
- **Vacuum Seal:** The central location of the vacuum chuck within the machine required the vacuum supply port to be routed to the user-facing side of the enclosure. Because the plumbing nearest to the chuck is exposed to elevated operating temperatures, thermally compatible materials were specified throughout the vacuum line: brass elbows sealed with PTFE thread-seal tape, a PTFE-lined flexible tube, and silicone adhesive were used to form a leak-free circuit between the chuck outlet and the panel-mounted vacuum fitting.
- **Doctor Blade:** The Doctor Blade assembly employs two complementary structural materials. The U-frame, which carries the micrometer adjustment mechanisms and serves as the kinematic datum for the blade gap, was fabricated from aluminium alloy, exploiting its low density, straightforward machinability, and the high dimensional accuracy achievable at its numerous connection interfaces. The blade itself was machined from 5 mm-thick stainless steel plate, selected for its superior wear resistance and chemical inertness against the volatile organic solvents present in NASICON ceramic casting slurries.
- **Enclosure:** The outer enclosure was fabricated from 1 mm-thick stainless steel sheet metal. Because the enclosure was to be shaped manually using conventional workshop tools rather than laser cutting or CNC press-braking, sheet metal in this thickness offered a practical compromise between formability and structural rigidity. The inherent chemical resistance of stainless steel to the organic solvents employed in ceramic slurry preparation simultaneously satisfies the corrosion-resistance requirement and ensures that accidental spillage does not degrade the protective housing over the service life of the machine.

Several structural interface components, including brackets, spacers, and connectors, were produced in the university machine workshop. Because these parts serve primarily to define relative positions and transmit moderate assembly loads, material selection was governed principally by workshop stock availability. In addition, a number of functional and aesthetic components — including the user interface panel housing, the fan guard grille, and the micrometer connector bracket — were produced by fused deposition modelling (FDM) using ABS filament, a widely used thermoplastic known for its toughness, impact resistance, and adequate heat tolerance for this application.

3.2 Selection of Manufacturing Processes

3.2.1 Vacuum chuck manufacturing

The vacuum chuck comprised two precision-machined components: a top plate and a bottom plate. Dimensional accuracy of the top plate surface was paramount, since the flexible polymer carrier substrate on which the ceramic slurry is cast rests directly upon it; any surface undulation translates directly into a thickness variation in the green tape. Equally, the bottom plate required precise production to minimise thermally induced distortion arising from differential expansion between the two dissimilar-alloy plates.

The top plate was obtained by cutting a 250 mm × 300 mm × 5 mm blank from a larger Al 2045 sheet using a jigsaw, after which all sawn edges were deburred on a bench sander. The 1 mm-diameter vacuum distribution holes specified in the design were pierced using wire electrical discharge machining (EDM), as shown in Figure 22, a process selected to avoid the burr formation and surface damage associated with conventional drilling at this hole diameter. Four 4.5 mm-diameter clearance holes for M5 fasteners were subsequently added on a vertical drill press to secure the top plate to the bottom plate. Although lapping was considered to remove microscopic surface undulations introduced during the original plate casting, the high cost of the operation led to its omission; the completed top plate is shown in Figure 23.



Figure 22: Production of top plate with wire EDM

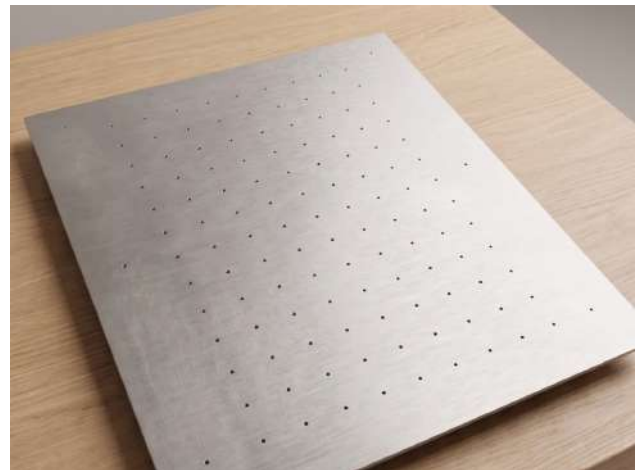


Figure 23: Top plate after production

The bottom plate was produced from a 250 mm × 290 mm × 20 mm Al 5083 H111 blank; since the as-purchased plate was already surface-polished, no additional surface treatment was required. The primary functions of the bottom plate are to replicate the vacuum distribution network of the top plate and to conduct heat uniformly from the mica heater to the casting surface. A network of internal vacuum channels, aligned with the EDM-drilled holes in the top plate and terminating at a quarter-inch NPT outlet, was machined by CNC milling, as can be seen in Figure 25. Six M5 tapped holes were machined into the top face and eight into the bottom face using a CNC drill to achieve precise fastener positioning. To ensure intimate thermal contact between the two mating faces, the top surface flatness was maintained to close tolerance throughout machining. Finally, a peripheral groove 2.8 mm deep and 4.2 mm wide was machined by CNC to seat a 3 mm-diameter FKM Viton O-ring, which provides the vacuum seal between the assembled plates. The completed bottom plate is shown in Figure 24.

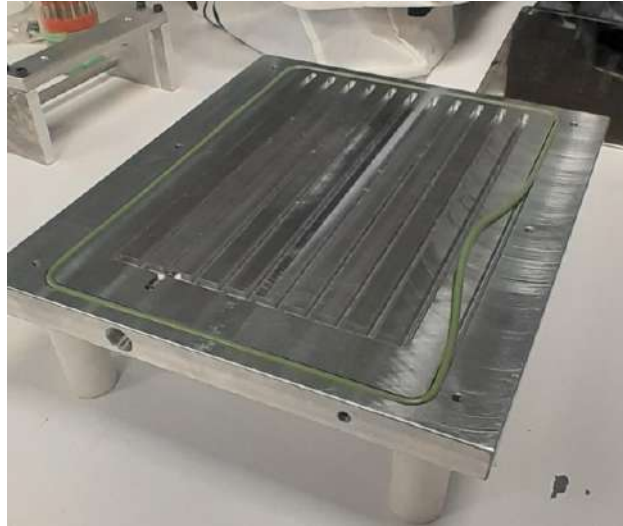


Figure 24: Finalized bottom plate of the vacuum chuck

3.2.2 Doctor Blade manufacturing

The Doctor Blade assembly comprises five components in total, subdivided into the aluminium U-frame and the stainless-steel blade. The front cross-member of the U-frame was rough-cut with a saw and finish-turned on a lathe to its final length of 146 mm. The two side arms were first cut to a square profile, subsequently profiled to their final geometry, and the longitudinal blade-retention slots — 2 mm deep and 5.1 mm wide — were machined on a CNC milling centre. Mounting holes and micrometer seating features in the upper section of the U-frame were produced in the same CNC setup to maintain positional accuracy between all critical interfaces. The 5 mm-thick stainless steel blade was first laser-cut with 0.2 mm of additional material on each edge, then brought to its final dimensions of 40 mm × 150 mm on a CNC milling machine. After the spring-attachment screw holes were drilled, the part was transferred to a vertical chamfering mill, where the blade tip was given its characteristic angled profile, as shown in Figure 26.



Figure 25: CNC milling machine



Figure 26: Vertical chamfer milling

3.2.3 Engineering workshop manufacturing

A range of structural interface components — including brackets, connectors, and kinematic supports — was produced in the Boğaziçi University Mechanical Engineering workshop. Although the available equipment was conventional and most operations were performed manually, the dimensional tolerances demanded of these parts were sufficiently relaxed that workshop-level accuracy did not compromise the overall system performance.

The MDF base plate was first laid out manually and then drilled on a vertical drill press to produce clearance holes compatible with M5, M6, M8, and M10 fasteners. Threaded wood insert nuts were press-fitted into the holes with controlled hammer blows to provide reliable pull-out resistance in the fibreboard substrate.

Intermediate rail–screw connectors, to which the KP08 pillow-block bearings are mounted on the JJ linear rails, were milled from stock material and are visible within the red-bordered insets in Figure 27. A dimensional tolerance of 0.2–0.5 mm was applied to these parts to prevent over-constrained fits during final assembly.

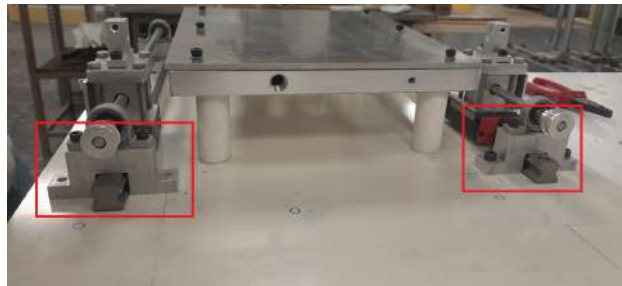


Figure 27: Machined rail lead screw connector

Because the standard T8 lead screw cube bracket is not directly compatible with the JJ linear carriage profile, a bespoke two-piece, four-screw mounting interface was designed. The L-shaped lower bracket, which bridges the cube bracket and the carriage bed, was cut from a pre-prepared section and drilled to the requisite hole pattern on the vertical drill press. The upper clamping piece, which captures the cube bracket from above and transmits its motion to the carriage, was rough-cut on a band saw, then held in a machine vice while screw holes compatible with the carriage were drilled. M5 tapped holes for the SK08 spindle-holder were produced in the same setup. The finished pieces are shown in Figure 28, where the L-bracket occupies the blue-bordered inset and the upper clamping piece occupies the red-bordered inset.

A four-piece NEMA 23 motor bracket was fabricated specifically for this project to secure the stepper motor rigidly to the MDF base plate. Two adjustable idler-pulley sliding joints were incorporated on the front face, flanking the motor-shaft aperture, to permit GT2 belt tensioning. All bracket components were cut to length on a bench saw, and the M4 screw holes were produced on the milling machine.

The perforated metal grid panel used to mount the electronic assemblies was cut to size from a commercial sheet on a bench saw, and all shear burrs were subsequently removed with a rotary grinding attachment.

Four thermal-break standoffs were produced from a solid 20 mm-diameter PTFE rod by first rough-sawing four blanks each 65 mm long, then finish-turning each piece to a precise 60 mm length on the lathe, as shown in Figure 29. M10 tapped blind holes 15 mm deep were produced at both ends of each standoff on the lathe, and the threads were chased with a hand tap to ensure clean engagement with the mating fasteners.

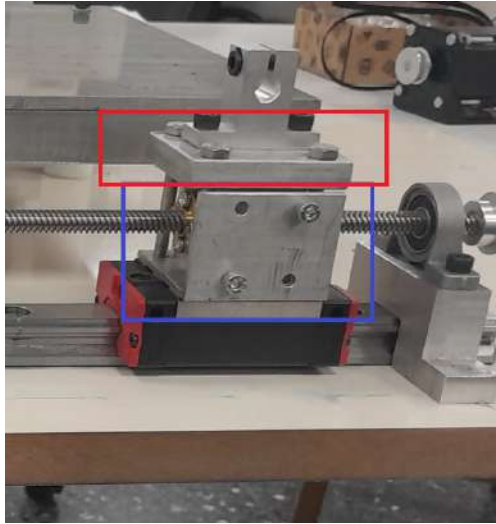


Figure 28: Manufactured L bracket inside blue and top bracket inside red



Figure 29: Teflon standings machined on lathe

3.2.4 Enclosure manufacturing and 3D printing

The stainless steel enclosure panels were fabricated in the university's electronics workshop. Flat blank developments were drawn directly onto 1 mm sheet metal stock, and the primary straight edges were sheared on a bench guillotine. Internal corner reliefs were produced with a dedicated corner-cutter, and the inward-pointing triangular tabs at folded edges were trimmed with a hand guillotine. All panels were then formed to their final geometry on a sheet-metal bending tool equipped with a variable-length holder tip. Prior to bending, rectangular apertures for the circuit breaker, IEC C14 power inlet, and ATX fan outlet were produced on the rear panel, while corresponding openings for the user interface and vacuum fitting were cut in the front panel; each aperture was initiated by drilling corner holes and completed with a jigsaw. Assembly rivet-nut holes of 3 mm and 5 mm diameter were drilled manually, with the sheet supported on a sacrificial timber block and the drill bit lubricated with cutting grease to prevent overheating. Remaining burrs at all cut edges were removed with a rotary grinding attachment.

Several exterior components, including the user interface panel housing, the power-unit fan guard grille, the circuit breaker retainer, and the micrometer connector bracket, were produced by fused deposition modelling using ABS filament. Because ABS contracts slightly during cooling, dimensional compensation was applied in the slicer: prismatic features were modelled 0.1 mm oversize, and cylindrical features 0.2 mm oversize in diameter. Parts that nonetheless fell outside

tolerance after printing were finished to dimension with a hand file.

3.3 Steps of Assembly Processes

The physical assembly of the Automated Tape Casting Machine was conducted in a strict, phase-by-phase hierarchy to ensure kinematic alignment, thermodynamic isolation, and electrical safety.

3.3.1 Linear driving system assembly

Assembly commenced from the MDF base plate outward. Metal threaded insert nuts were pressed into all pre-drilled MDF fastener holes to provide reliable pull-out resistance without stripping the wood fibres. The aluminium extrusion frame was then positioned, and squareness and perpendicularity were verified continuously with a machinist's square before any fasteners were fully torqued. Once alignment was confirmed, each linear guide rail was secured from the underside of the board with three M5 screws and nuts. JJ linear carriages were subsequently slid onto the secured rails. The anti-backlash nut and cube bracket, which together convert lead-screw rotation into carriage translation, were joined with M3 screws and mounted onto the T8 shaft. The workshop-produced rail-screw connector brackets were attached to the rails using M8 Allen-head cap screws, providing support for the KP08 pillow-block bearings at each shaft end; the assembly at this stage is shown in Figure 30. The bearing block at the driven end of each lead screw was preloaded to zero axial play, while the bearing at the free end was left with deliberate clearance to accommodate thermal elongation of the shaft and to provide space for the 30-tooth GT2 pulley.



Figure 30: Initial linear driving mechanism assembly

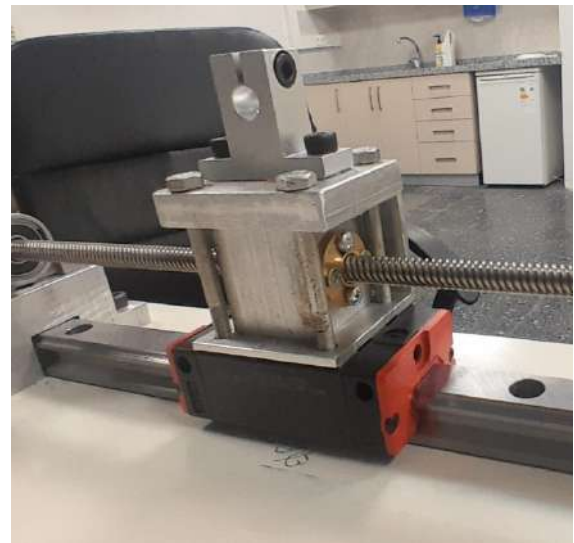


Figure 31: Lead screw bracket and linear rail car assembly

The 30-tooth GT2 pulleys were fitted to the protruding end of each lead screw with a 2 mm Allen key, ensuring a torque-free seating so that the pulley could drive the shaft without introducing parasitic preload. The L-bracket was then inserted between the cube bracket and the carriage bed, providing a direct vertical load path from the nut assembly to the carriage. Two M3 bolts connected the cube bracket to the L-bracket, while the upper clamping piece was fastened to the carriage along its longitudinal axis with 50 mm bolts; to distribute the clamping stress uniformly across the cube bracket faces, the fasteners were tightened in equal incremental steps in a cross-pattern sequence. The SK08 shaft support holder was bolted to the upper clamping piece with two M5

screws, oriented precisely parallel to the shaft axis. The completed assembly at this stage is shown in Figure 31.

The completed lead-screw sub-assembly was verified by hand-turning the screw through the full travel range; any binding or uneven resistance was traced to individual bearing and connector fasteners, which were sequentially loosened and retorqued until smooth, drag-free travel was achieved. The finished first rail system is shown in Figure 32. An identical and symmetrically mirrored second rail system was then assembled on the opposite side of the MDF base plate, as shown in Figure 33.



Figure 32: Assembly of first linear rail lead screw mechanism

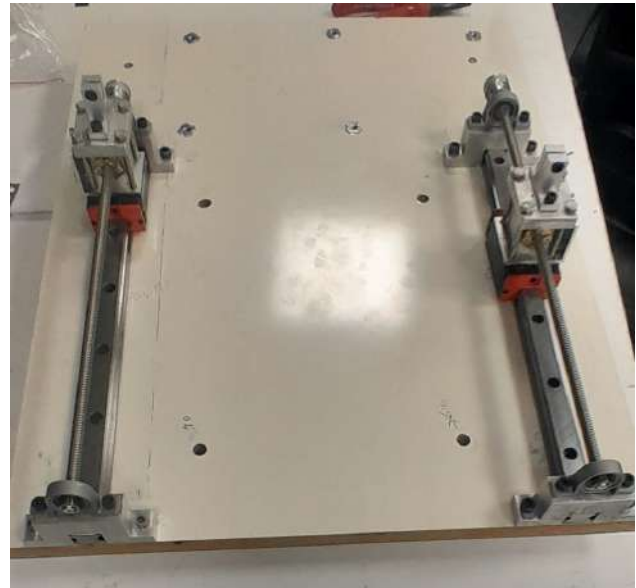


Figure 33: Assembly of both linear rail lead screw mechanism

3.3.2 Vacuum chuck assembly

With the linear drive mechanism in place, the heated vacuum chuck was assembled into the space between the two rail systems. The mica plate heater was first secured to the underside of the bottom plate with four M10 screws and washers, with a thin layer of thermal compound applied at the interface to minimise contact resistance. The four PTFE standoff feet were then attached to the bottom plate via set-screws and anchored to the MDF base from below using 30 mm M10 screws with washers, thermally isolating the heated assembly from the structural chassis. To establish the vacuum seal, the FKM Viton O-ring gasket was seated in its machined groove in the bottom plate before the top plate was lowered into contact. Six M5 Allen-head bolts with washers were installed and tightened in equal incremental steps in a cross-pattern to close the joint uniformly and prevent differential deflection between the plates. Front and side views of the assembled vacuum chuck are shown in Figures 34 and 35.



Figure 34: Front view of vacuum chuck assembly



Figure 35: Side view of vacuum chuck assembly

The quarter-inch NPT brass elbows at the bottom plate outlet were sealed with PTFE thread tape and further bonded with silicone to prevent vacuum leakage at elevated operating temperatures. An 8 mm-diameter PTFE tube was then connected via a 1/4 NPT-to-8 mm hose-end fitting and routed to the panel-mounted vacuum inlet on the user-facing enclosure wall. The K-type thermocouple was inserted into its dedicated hole in the bottom plate and secured with thermal paste to ensure consistent temperature measurement throughout the operating range.

3.3.3 NEMA23 motor bracket assembly

The four-piece NEMA 23 motor bracket was assembled using M4 Allen-head bolts and secured to the MDF base plate in its designated position. The NEMA 23 stepper motor was bolted to the bracket, and a 40-tooth GT2 pulley was clamped to the motor shaft using a 2 mm Allen key. The two idler pulleys were mounted in their adjustment slots on the front face of the bracket with M3 $\times$ 30 mm bolts, washers, and lock-nuts, oriented to rotate freely and align concentrically with the motor pulley. The closed 6 mm-wide, 754 mm-long GT2 belt was routed around all pulleys, after which each idler was translated in its slot and locked to achieve the specified belt tension. The completed drive assembly is shown in Figure 36.

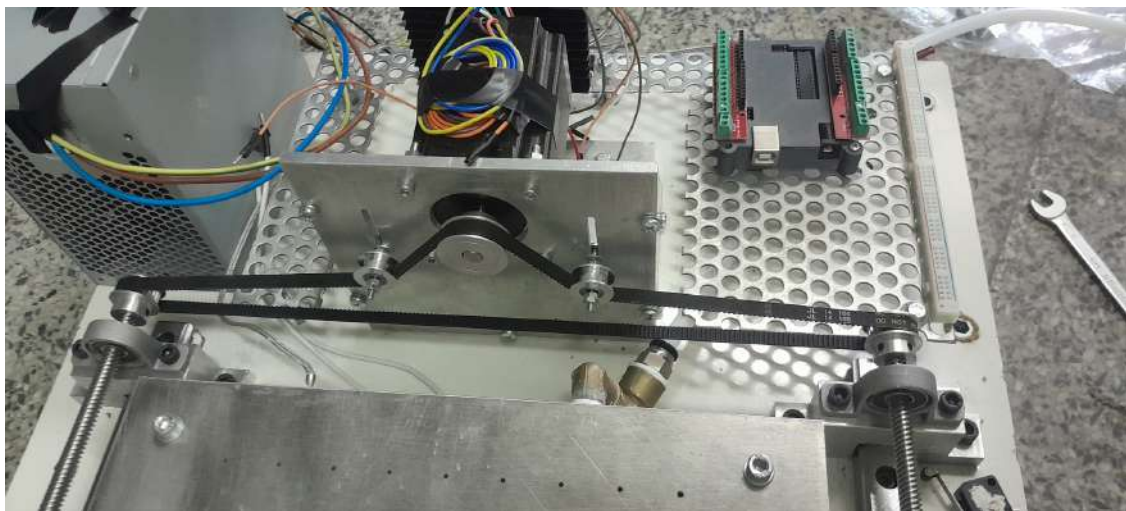


Figure 36: Belt pulley assembly

3.3.4 Doctor Blade assembly

Assembly of the Doctor Blade commenced by securing the U-frame around the blade using Allen-head bolts. The retention springs were anchored at their lower ends to the pre-drilled holes in the

blade body with M4 pan-head screws; their upper ends were fastened by threading a bolt through the base of the U-frame and tightening a nut from above, placing the blade in controlled preload against the gap-setting surfaces. The commercial micrometer heads, which were supplied with C-frame bodies, had their frames trimmed by saw-cutting and the residual protrusions ground flush. A custom ABS bracket, produced by FDM, was used to grip each micrometer spindle and locate it at the top of the U-frame; the parts were permanently bonded with structural adhesive, yielding the final assembly shown in Figure 37.

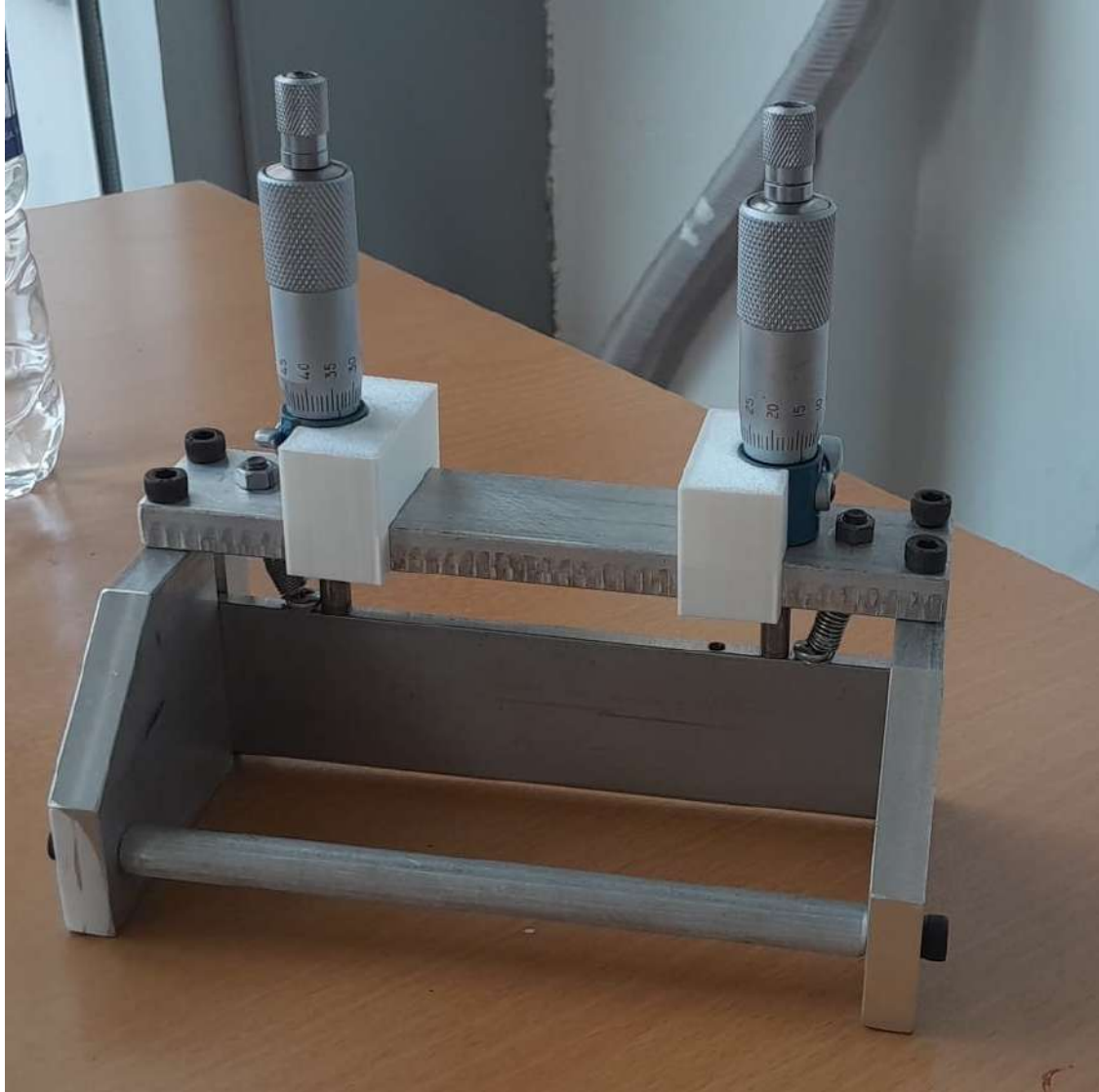


Figure 37: Doctor Blade assembly

3.3.5 Electronics system assembly

The electronic subsystem was assembled inside the stainless-steel enclosure, so that every high-voltage component remained shielded from accidental fluid spills. Assembly proceeded from the mains entry toward the loads, following the protective cascade described in the design phase. The phase conductor was routed from the C14 filtered inlet to the circuit breaker, then to the soldered terminals of the KCD switch, and finally to the common phase rail of the busbar; the neutral conductor was carried through the second switch pole to the common neutral rail. The ATX supply

was tapped from these rails, and its PS_ON line was bonded to ground to keep the unit permanently enabled.

On the direct-current side, the +12 V rail was distributed to the motor driver and to the Arduino regulator input, while the regulated +5 V line and the consolidated ground formed the common logic bus for the controller, the display, and the potentiometer. A consistent wire colour code, listed in Table 4, was adopted throughout to reduce assembly errors and to assist future maintenance.

During prototyping, the common-ground connections were initially made on a breadboard, which proved unreliable under the micro-stepping vibration of the motor; jumper wires worked loose and produced intermittent faults. The breadboard was subsequently removed. Its dedicated side power rails — the continuous positive and negative conductor strips running along the board edges — were detached from the carrier and permanently fixed in place with adhesive. These strips were repurposed as the common positive and common ground distribution buses, respectively, providing stable multi-point connection nodes for all low-voltage supply and return conductors. This modification eliminated the vibration-induced dropouts observed during the prototyping phase and produced a mechanically reliable power distribution arrangement.

Table 4: Wire colour code convention

Wire Colour	Circuit
Brown	Phase (220 V AC)
Blue	Neutral (220 V AC)
Yellow-Green	Earth
Yellow	+12 V DC
Red	+5 V DC
Black	DC ground / common return

3.3.6 Enclosure assembly

Rivet nuts were installed at all through-bolt locations on the manually formed enclosure panels before final assembly. The enclosure panels were secured to the MDF base plate using screw-type wood-insert rivets, bolts, and washers, while the inter-panel connections were made with M3 domed-head Allen-head bolts, selected for their clean aesthetic finish. On the rear control side of the enclosure, the EMI line filter and miniature circuit breaker were fixed to their pre-cut apertures with through-bolts. The 3D-printed fan guard grille was bonded over the ATX PSU fan outlet with silicone adhesive. The user interface panel was secured to the front enclosure face with four Phillips-head bolts and nuts. The vacuum routing was completed by passing 12 mm and 8 mm hose barbs through the sheet metal aperture, sealing each with PTFE tape and silicone; the resulting 8 mm PTFE tube run to the panel surface permits a standard 12 mm vacuum hose to be connected externally. The assembly at this stage is shown in Figure 38.

3.3.7 General assembly

Apart from those described, most of the remaining parts were electronic components. These included the Arduino, motor driver, SSR relay, power distribution busbar, grounding junction, and power supply. These parts were mounted onto a perforated mounting plate cut for the control unit of the machine. The power supply was secured to its designated area using strong silicone adhesive.



Figure 38: Enclosure assembly

3.4 Cost Analysis

The project was initiated at the Boğaziçi University Design Workshop, where a significant portion of the work and required equipment was made available through the department's facilities. The cost breakdown presented in Table 5 reflects the full list prices of all materials, components, and services employed in the project, rather than the amounts actually disbursed; this distinction is important because several high-value items — including machining operations, raw materials, and consumables — were obtained at no charge through institutional or industrial support. The *Covered by University or Labs* column indicates the list-price equivalent of items sourced through Boğaziçi University or its affiliated laboratories. Where this column reads zero, the item was procured via the university but at no direct cost to the project team, reflecting the institution's in-kind contribution of equipment access, workshop time, and existing inventory. The *List Price* column thus encompasses not only commercially purchased items but also the market-equivalent value of university-supplied and gratuitously donated goods and services, providing a realistic picture of the total resources consumed. The *Covered by Team* column records the amounts paid directly out-of-pocket by the project members. Finally, the *Gratis Support By* column identifies the firms and individuals who provided components or manufacturing services free of charge; ERK Rezistans supplied the mica heater and contributed CNC machining and EDM operations in collaboration with Güveniş Makine, NOTA Entegre Ahşap Orman Ürünleri donated the MDF base plate, and Selamet and Osman Bey provided a range of miscellaneous parts and fabrication support. The combined effect of these contributions substantially reduced the out-of-pocket expenditure relative to the total project value.

Table 5: Project cost breakdown

Product / Process Name	List Price (TL)	Covered by University or Labs	Covered by Team	Gratis Support By
2× Linear Rail Guides + 2× Lineer Car	3,000	3,000		
2× T8 Lead Screw + 2× Nut + 2× Bracket	932	932		
4× KP08 Pillow Block Bearing	320	320		
3× GT2 Pulleys + 1× Closed Loop Belt	400	400		
NEMA 23 Stepper Motor + Bracket + TB6600 Motor Driver	3,000	3,000		
Ardunio Uno R3	300	0		
Power Supply	750	0		
20×250×290 mm 5083H111 Al Plate + 5×250×300 mm 2045 Al Plate	2,000	0		
CNC Lathe and Milling + EDM Hole Popper	5,000			ERK Rezistans and Güveniş Firms
Vacuum Pump (Value 135N)	9,000	0		
2× Micrometer	900	900		
PID Temperature Controller	1,500	0		
Mica Heater	1,200			ERK Rezistans
Metal Case for Outside	0	0		
MDF Plate	300			NOTA Entegre Ahşap Orman Ürünleri
2× Fans	400	0		
Dr Blade Parts (Al and Stainless Steel)	600		600	
Other Expenses	2,500	0		
M3, M4, M6 Bolt Sets, Screws, Nuts and Washers	500		500	
Threaded Wood Insert and M3 Rivet Nut	400		400	
Butt Hinge, Cabinet Handle and Cabinet Door Gas Strut	250		250	
Spray Paint	140		140	
Heat Shrink Tubing Assortment Kit, Power Cords and Jumper Wires	350		350	
IEC Male Power Inlet with EMI Filter	400		400	
Buttons, Limit Switches	250		250	
Other Parts (some fasteners/connection components, aluminum for the NEMA 23 stepper motor bracket and its machining, casing materials and bending operations, other cutting and turning/lathe operations, 3D printer parts)	0			Selamet and Osman Bey
Ardunio Uno R3 Mounting Case	350		350	
TOTAL	34,392	8,552	3,240	

4 Project Management

Normally spread over two semesters as ME429 Mechanical and Thermal Design and ME492 final project, this project involved taking the two-semester portion of the course from scratch and completing it solely in the ME492 semester. Therefore, project management was crucial in the systematic progress of the automatic tape casting machine, from its initial design to the fully functional prototype.

The project scope was divided into structured Work Packages, clearly defining subtasks, milestones, and resources. The timeline, task dependencies, and execution of these packages were tracked using a 15-week Gantt Chart (see Figure 39 for a visual timeline).

4.1 Work Packages and Subtasks

The project workflow was divided into the following sequential and parallel Work Packages:

WP1: Project Definition and Literature Review (ME 429) This package encompassed the initial research and theoretical framing of the project.

- *Subtask 1.1:* Literature review on NASICON solid-state electrolytes, non-Newtonian slurry rheology, and the thermodynamics of drying kinetics.
- *Subtask 1.2:* Market analysis of existing commercial coaters (e.g., MTI MSK-AFA-MR200, Ossila) to benchmark current limitations in open-architecture design and affordability.

WP2: Conceptual Design and Selection (ME 429) This package focused on establishing the engineering constraints and evaluating potential architectures.

- *Subtask 2.1:* Establishment of Product Design Specifications (PDS) regarding safety, velocity stability, precision gap control, and other criteria.
- *Subtask 2.2:* Evaluation of six distinct coating concepts utilizing a Binary Dominance Matrix and a weighted Decision Matrix, resulting in the selection of the custom open-architecture tape caster.

WP3: Detailed Mechanical and Electrical Design (ME 429) This package translated the selected concept into an actionable engineering model.

- *Subtask 3.1:* 3D CAD modeling of the bimetallic vacuum chuck (Al 2045 top, Al 5083 bottom), U-frame Doctor Blade, and chassis.
- *Subtask 3.2:* Theoretical calculations for fluid dynamics (wet tape thickness vs. velocity), convective mass-transfer for the cooling fans, and NEMA 23 stepper motor torque requirements.
- *Subtask 3.3:* Design of the electrical safety hierarchy and the TB6600 driver / Arduino control schematic.

WP4: Material Selection and Procurement (ME 492 Transition) This package managed the supply chain logistics between the two semesters.

- *Subtask 4.1:* Sourcing commercial off-the-shelf (COTS) components (linear guide rails, trapezoidal lead screw, mica heater, PID controller).
- *Subtask 4.2:* Procurement of raw materials for custom manufacturing (MDF, stainless steel sheet metal, aluminum blocks, Teflon, ABS filament).

WP5: Manufacturing of Custom Components (ME 492) This package involved the physical fabrication of parts within the university workshops.

- *Subtask 5.1:* CNC milling the vacuum channels and wire EDM drilling the 117 micro-perforated holes into the Al 2045 top plate.
- *Subtask 5.2:* Machining the Doctor Blade U-frame, stainless steel knife, and NEMA23 motor brackets.
- *Subtask 5.3:* Manual cutting, drilling, and CNC press-braking of the 1mm stainless steel outer enclosure.
- *Subtask 5.4:* 3D printing (FDM) the ABS user interface, fan guards, and micrometer connectors.

WP6: System Assembly and Integration (ME 492) This package focused on marrying the mechanical, electrical, and pneumatic subsystems.

- *Subtask 6.1:* Anchoring the aluminum skeleton to the MDF base and assembling the linear drive mechanism (rails, carriages, lead screw).
- *Subtask 6.2:* Assembling the bimetallic vacuum chuck, securing the Viton seal, and routing the high-temperature pneumatics using brass fittings and Teflon tubing.
- *Subtask 6.3:* Wiring the AC protective cascade (circuit breaker, KCD switch, SSR) and the DC logic bus.

WP7: Software Development and Calibration (ME 492) This package developed the control logic of the machine.

- *Subtask 7.1:* Writing the Arduino C++ firmware, implementing the four-state finite-state machine, and tuning the step-delay constants.
- *Subtask 7.2:* Calibrating the Doctor Blade gap zero-point using a feeler gauge and aligning the linear rails.

WP8: Testing, Optimization, and Documentation (ME 492) This package validated the machine against the initial PDS and finalized academic deliverables.

- *Subtask 8.1:* Verifying kinematic stability, PID thermal uniformity up to 150°C, and active vapor extraction efficiency.
- *Subtask 8.2:* Compilation of the final thesis report, cost analysis, and preparation for the jury defense.

4.2 Roles of team members

The successful execution of this multidisciplinary project required close collaboration between the two team members, Muhammet Ali Doğan and Buğrahan Güreşçi. While both members participated equally in the conceptualization, literature review, and general assembly phases, primary responsibilities were divided according to technical domains to maximize efficiency:

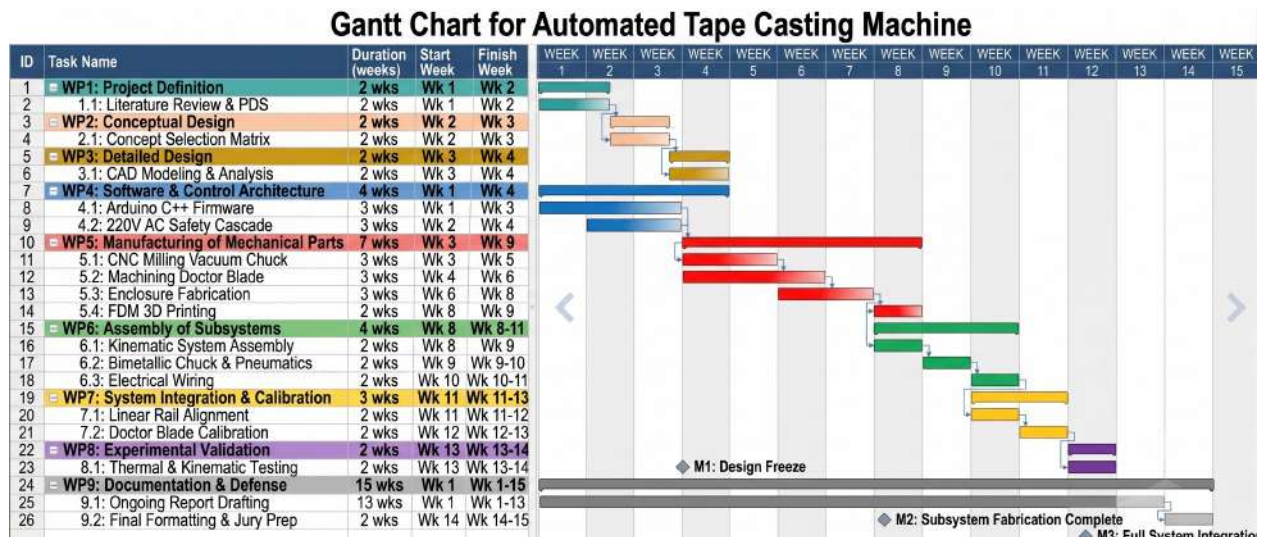


Figure 39: Gantt Chart

- **Mechanical Design and Manufacturing:** Focused primarily on the CAD modeling of the subsystems, fluid dynamics analysis, and the physical workshop fabrication of the custom aluminum and stainless steel components (e.g., CNC milling the vacuum chuck and machining the Doctor Blade).
- **Mechatronics, Controls, and Thermodynamics:** Focused primarily on the electrical architecture, the protective AC cascade, writing the Arduino firmware (finite-state machine), and analyzing the thermodynamic drying kinetics and convective mass transfer of the cooling fans.

Note: Both members collaboratively executed the final system integration, troubleshooting, and preparation of the final ME 492 report.

4.3 Project milestones and realizations

To benchmark progress, three major project milestones were established and successfully realized:

1. **Milestone 1: Design Freeze (First Month of Spring Term - ME492):** Realized upon the completion of all theoretical calculations, 3D CAD models, and the finalization of the Bill of Materials (BOM).
2. **Milestone 2: Subsystem Fabrication Complete (Mid Spring Term - ME 492):** Realized when all custom components (bimetallic chuck, U-frame, enclosure) were successfully machined and verified against dimensional tolerances.
3. **Milestone 3: Full System Integration (Late Spring Term - ME 492):** Realized when the mechanical, electrical, and pneumatic systems were assembled into a single unit, and the Arduino firmware successfully executed a stutter-free casting cycle.

5 Discussions

5.1 Interpretation of the Design Process

The design and manufacturing process of the Automatic Micrometer-Controlled Tape Casting Machine has demonstrated that, even within the limited resource context of a license project, a theo-

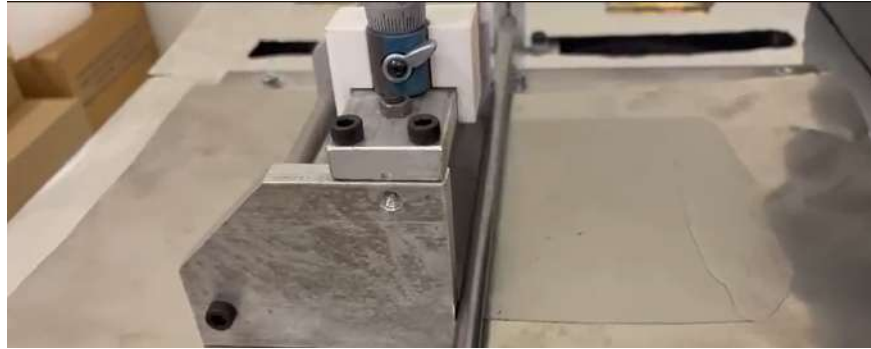


Figure 40: Coating by Automated Tape Casting Machine

retically rigorous design approach can produce a functional and far more cost-effective laboratory instrument that, in several key aspects, surpasses and even catches up with its commercial competitors. Several design decisions, in particular, deserve critical scrutiny in light of their practical performance. Figure shows a moment while coating.

The selection of MDF as the main base material, initially an unusual choice for a precision instrument, was one of the most cost-effective decisions in the design, shortening and simplifying assembly time. Furthermore, it significantly dampened the vibration caused by the stepper motor, ensuring the table remained stable; however, it is estimated that the microstepping regime with 1600 pulses per motor revolution essentially produced imperceptible surface distortion. Additionally, while a metal chassis might have been preferable for aesthetic reasons and its potential for longevity, it would have transmitted the motor's 1.8-degree step pulses directly to the liquid film. Considering this outcome, the choice can be considered the right one.

Theoretically, the motor rotational speed was studied, and its effect on the shaft's traverse speed via the belt-pulley system was experimentally tested by determining two fixed points and timing them. The results of this test showed no difference between the theoretically calculated and the actual shaft traverse speed.

The bimetallic vacuum chuck design —combining an Al 2045 top plate for hardness and surface quality with an Al 5083 bottom plate for machinability and groove depth integrity— showed no noticeable buckling when tested by raising the chuck to different temperatures. However, the difference in thermal expansion coefficients between the two alloys was a sound choice: a larger difference would have caused thermally induced gap change between the plates, progressively degrading the Viton seal during the operating temperature cycle. Tests with a heater at different temperatures further confirmed that the PTFE support feet provided effective thermal insulation between the heated chuck and the MDF base, protecting both the structural plate and the electronics from continuous conductive heat ingress.

The first problem with the developed Doctor Blade prototype was the relatively low quality of the purchased micrometers. The measuring extensions of the micrometers used to cut the C-frames differed by 0.8 mm, preventing the two micrometers from calibrating and moving equally when adjusting the Doctor Blade gap. Therefore, a more manual method was necessary to adjust the gap by monitoring how much the two micrometers extended and shortened. In addition, the hand-crafted U-frame, due to its cost, prevented the side feet from being exactly the same and of the correct dimensions. On a micro-scale, when the Doctor Blade was placed on a flat surface, it could be seen to limp when its left foot was pressed from the front. However, since the spindle pushed the Doctor Blade from the back, it was observed during the tests that this micro-height difference at the front did not affect the blade height or cause slurry leakage.

Since homogeneous heat distribution is crucial in the system, the bottom plate was chosen as a bulky material that would facilitate heat diffusion. A 2D heat transfer analysis revealed how

the mica plate heater heated the bottom of the bottom plate and how this heat was reflected onto the top surface. Based on this analysis, a point on the edge of the bottom plate closest to the top plate's temperature was chosen for thermocouple placement. After calibration, tests showed that the difference between the thermocouple's measured temperature and the top plate temperature did not exceed 1 degree Celsius. Similarly, the difference between the coldest and hottest points of the top plate was found to be approximately 0.5 degrees Celsius on average. Since these results aligned with the analysis, it was concluded that the placement was the correct decision.

Lastly, to evaluate whether the device was functioning correctly, the mixture, which constitutes the basic building block of the electrolyte slurry, was prepared in the battery laboratory using PE, ACN, and NaClO_4 , as described in Uğur Cucu's article [31]. Different tape castings were performed on aluminum foil using a Doctor Blade with gap clearances of 100 microns and 250 microns. Photographic evidence of these castings is presented in Figure 41 and 42. As anticipated, the vacuum system held the aluminum foil stable, while the evenly distributed heat effectively prevented scaling, and no bubble formation was observed. When the samples were examined under a light microscope, the thickness of the aluminum foil substrate was determined to be 60 micrometers, as shown in Figure 43. Because the mixture contained a high percentage of ACN, the layer thickness decreased significantly after drying. While no measurable layer was obtained from the 100-micrometer gap castings, the final thickness decreased to 80 microns for the 250-micrometer gap castings, as can be seen in Figure 44.

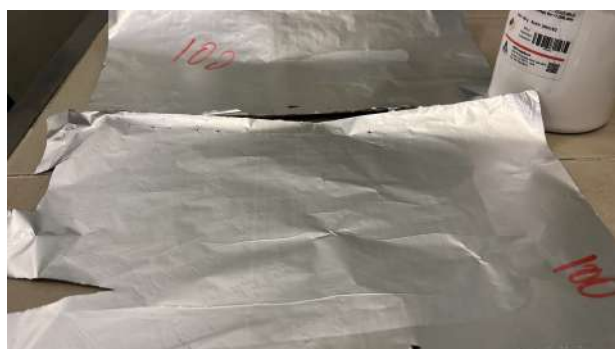


Figure 41: 100 micrometer tape casting on aluminium sheet



Figure 42: 250 micrometer tape casting on aluminium sheet



Figure 43: Sheet aluminium thickness photographed in a light microscope

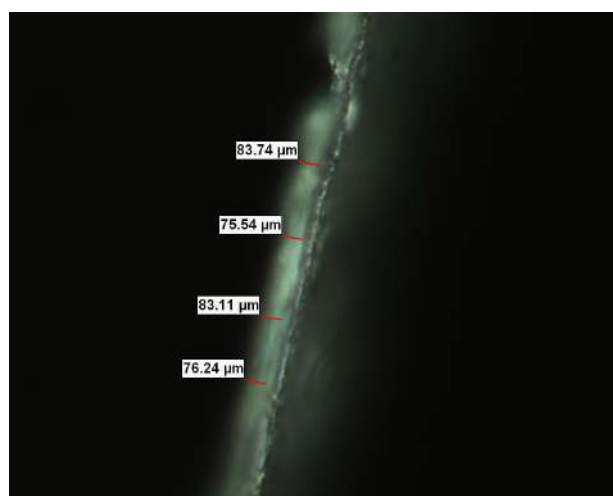


Figure 44: 250 micrometer tape casting result photographed in a light microscope

5.2 Manufacturing Challenges, Failures, and Corrective Measures

During the production phase, several significant challenges emerged that required adaptive engineering reasoning under the limitations of the university's workshop resources. The most critical mechanical issue was ensuring the dimensional compatibility of the manually fabricated components. Deviations between the theoretical CAD models and the physically manufactured parts necessitated real-time design adjustments during assembly. Consequently, adaptive tolerances were introduced at the connection interfaces throughout every step of the production process. This methodology effectively minimized the stack-up errors inherent to manual manufacturing, which was primarily necessitated by strict budget constraints.

For highly critical components requiring strict dimensional precision, external industrial partnerships were established after extensive research. *Güveniş*, located in the İkitelli Organized Industrial Zone—recognized for supporting student engineering initiatives—sponsored the project by utilizing their precision CNC milling and lathe centers to manufacture the stainless steel Doctor Blade and the bottom aluminum plate. Similarly, *Han Metal*, located in Bayrampaşa, provided the precision machining for the top vacuum plate. These industry collaborations allowed the project to achieve the necessary micrometer-level precision while strictly adhering to a low-budget framework.

A second series of notable challenges arose within the high-voltage (220V AC) electrical subsystem, primarily stemming from integration errors during the early prototyping phase. The first incident involved the PID temperature controller; inadequate insulation and exposed wire leads at the power input terminals resulted in an accidental dead short circuit, leading to the catastrophic failure of the component. Similarly, during the preliminary load testing of the mica plate heater, an under-sized wire gauge was inadvertently utilized. The high current draw of the resistive heating element exceeded the wire's ampacity, causing the PVC insulation to melt rapidly due to severe Joule heating.

Furthermore, an incorrect pin configuration during the wiring of the 3-pole illuminated actuator switch caused an internal short circuit. However, this specific failure successfully validated the machine's electrical safety architecture: the integrated Siemens miniature circuit breaker (MCB) tripped instantaneously, isolating the fault and preventing any propagation of damage to the downstream electronics or the Arduino logic board.

These early-stage failures served as critical learning experiences, prompting a strict revision of laboratory safety protocols and standard operating procedures (SOP). The corrective measures included utilizing appropriate AWG wire gauges calculated for specific current loads, using heat-shrink tubing to eliminate exposed conductive surfaces, and mandating the use of dielectric (non-conductive) safety gloves during all electrical integration. Most importantly, rigorous continuity checks for robust Earth/Ground (PE) bonding were mandated before any live-line testing. Ultimately, these stringent practices transformed initial prototyping errors into a permanently reliable, industrially safe electrical distribution network.

A deliberate omission driven by strict budget constraints was the precision surface lapping of the aluminum vacuum chuck. Although the exact peak-to-valley surface deviation was not empirically measured during testing, inherent microscopic undulations and warping in the raw aluminum stock theoretically limit the absolute thickness uniformity of the cast tape. Consequently, the machine's true precision is currently governed by the unlapped surface flatness rather than the 10 μm resolution of the micrometer heads. Precision surface grinding remains a necessary future upgrade for metrology-grade casting.

Finally, the linear drive exhibited a torque-collapse stall near the upper limit of 45 mm/s. This kinematic limitation stems from powering the TB6600 stepper driver exclusively via the +12V ATX rail. At higher pulse rates, 12V is insufficient to overcome the motor's winding inductance, preventing the phase current from rising fast enough and causing a rapid torque drop. While this

configuration successfully satisfies the baseline PDS requirement (up to 45 mm/s), the operational margin is narrow. Future applications requiring higher traverse rates for ultra-thin films will necessitate a dedicated 24V–36V power supply.

5.3 Future Work

The machine as constructed represents a highly capable first-generation prototype. However, to transform the system from a developmental research instrument into an industrial-grade metrology tool, several structural, electrical, and process-oriented refinements are necessary.

Immediate Structural and Electrical Refinements

- **Enclosure Rigidity and Component Accessibility:** Due to immediate material availability constraints, the front enclosure surrounding the linear shafts was fabricated from 0.5 mm sheet metal, making it susceptible to impact deformation. Future iterations must upgrade this to a rigid 1 mm standard. Furthermore, the top cover housing the fans suffers from dimensional cutting oversights, leading to lateral overextension and inadequate vapor sealing. Redesigning this enclosure with a hinged, quick-release mechanism would optimize vapor containment while vastly improving maintenance access to the underlying electronics bay.
- **Permanent Power Distribution:** Managing the high density of Earth/Ground (PE) and logic wires proved challenging. The current prototype relies on repurposed breadboard distribution strips to manage this wiring density. A permanent, long-term solution requires replacing these strips with a custom-soldered Printed Circuit Board (PCB) or a dedicated industrial copper busbar block.
- **Chassis and Surface Flatness:** Over extended periods of laboratory use, the MDF base plate is prone to wear and moisture degradation; replacing it with an anodized aluminum or solid composite chassis would ensure lifetime durability. Additionally, precision lapping of the vacuum chuck to a flatness tolerance of 5 μm is an outstanding requirement to ensure the micrometer resolution (10 μm) is the sole source of thickness uncertainty.

Long-Term Process and Research Extensions

- **Rheological Validation and Electrochemical Testing:** The theoretical casting model should be validated by experimentally characterizing the exact NASICON slurry utilizing a rotational rheometer. Ultimately, a complete process run—from casting and sintering a ~ 100 μm green tape to measuring its ionic conductivity via electrochemical impedance spectroscopy—is required as final proof of the machine’s utility.
- **Kinematic and Control Enhancements:** Upgrading the TB6600 driver supply to a dedicated 24V–36V source would prevent high-speed torque collapse, enabling casting velocities up to 100 mm/s for ultra-thin films. Additionally, integrating a non-contact laser displacement gauge downstream of the blade would enable closed-loop thickness monitoring.
- **Safety and Metrology:** To improve occupational safety, a compact activated-carbon filter cartridge should be integrated into the fan exhaust path to physically capture evaporated ethanol/toluene. Finally, integrating an overhead camera with a machine-vision pipeline (e.g., YOLOv5) would allow for the automated, real-time detection of pinholes and streak defects.

5.4 Key Lessons Learned and Academic Contribution

The transition from theoretical design to physical manufacturing provided the most critical engineering lessons, particularly regarding the discrepancy between ideal CAD environments and workshop realities. Unanticipated dynamic factors—such as the micro-stepping vibrations of the NEMA 23 motor causing solderless breadboard failures, and the strict machining tolerances required for the bimetallic vacuum chuck—demonstrated the absolute necessity of robust, industrial-grade assembly methods. Overcoming these mechanical and financial constraints was only possible through proactive supply chain management and invaluable machining sponsorships from industry partners like *Güveniş* and *Erk Resistance* for university projects. This experience underscored the importance of establishing industry collaborations early in the prototyping phase rather than relying solely on university workshop availability.

Furthermore, the project reinforced the necessity of a strict systems-engineering perspective. Successfully developing the tape casting machine required simultaneously balancing thermodynamics (bottom-up PID heating), fluid kinematics (constant velocity for non-Newtonian slurries), and mechatronic safety (220V AC protective cascade). The realization that isolated subsystem optimization could compromise the entire machine—such as initially viewing the DC cooling fans merely as a thermal measure rather than a safety-critical ACN vapor extraction system—was a defining academic takeaway that classroom instruction alone cannot fully convey.

Ultimately, executing this capstone project across this course served as the true culmination of the undergraduate engineering curriculum. It forced the practical application of theoretical courses onto a single physical system, bridging the gap between textbook physics and mechatronic reality. Managing the iterative documentation via LaTeX ensured mathematical and academic rigor, while the hands-on experience in precision mechanical design, embedded control, and process modeling has directly prepared the team for future professional endeavors in advanced manufacturing and solid-state energy systems.

6 Conclusions

This project successfully designed, analysed, and manufactured a fully functional automated tape casting machine for the production of NASICON ($\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$) ceramic solid electrolyte green tapes. The completed prototype satisfies the dimensional precision, thermal uniformity, velocity stability, solvent safety, and cost-effectiveness criteria defined in the Product Design Specifications, and its operational capability was confirmed by physical casting trials.

The kinematic design demonstrated that the NEMA 23 stepper motor, TB6600 driver at 1600 micropulses per revolution, and the GT2 belt–lead screw transmission deliver a stutter-free traverse velocity over 2–45 mm/s. Direct timing experiments confirmed no measurable deviation between theoretically predicted and experimentally observed carriage speeds. The velocity-locking mechanism in the Arduino firmware preserves a constant shear rate throughout each casting stroke, which the fluid dynamics analysis identifies as the dominant requirement for lateral thickness uniformity in shear-thinning ceramic slurries. The four-state finite-state machine structure produced deterministic and repeatable cycle behaviour, eliminating the motor stutter that affected earlier prototype versions.

The reduced-order hydrodynamic model, incorporating power-law rheology and Arrhenius temperature dependence, established the usable blade-gap window as 0.20–0.55 mm for wet tape thicknesses of 100–300 μm , which sinter to the sub-100 μm NASICON electrolyte membranes required by solid-state cell design constraints. The 10 μm adjustment resolution of the micrometer heads is commensurate with this window, confirming that the precision adjustment mechanism is a functional requirement. A velocity sensitivity analysis further showed that a 5% speed fluctuation

introduces thickness errors exceeding $\pm 5\text{ }\mu\text{m}$ at low traverse rates, which quantitatively justifies the microstepping drive configuration as the mechanism by which the dominant source of thickness scatter is suppressed.

Substrate temperature uniformity was identified as a critical secondary control variable, because the Arrhenius sensitivity of slurry consistency implies that even a few degrees of spatial non-uniformity would produce local thickness gradients indistinguishable from a blade-gap error during inspection. The two-dimensional finite-difference thermal simulation and subsequent experimental calibration confirmed that the thermocouple placement strategy keeps the measurement error below 1°C and the peak-to-valley surface non-uniformity below approximately 0.5°C , validating the design of the bimetallic heated vacuum chuck. The thermal elongation analysis confirmed that the close match in thermal expansion coefficients between Al 2045 and Al 5083 prevents progressive interface gap degradation, and PTFE support feet provided effective thermal isolation of the MDF base from the heated assembly.

The forced-convection drying analysis established that the two PL80S12M fans are a process requirement of the tape casting cycle rather than an auxiliary feature. Without active ventilation, ethanol vapour accumulates in the sealed chamber and arrests evaporation at approximately 24% residual solvent content regardless of drying time. With the fans operating, the mass-transfer coefficient increases by approximately an order of magnitude and the film dries to near completion within roughly 400 seconds. The analysis also confirmed that the unventilated enclosure would exceed the lower explosive limit for ethanol vapour within approximately 90 seconds, making active extraction a safety-critical design requirement in addition to a process necessity.

The electrical safety architecture functioned correctly under fault conditions encountered during prototyping. Accidental short circuits were successfully contained at the circuit-breaker level without propagation to downstream electronics, validating the cascaded protective hierarchy. Optical isolation between the ITC-100VH PID controller and the 220 V heater circuit through the solid-state relay ensured that mains-side faults could not propagate into the low-voltage logic domain.

Experimental casting trials on aluminium foil substrates confirmed that the machine operates within the geometrically relevant range for NASICON electrolyte research: the $250\text{ }\mu\text{m}$ blade gap yielded a dried tape of $80\text{ }\mu\text{m}$ thickness as verified by light-microscopy cross-section imaging. The total prototype resource value of approximately TL 34,392 against a direct team expenditure of TL 3,240 demonstrates that the open-architecture design strategy, supported by institutional and industrial contributions, can deliver laboratory-grade tape casting capability at a small fraction of the cost of equivalent commercial instruments.

The primary limitations of the current prototype are the unlapped vacuum chuck surface, which means casting precision is bounded by raw aluminium flatness rather than by the $10\text{ }\mu\text{m}$ micrometer resolution, and the 12 V supply voltage of the stepper driver, which restricts the usable speed range to 45 mm/s. These constraints, along with the need for rheological characterisation of the actual NASICON slurry and closed-loop thickness monitoring, define the most impactful directions for future development of the platform.

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Appendices

Appendix A: Material Property Data

Table 6: Thermomechanical properties of selected subsystem materials.

Component (Material)	Thermal Cond. (k , W/m · K)	Thermal Exp. (α , $10^{-6}/K$)	Spec. Heat Cap. (c_p , J/kg · K)	Yield Strength (σ_y , MPa)	Density (ρ , g/cm ³)	Max Service Temp. (°C)
Metals & Alloys						
Doctor Blade & Enclosure Panels^a (304 Stainless Steel)	16.2	17.2	500	215	8.00	870
Chuck Top Plate (Al 2045)	121	23.2	875	325	2.78	$\geq 200^b$
Chuck Bottom Plate (Al 5083-H111)	117	23.8	900	145 ^c	2.66	$\geq 200^b$
Vacuum Routing Fittings (Brass)	109	20.3	385	~ 125	8.50	900
Polymers, Elastomers & Insulators						
Base Plate (MDF)	0.13	—	1700	—	0.75	~ 100
Thermal Standoffs (PTFE / Teflon)	0.25	~ 130	1000	20	2.20	260
Insulation Pads (Mineral Wool)	0.04	—	840	—	0.15	1,000+
3-D-Printed Components (ABS)	0.17	~ 90	1400	40	1.04	105 (T _g)
Vacuum Chuck O-Ring Seal (Viton / FKM)	0.25	159	1050	— (elas.)	1.85	200

^a The stainless-steel blade (5 mm plate, CNC-milled) and the enclosure panels (1 mm sheet, hand-formed) are both 304 grade; their properties are identical.

^b Both aluminium alloys retain adequate mechanical properties throughout the machine’s 0–150 °C operating range. Sustained exposure above ~ 150 °C promotes overageing in 2xxx-series alloys (Al 2045) and sensitisation to intergranular corrosion in 5xxx-series alloys (Al 5083-H111); the 150 °C upper setpoint of the ITC-100VH controller therefore constitutes the design limit for both plates. Note: the MATLAB code of Appendix E uses $k_t = 130$ W/m K for Al 2045 and the code of Appendix F uses $\alpha_{\text{top}} = 22.5 \times 10^{-6} \text{ K}^{-1}$; both deviate slightly from the literature values listed above and represent approximations in the simulation scripts.

^c Yield strength quoted for the H111 temper of the as-purchased blank (Section 3.2.1). H321 temper would give $\sigma_y \approx 228$ MPa; H111 was specified to ease CNC milling.

^d Antoine constants from the NIST WebBook (ethanol, vapour-pressure equation valid 20–93 °C).

Appendix B: Technical Drawings

This part consists of technical drawings that are manufactured for this project.

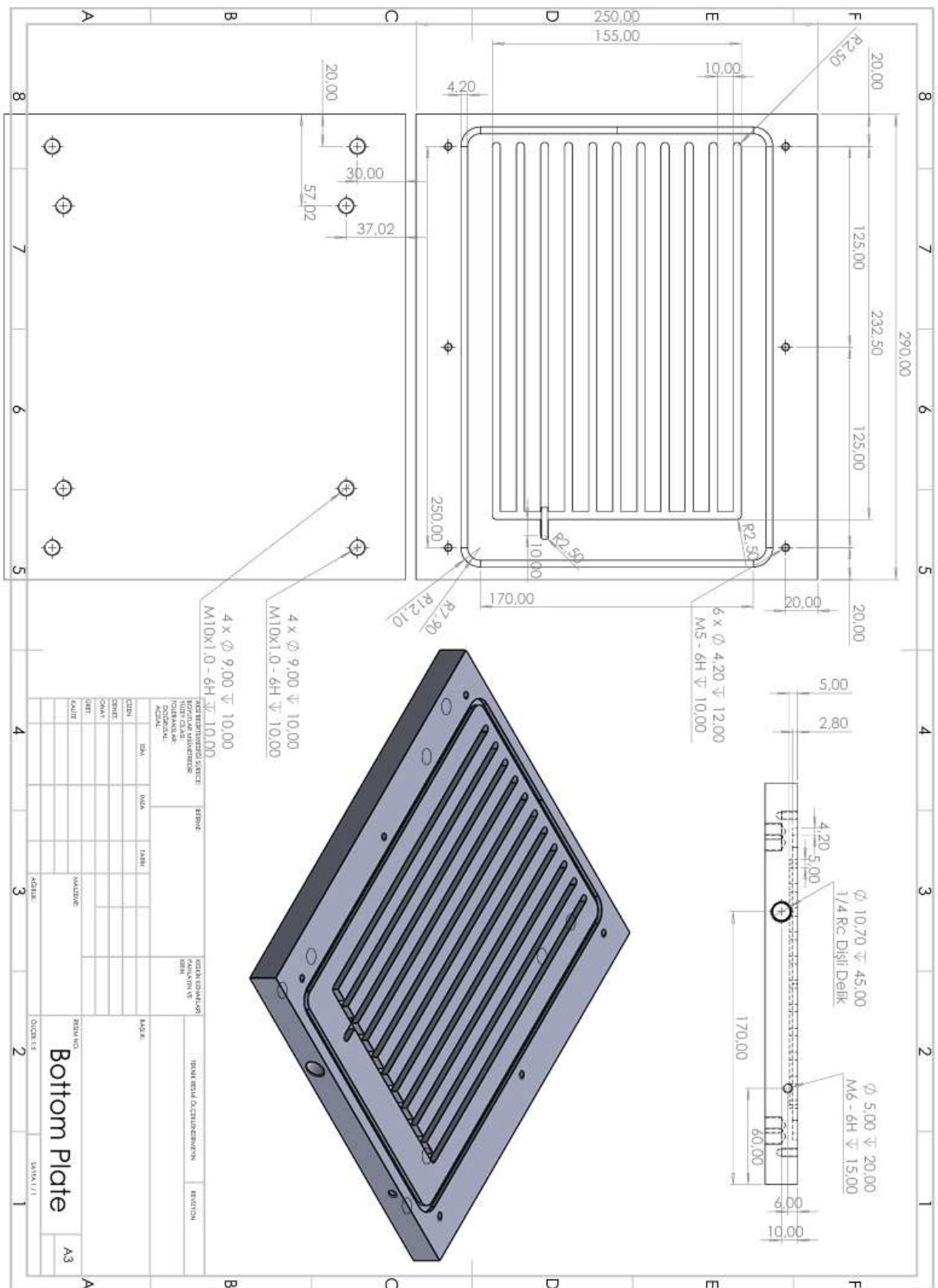


Figure 46: Vacuum chuck bottom plate (Al 5083) detailed manufacturing drawing.

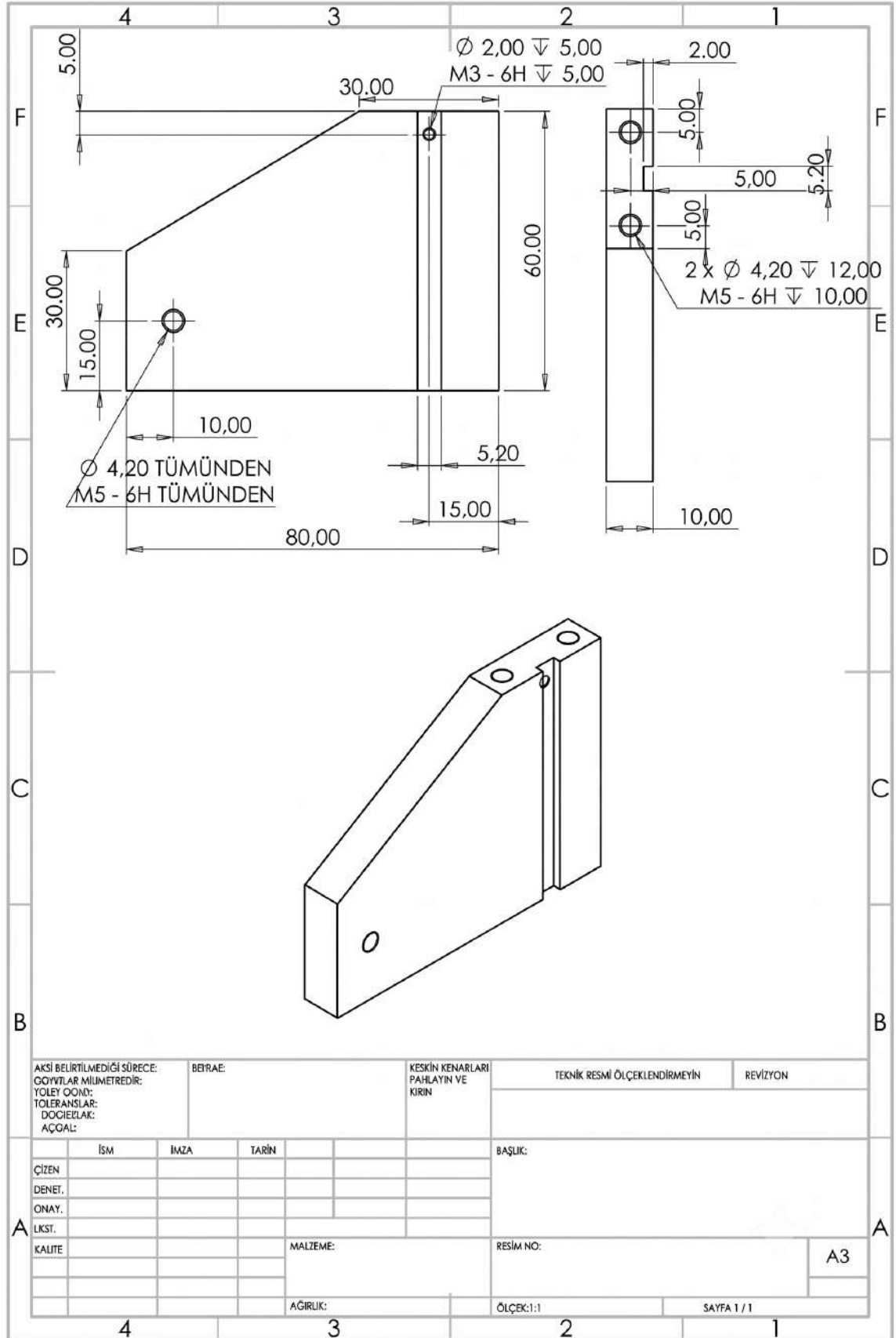


Figure 47: Right side of the Doctor Blade assembly.

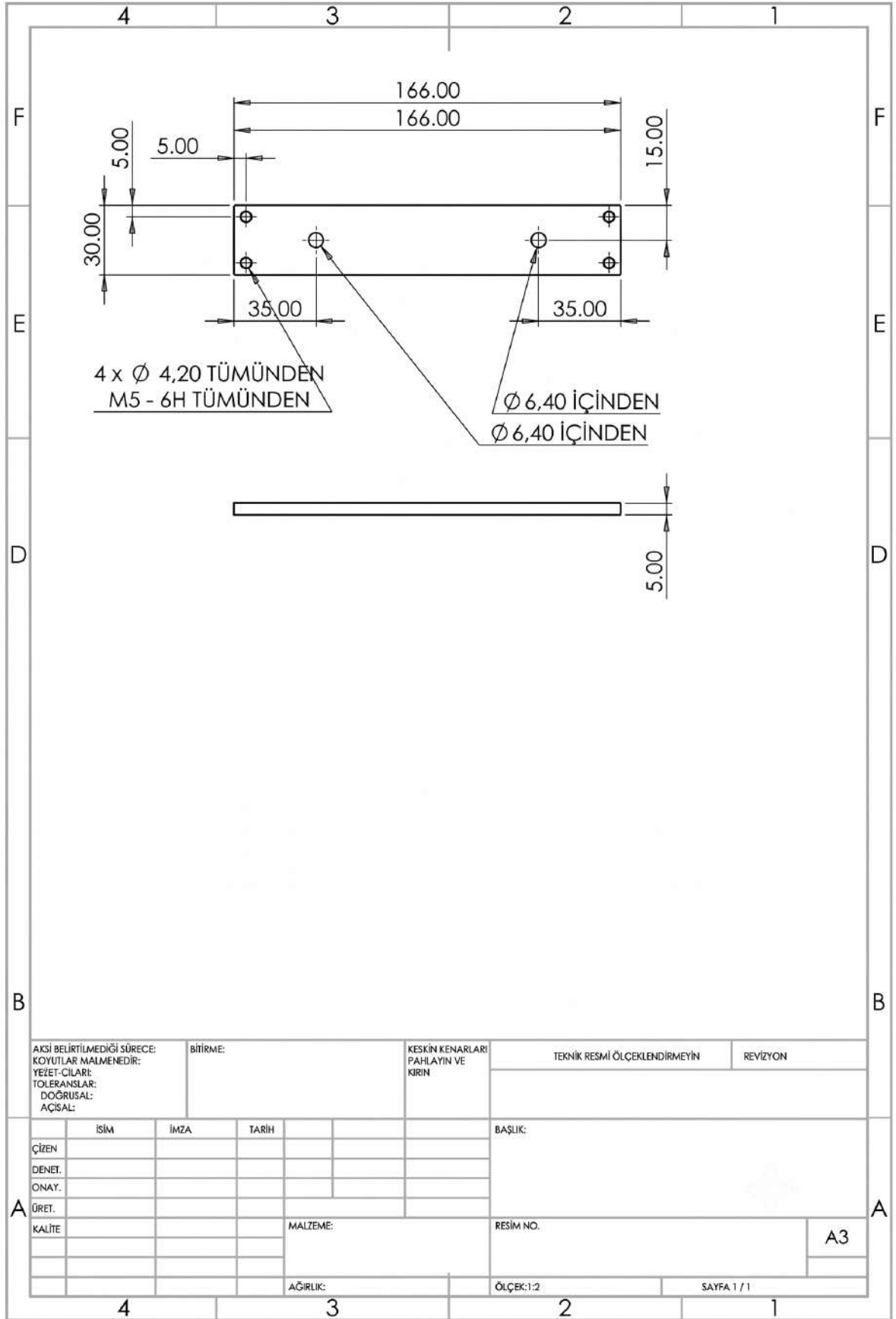


Figure 49: Top of the Doctor Blade assembly.

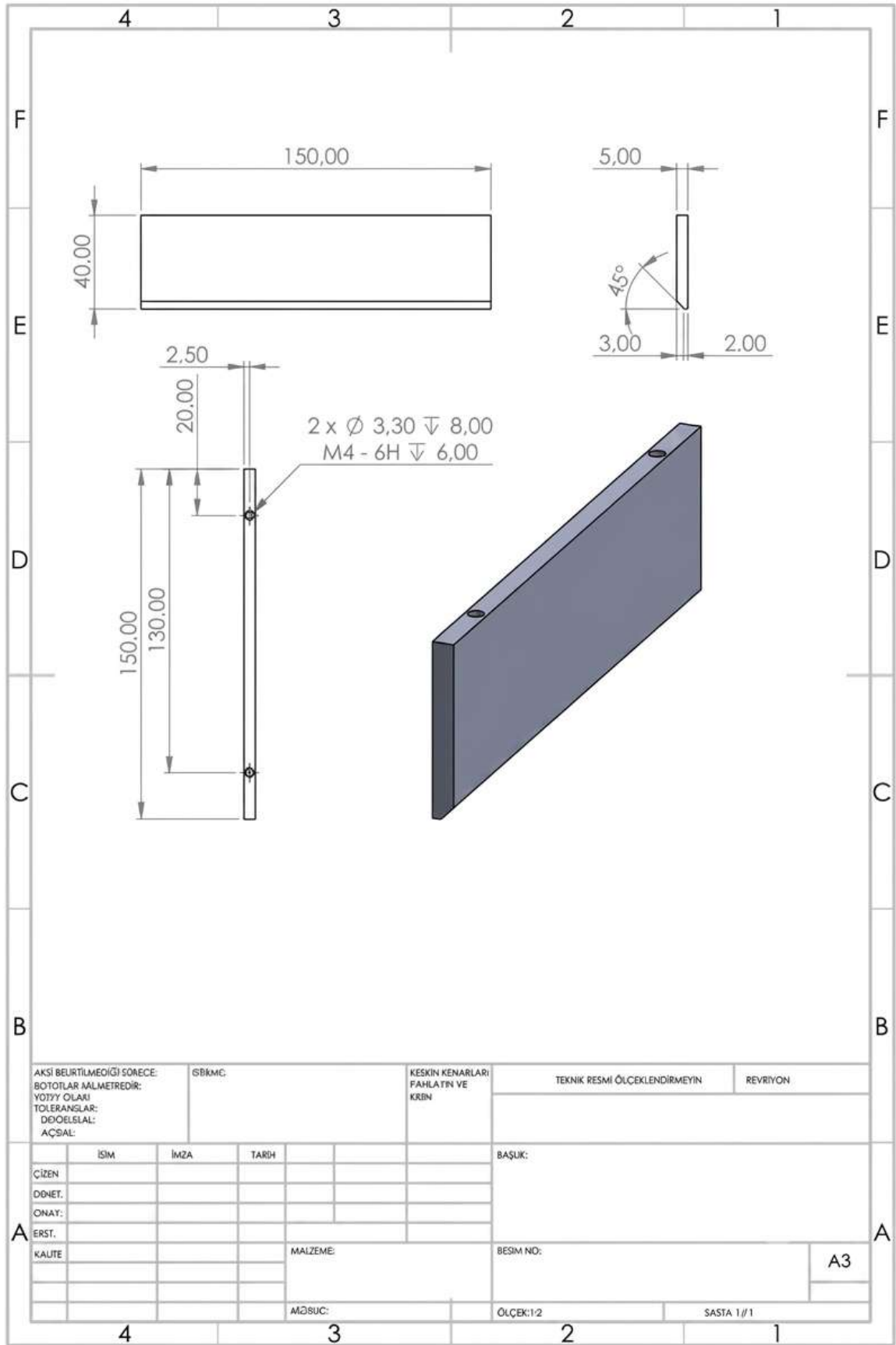


Figure 51: Knife of the Doctor Blade assembly.

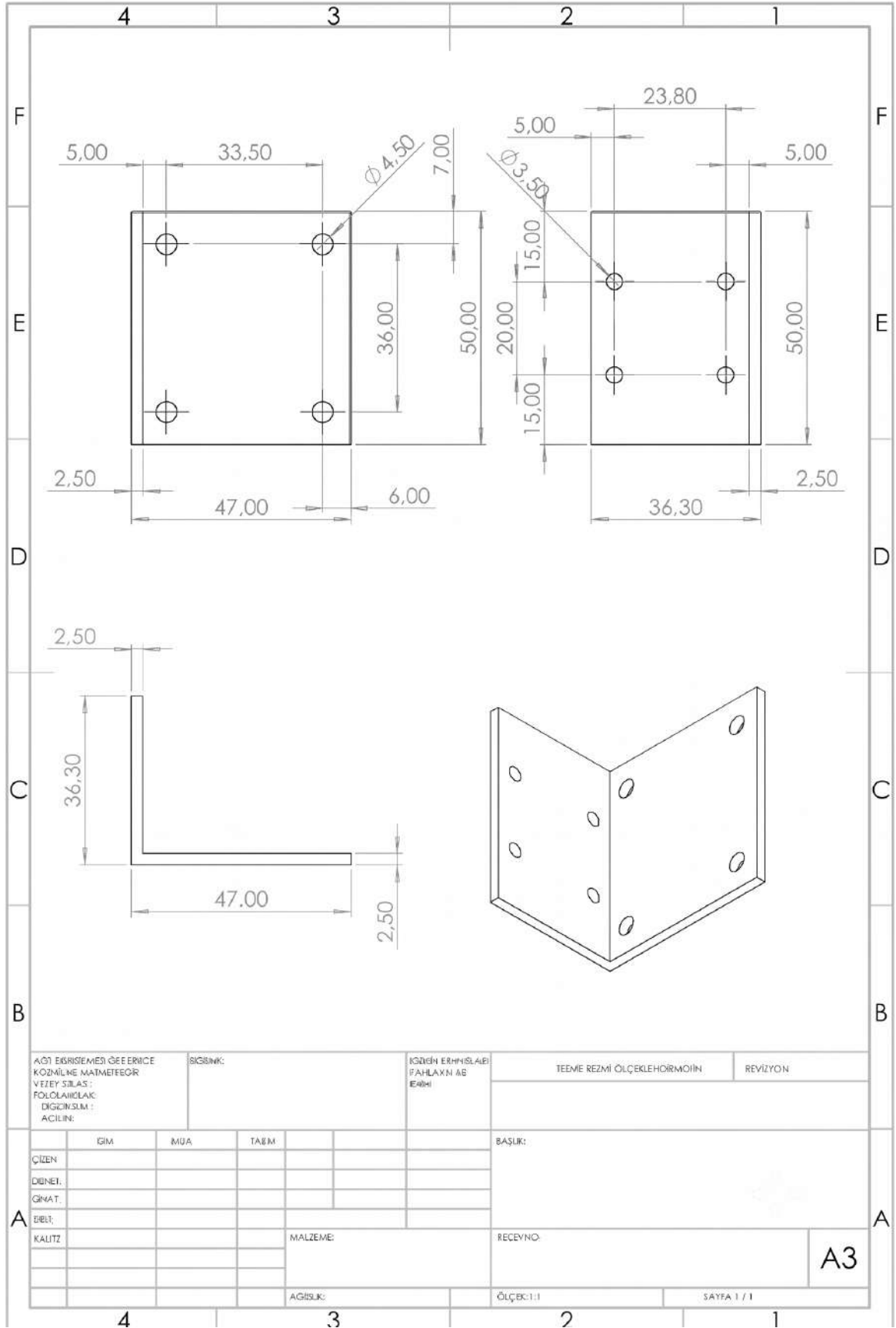


Figure 52: L-bracket that connects linear car and lead screw bracket.

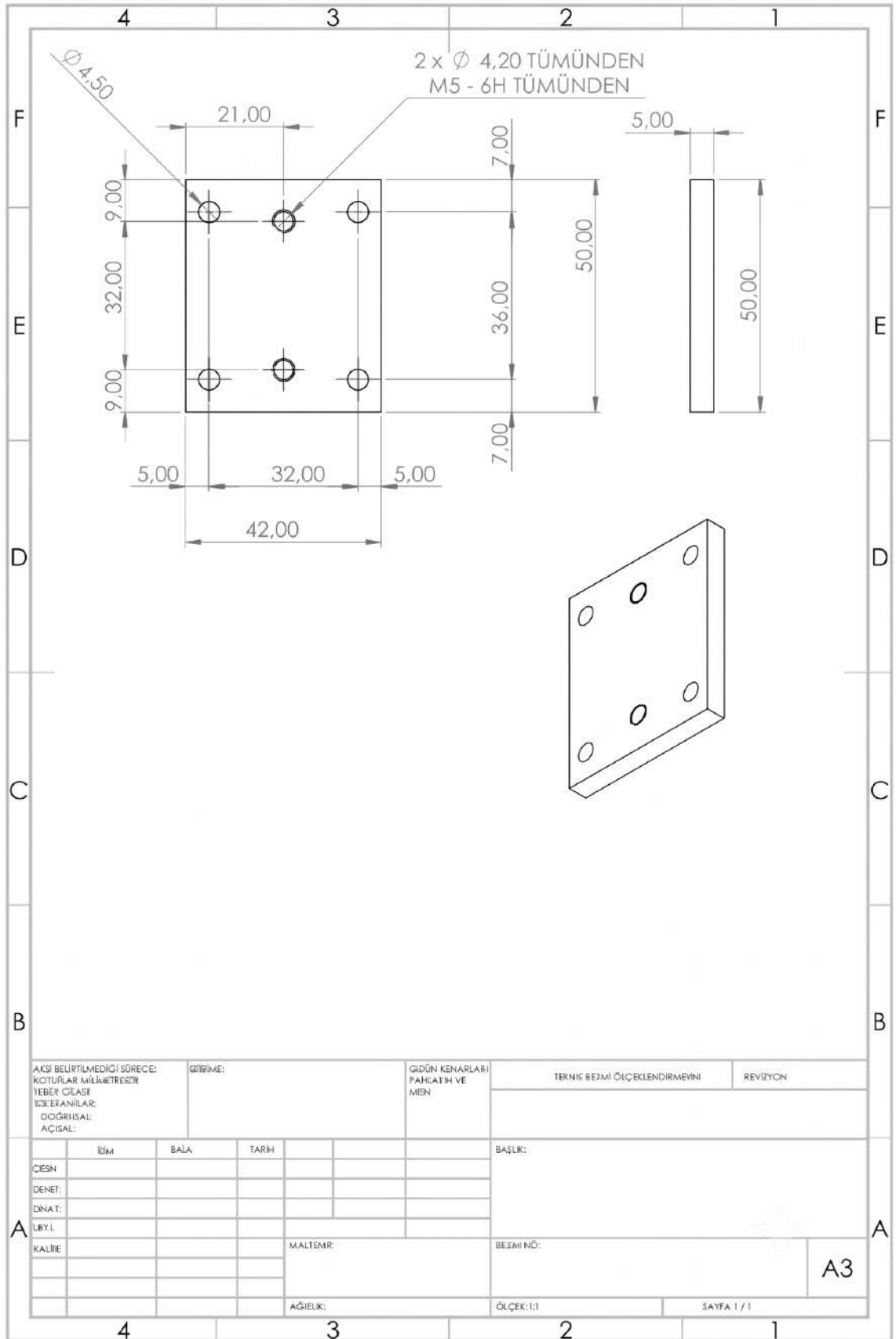


Figure 53: Top bracket that compresses the lead screw bracket to the linear car.

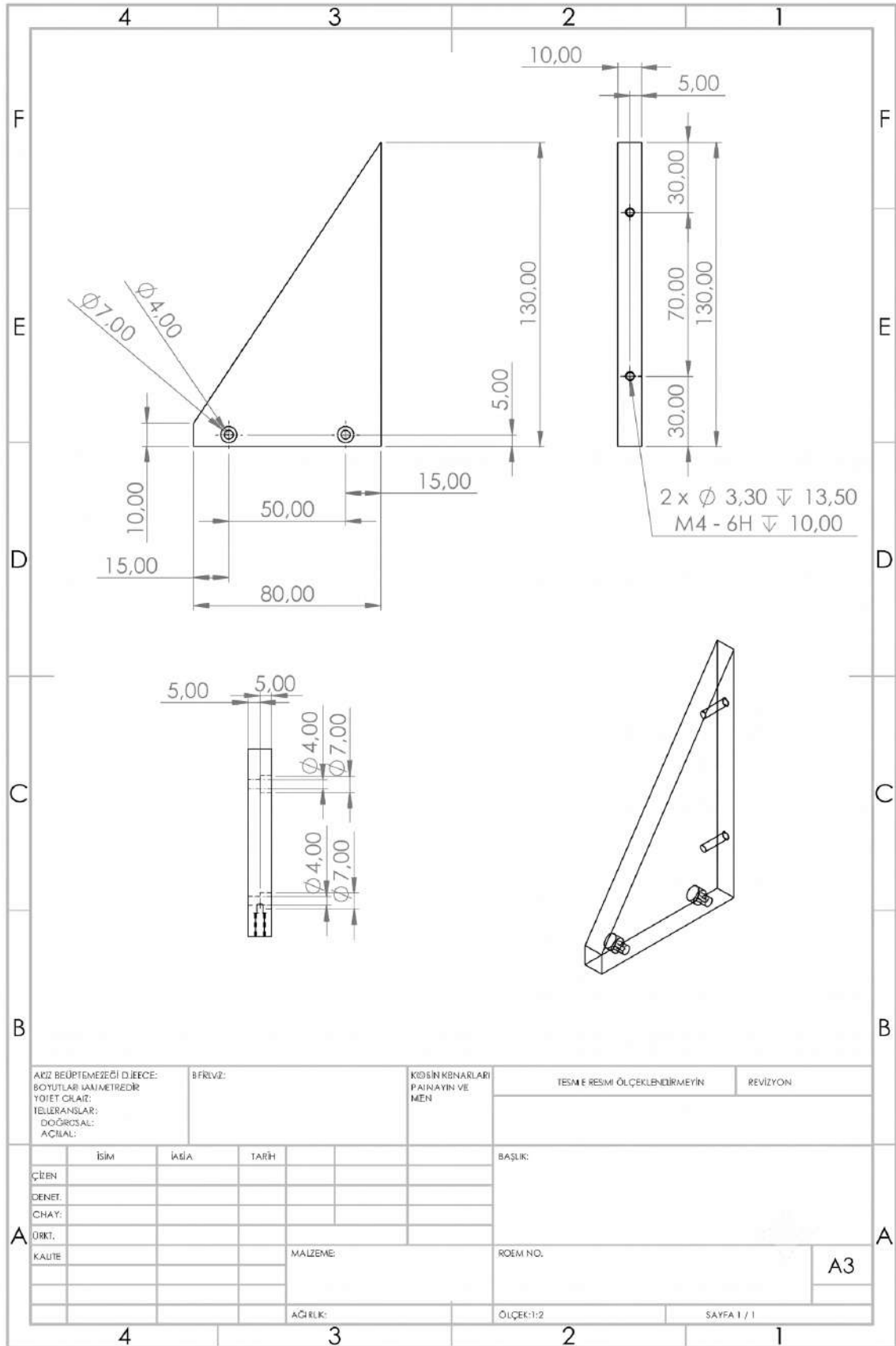


Figure 54: Two sides of the NEMA23 motor bracket.

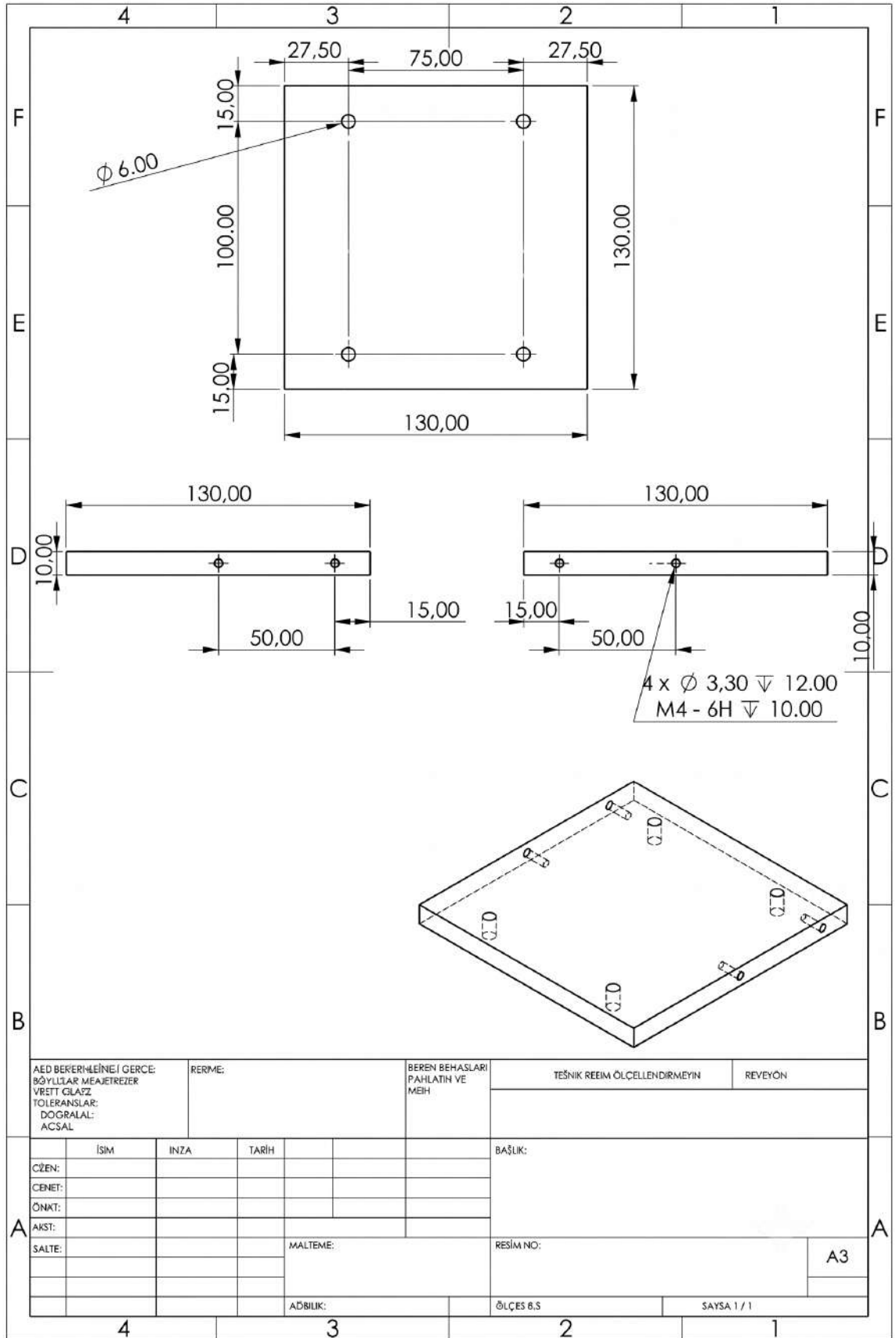


Figure 55: Bottom of the NEMA23 motor bracket.

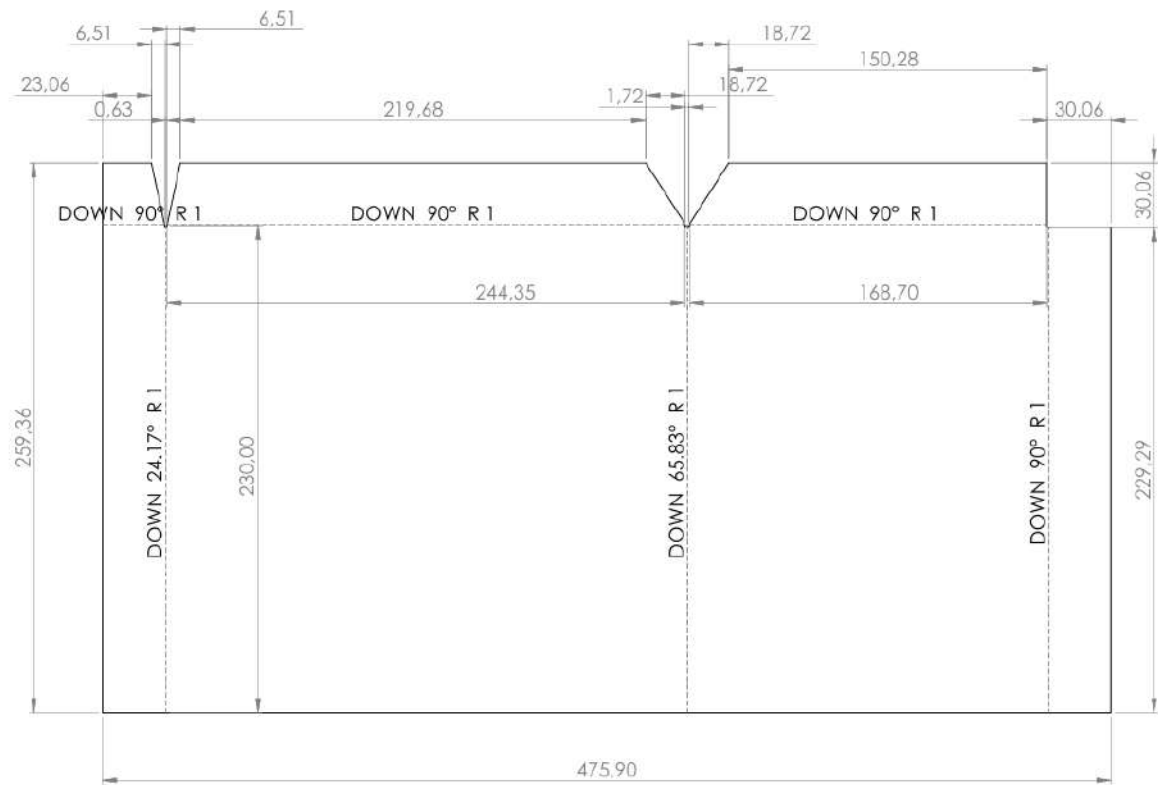


Figure 58: Front face of the enclosure

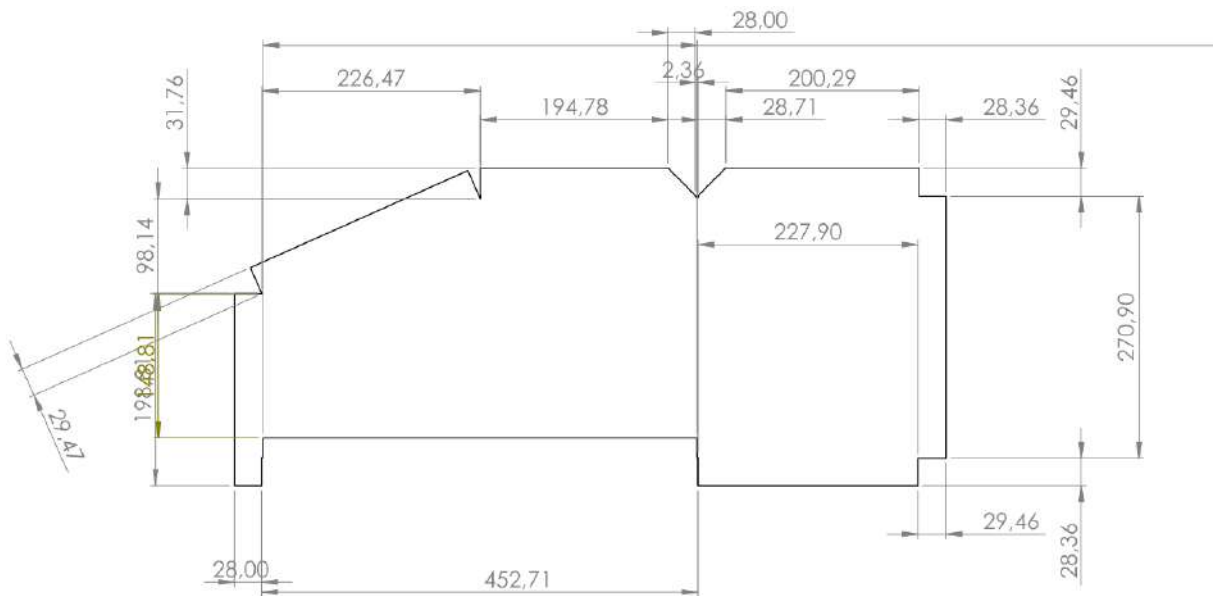


Figure 59: Left face of the enclosure

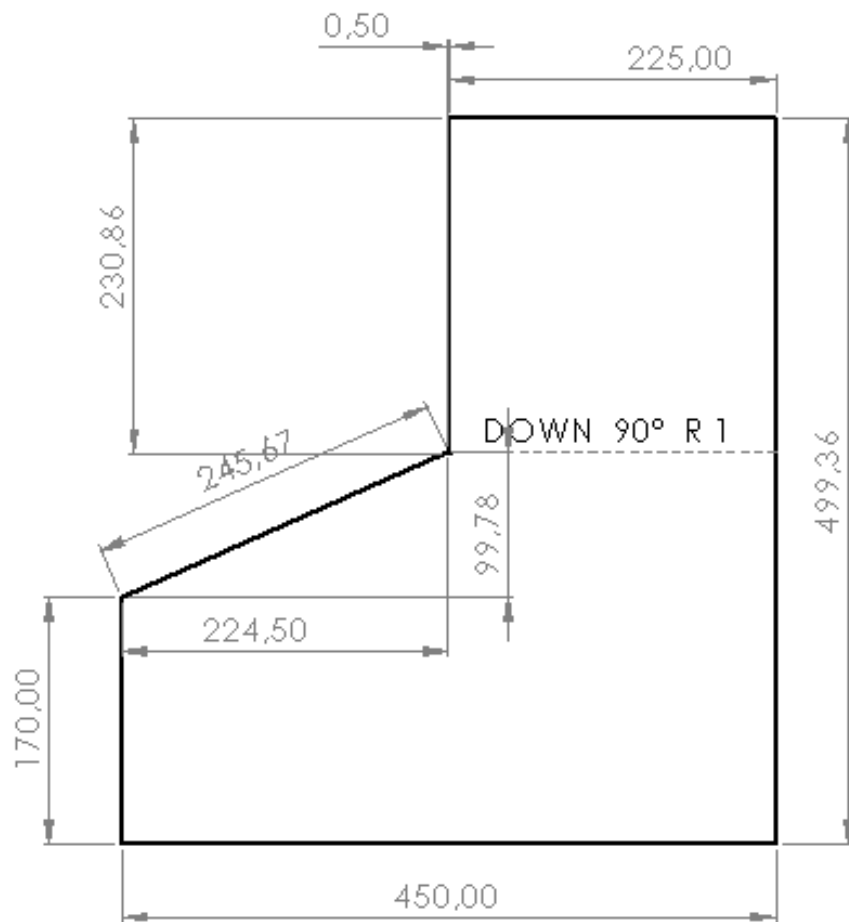


Figure 60: Removable cover of the enclosure

Since the left and right of the front faces of enclosure are symmetrical, only the left face technical drawings of these faces are shown.

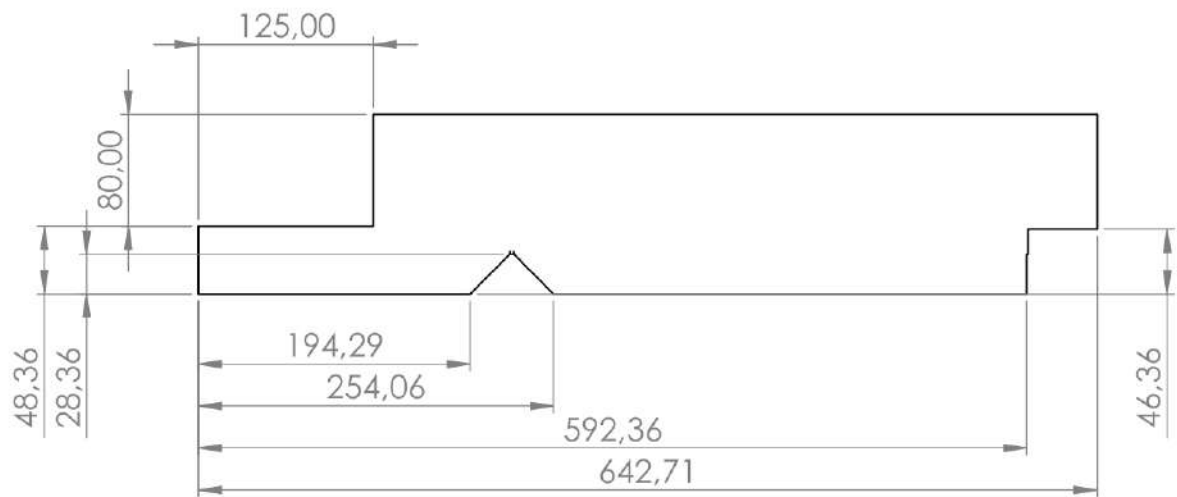


Figure 61: Left front face of the enclosure

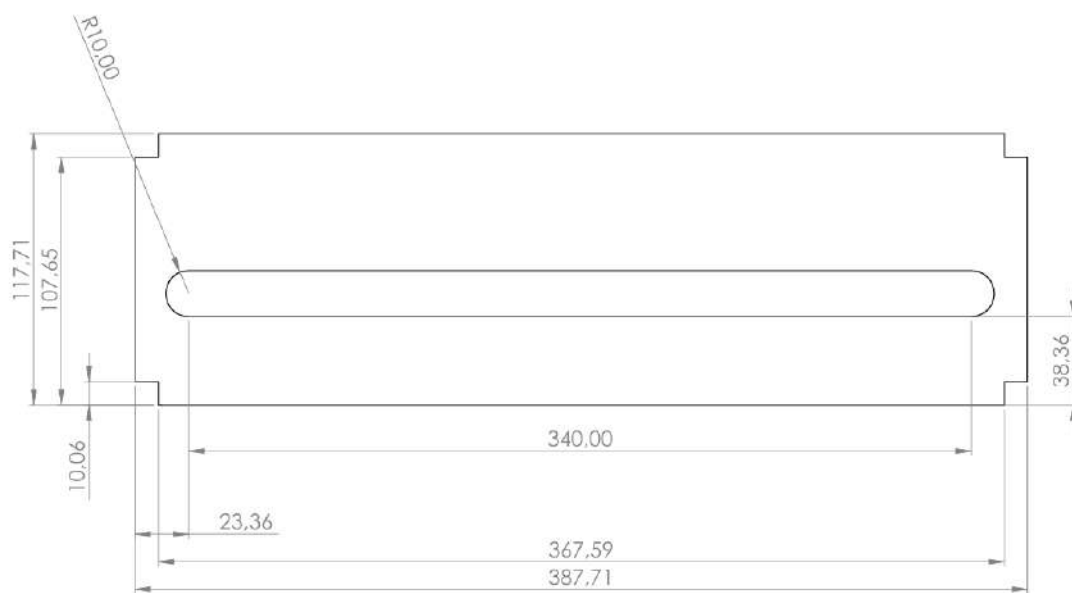


Figure 62: Top Left front face of the enclosure

Appendix C: MATLAB Script for Doctor Blade Fluid Dynamics Analysis

The script listed below was used to produce the three figures presented in Section 2.3.3. It evaluates the reduced-order drag-plus-pressure casting model (Equations (9)–(11)) over the machine's operating envelope and exports publication-resolution PNG files directly.

```

1 % Fluid Dynamics Analysis of the Micrometer-Adjustable Doctor Blade
2 % ME 492 - Automated Tape Casting Machine (Group 17)
3 % Model:  $\delta = h/2 + \rho \cdot g \cdot H_0 \cdot h^{(2+n)} / (12 \cdot k \cdot v_0^n \cdot L_b)$ 
4 clear; clc; close all;
5
6 % ---- Assumed slurry properties (ethanol-based NASICON, lit.-consistent)
7 rho = 1700; % kg/m^3 slurry density
8 n = 0.45; % - power-law index
9 k_ref = 12; % Pa.s^n consistency index at T_ref = 40 C
10 T_ref = 40 + 273.15; % K
11 Ea = 30e3; % J/mol Arrhenius activation energy
12 Rgas = 8.314; % J/mol/K
13
14 % ---- Machine geometry
15 g = 9.81;
16 H0 = 10e-3; % m reservoir head
17 Lb = 2.0e-3; % m blade land length
18 h_min = 0.05e-3; h_max = 5.0e-3; % m gap range (0-5000 um)
19 v_min = 2e-3; v_max = 45e-3; % m/s speed range
20 T_map = 60; % C baseline temperature
21 tgt_lo = 100; tgt_hi = 300; % um target wet-thickness window
22
23 % ---- Helper functions
24 kT = @(Tc) k_ref.*exp(Ea/Rgas.*(1./(Tc+273.15)-1/T_ref));
25 delta = @(h,v0,Tc) h/2 + (rho*g*H0.*h.^(2+n))./(12*kT(Tc).*v0.^n*Lb);
26
27 % ===== Fig A : wet-thickness operating window
28 hv = linspace(h_min,h_max,400); vv = linspace(v_min,v_max,400);
29 [H,V] = meshgrid(hv,vv);
30 D = delta(H,V,T_map)*1e6;
31 figA = figure('Color','w','Position',[100 100 720 540]);
32 contourf(V*1e3,H*1e3,D,0:50:1000,'LineColor','none'); hold on;
33 clim([0 1000]); colormap(parula);
34 cb = colorbar; cb.Label.String = 'Wet tape thickness \delta (\um)';
35 [c1,h1] = contour(V*1e3,H*1e3,D,[tgt_lo tgt_lo],'w','LineWidth',2.5);
36 [c2,h2] = contour(V*1e3,H*1e3,D,[tgt_hi tgt_hi],'w','LineWidth',2.5);
37 clabel(c1,h1,'Color','w','FontSize',11,'FontWeight','bold');
38 clabel(c2,h2,'Color','w','FontSize',11,'FontWeight','bold');
39 plot(20,0.30,'pentagram','MarkerSize',16,'MarkerFaceColor','r',...
40 'MarkerEdgeColor','k','LineWidth',1.2);
41 text(21,0.34,'operating point','Color','w','FontSize',10,'FontWeight','bold');
42 xlabel('Casting speed v_0 (mm/s)','FontSize',12);
43 ylabel('Blade gap h (mm)','FontSize',12);
44 title(sprintf('Predicted wet-tape thickness at %d C',T_map),'FontSize',12);
45 ;
46 set(gca,'FontSize',11,'Layer','top','TickDir','out'); box on;

```

```

46 exportgraphics(figA,'fluid_thickness_map.png','Resolution',300);
47
48 % ===== Fig B : flow curves
49 % =====
50 gdot = logspace(-1,3,300); Tlist = [40 50 60 70 80];
51 cmap = lines(numel(Tlist));
52 figB = figure('Color','w','Position',[100 100 720 540]); hold on;
53 fill([v_min/h_max v_max/h_min v_max/h_min v_min/h_max],...
54       [1e-2 1e-2 1e3 1e3],[0.85 0.85 0.85],'EdgeColor','none','FaceAlpha'
55       ,0.6);
56 for i = 1:numel(Tlist)
57     loglog(gdot,kT(Tlist(i)).*gdot.^(n-1),'LineWidth',2,'Color',cmap(i,:),
58           ...
59           'DisplayName',sprintf('%d C',Tlist(i)));
60 end
61 set(gca,'XScale','log','YScale','log','FontSize',11,'TickDir','out');
62 xlim([1e-1 1e3]); ylim([5e-2 6e1]); box on;
63 xlabel('Shear rate \gamma-dot (s^{-1})','FontSize',12);
64 ylabel('Apparent viscosity \mu_a (Pa.s)','FontSize',12);
65 title('Flow curves of the ethanol-based NASICON slurry','FontSize',12);
66 legend('Location','northeast','FontSize',10);
67 exportgraphics(figB,'fluid_flow_curves.png','Resolution',300);
68
69 % ===== Fig C : velocity sensitivity
70 % =====
71 v = linspace(v_min,v_max,300); hsel = [0.3e-3 0.4e-3 0.5e-3];
72 cmap2 = lines(numel(hsel));
73 figC = figure('Color','w','Position',[100 100 720 540]); hold on;
74 for i = 1:numel(hsel)
75     hh = hsel(i);
76     A = (rho*g*H0*hh^(2+n))/(12*kT(T_map)*Lb);
77     err = n*A.*v.^(-n-1).*(0.05*v)*1e6;
78     plot(v*1e3,err,'LineWidth',2,'Color',cmap2(i,:),...
79         'DisplayName',sprintf('h = %.1f mm',hh*1e3));
80 end
81 yline(5,'--r','LineWidth',1.5,'DisplayName','\pm 5 \mu m tolerance');
82 xlabel('Casting speed v_0 (mm/s)','FontSize',12);
83 ylabel('Thickness error from 5% speed wobble (\mu m)','FontSize',12);
84 title('Sensitivity of wet thickness to velocity fluctuation','FontSize'
85       ,12);
86 set(gca,'FontSize',11,'TickDir','out'); box on; grid on;
87 legend('Location','northeast','FontSize',10);
88 exportgraphics(figC,'fluid_velocity_sensitivity.png','Resolution',300);

```

Listing 1: MATLAB script for the Doctor Blade fluid dynamics parametric analysis (Section 2.3.3).

Appendix D: MATLAB Code for Forced-Convection Drying Analysis

```

1 % ME 492 Group 17 - Section 2.3.4 Forced-Convection Drying Analysis
2 % Fans: 2x Power Logic PL80S12M (80mm, 12V, 0.14A)
3 clear; clc; close all;
4
5 % --- Fan and chamber ---
6 Q_total = 2 * 25 * 4.719e-4; % [m3/s] two fans at 25 CFM each
7 V_ch     = 0.400 * 0.350 * 0.130; % [m3] chamber volume

```



```

8
9 % --- Cast film ---
10 A_f = 0.200 * 0.150; % [m2]
11 m0 = A_f * 250e-6 * 0.45 * 789; % [kg] initial ethanol mass
12 M = 0.04607; % [kg/mol]
13
14 % --- Thermodynamics at 48 C ---
15 T_C = 48; T = T_C + 273.15; R = 8.314;
16 P_sat = 10^(8.20417 - 1642.89/(230.300 + T_C)) * 133.322; % [Pa]
17 C_sat = (P_sat/(R*T)) * M; % [kg/m3]
18 D = 1.20e-5 * (T/298.15)^1.75; % ethanol-air diffusivity
19 nu = 17.9e-6; % air kinematic viscosity
20 Sc = nu / D;
21
22 % --- Mass-transfer coefficients ---
23 u = Q_total / (0.400*0.350);
24 Re = u * 0.200 / nu;
25 k_f = 0.664 * Re^0.5 * Sc^(1/3) * D / 0.200; % forced convection
26 k_n = 2.0e-4; % natural convection
27
28 % --- Transient integration (Euler, dt = 0.5 s) ---
29 dt = 0.5; t = (0:dt:600)'; N = numel(t);
30 Q_v = [0, Q_total]; k_m = [k_n, k_f];
31 C = zeros(N,2); m = zeros(N,2); rate = zeros(N,2);
32 for c = 1:2
33     m(1,c) = m0;
34     for i = 1:N-1
35         x = m(i,c)/m0;
36         ev = k_m(c)*A_f*max(0, x*C_sat - C(i,c));
37         rate(i,c) = ev;
38         m(i+1,c) = max(0, m(i,c) - ev*dt);
39         C(i+1,c) = max(0, C(i,c) + (ev - Q_v(c)*C(i,c))/V_ch*dt);
40     end
41     rate(N,c) = rate(N-1,c);
42 end
43 m_frac = m./m0.*100;
44 rate_min = rate.*1000.*60;
45 res_eq = (m0/V_ch)/(C_sat/m0 + 1/V_ch)/m0*100;
46
47 % --- Figure 1: residual solvent ---
48 figure('Position',[100 100 700 420]); hold on; box on; grid on;
49 plot(t,m_frac(:,1),'r-','LineWidth',2);
50 plot(t,m_frac(:,2),'b-','LineWidth',2);
51 yline(res_eq,'r:','LineWidth',1.2); yline(10,':','Color',[0.4 0.4 0.4]);
52 xline(206,'b-.','LineWidth',0.9);
53 legend({'No ventilation (fans off)','Forced ventilation (2x PL80S12M)'},
54     ...
55     'Location','northeast');
56 xlabel('Time (s)'); ylabel('Remaining solvent fraction (%)');
57 title('Green tape drying: residual solvent fraction vs. time');
58 xlim([0 600]); ylim([0 105]);
59 exportgraphics(gcf,'drying_curve.png','Resolution',300);
60
61 % --- Figure 2: evaporation rate ---
62 figure('Position',[100 100 700 420]); hold on; box on; grid on;
63 plot(t,rate_min(:,1),'r-','LineWidth',2);
64 plot(t,rate_min(:,2),'b-','LineWidth',2);
65 legend({'No ventilation (fans off)','Forced ventilation (2x PL80S12M)'},
66     ...

```

```

65         'Location','northeast');
66 xlabel('Time (s)'); ylabel('Solvent evaporation rate (g/min)');
67 title('Instantaneous solvent evaporation rate vs. time');
68 xlim([0 600]); ylim([0 2.0]);
69 exportgraphics(gcf,'evaporation_rate.png','Resolution',300);

```

Listing 2: MATLAB script for Section 2.3.5: forced-convection solvent drying analysis. Outputs drying_curve.png and evaporation_rate.png.

Appendix E: MATLAB Code for 2D Heat Transfer Simulation

```

1  %
2  % =====
3  % Precision Vacuum Chuck 2D Heat Transfer Simulation
4  % Finite Difference Method with Simulated PID Control
5  % =====
6
7
8  clear; clc; close all;
9
10 %% 1. Parameters & Geometry
11 % Grid resolution (5mm per cell)
12 dx = 0.005;
13
14 % Dimensions (using 250mm x 290mm contact area for the grid)
15 Lx = 0.250;
16 Ly = 0.290;
17
18 % Number of grid cells
19 nx = round(Lx / dx) + 1; % 51 cells
20 ny = round(Ly / dx) + 1; % 59 cells
21
22 % Material Properties
23 k_b = 120; % Thermal conductivity Bottom Plate Al 5083 (W/mK)
24 k_t = 130; % Thermal conductivity Top Plate Al 2045/2024 (W/mK)
25 t_b = 0.020; % Thickness Bottom Plate (m)
26 t_t = 0.005; % Thickness Top Plate (m)
27
28 % Thermal Resistances
29 h_c = 1500; % Contact conductance between plates (W/m^2K)
30 h_conv = 10; % Natural convection to ambient air (W/m^2K)
31 T_amb = 22; % Ambient temperature (C)
32
33 % Heater Properties (200mm x 200mm centered)
34 heater_W = 800; % Max heater power in Watts
35 heater_area = 0.200 * 0.200; % m^2
36 max_q_flux = heater_W / heater_area; % Max Heat flux (W/m^2)
37
38 % Define heater location on the grid (Centered)
39 % 200mm / 5mm = 40 cells wide/long
40 heater_mask = zeros(nx, ny);
41 x_start = round((nx - 40)/2); x_end = x_start + 40;
42 y_start = round((ny - 40)/2); y_end = y_start + 40;
43 heater_mask(x_start:x_end, y_start:y_end) = 1;
44

```

```

43 %% 2. Initialization & PID Setup
44 Tb = ones(nx, ny) * T_amb; % Bottom plate temp matrix
45 Tt = ones(nx, ny) * T_amb; % Top plate temp matrix
46
47 % Thermocouple Location (Centered on the 250mm edge, y=1)
48 tc_x = round(nx / 2);
49 tc_y = 1;
50
51 % Simulation Settings
52 max_iter = 15000; % Number of iterations to reach steady state
53 target_temp = 105; % The temperature you set on your PID controller (C
54 )
55 Kp = 500; % Proportional gain for simulated controller
56
57 % Pre-calculate constants for the FDM equations to speed up the loop
58 alpha_b = (dx^2) / (k_b * t_b);
59 alpha_t = (dx^2) / (k_t * t_t);
60 denom_b = 4 + alpha_b * (h_c + h_conv);
61 denom_t = 4 + alpha_t * (h_c + h_conv);
62
63 %% 3. Finite Difference Loop
64 disp('Running simulation... Please wait.');
```

```

65 for iter = 1:max_iter
66
67     % --- PID Controller Simulation ---
68     % Read the thermocouple and adjust heater power
69     current_tc_temp = Tb(tc_x, tc_y);
70     error = target_temp - current_tc_temp;
71
72     % Calculate power and clamp it between 0 and max_q_flux
73     current_q_flux = Kp * error;
74     if current_q_flux > max_q_flux
75         current_q_flux = max_q_flux;
76     elseif current_q_flux < 0
77         current_q_flux = 0;
78     end
79
80     % Apply active heat flux to the heater area
81     Q_in = heater_mask * current_q_flux;
82
83     % --- Heat Transfer Equations ---
84     Tb_old = Tb;
85     Tt_old = Tt;
86
87     % Update inner grid points
88     for i = 2:nx-1
89         for j = 2:ny-1
90             % Bottom Plate Eq
91             Tb(i,j) = (Tb_old(i+1,j) + Tb_old(i-1,j) + Tb_old(i,j+1) +
92                 Tb_old(i,j-1) + ...
93                 alpha_b * (h_c * Tt_old(i,j) + Q_in(i,j) + h_conv *
94                     T_amb)) / denom_b;
95
96             % Top Plate Eq
97             Tt(i,j) = (Tt_old(i+1,j) + Tt_old(i-1,j) + Tt_old(i,j+1) +
98                 Tt_old(i,j-1) + ...
99                 alpha_t * (h_c * Tb_old(i,j) + h_conv * T_amb)) /
100                 denom_t;

```

```

97         end
98     end
99
100    % Apply Boundary Conditions
101    Tb(1,:) = Tb(2,:); Tb(nx,:) = Tb(nx-1,:);
102    Tb(:,1) = Tb(:,2); Tb(:,ny) = Tb(:,ny-1);
103    Tt(1,:) = Tt(2,:); Tt(nx,:) = Tt(nx-1,:);
104    Tt(:,1) = Tt(:,2); Tt(:,ny) = Tt(:,ny-1);
105
106    end
107    disp('Simulation complete!');
108
109    %% 4. Data Extraction & Visualization
110    center_bottom = Tb(round(nx/2), round(ny/2));
111    center_top = Tt(round(nx/2), round(ny/2));
112    tc_reading = Tb(tc_x, tc_y);
113
114    fprintf('\n--- STEADY STATE RESULTS ---\n');
115    fprintf('Thermocouple Reading (PID Target): %.1f C\n', tc_reading);
116    fprintf('Bottom Plate Center Temp: %.1f C\n', center_bottom);
117    fprintf('Top Plate Center Temp: %.1f C\n', center_top);
118    fprintf('Max Temperature Difference on Top Plate: %.1f C\n', max(Tt(:)) -
        min(Tt(:)));
119
120    figure('Position', [100, 100, 900, 400]);
121    subplot(1,2,1);
122    imagesc([0 290], [0 250], Tb);
123    set(gca, 'YDir', 'normal');
124    colormap('hot'); colorbar;
125    title('Bottom Plate Temp ( C )');
126    xlabel('Length (mm)'); ylabel('Width (mm)');
127    hold on;
128    plot(0, 125, 'g*', 'MarkerSize', 10, 'LineWidth', 2);
129    legend('Thermocouple', 'Location', 'best');
130
131    subplot(1,2,2);
132    imagesc([0 290], [0 250], Tt);
133    set(gca, 'YDir', 'normal');
134    colormap('hot'); colorbar;
135    title('Top Plate Temp ( C ) [Film Coating Surface]');
136    xlabel('Length (mm)'); ylabel('Width (mm)');

```

Appendix F: MATLAB Code for Thermal Elongation Analysis

```

1  %
2  % =====
3  % Thermal Elongation Analysis for Precision Vacuum Chuck
4  % =====
5
6
7  clear; clc; close all;
8
9  %% 1. Parameters
10 T_amb = 22; % Ambient temperature (C)
11 T_target_max = 120; % Maximum temperature for the graph (C)

```

```

10 T = linspace(T_amb, T_target_max, 100); % Temperature array
11
12 % Coefficients of Thermal Expansion (1/C)
13 alpha_bottom = 23.8e-6; % Al 5083
14 alpha_top = 22.5e-6; % Al 2045/2024 (approximate)
15
16 % Original Dimensions at 22 C (in millimeters)
17 L0_bottom_x = 250; % Bottom width
18 L0_bottom_y = 290; % Bottom length
19 L0_top_x = 250; % Top width
20 L0_top_y = 300; % Top length
21
22 %% 2. Calculate Elongation (Delta L)
23 deltaT = T - T_amb;
24
25 dL_bottom_x = L0_bottom_x * alpha_bottom * deltaT;
26 dL_bottom_y = L0_bottom_y * alpha_bottom * deltaT;
27
28 dL_top_x = L0_top_x * alpha_top * deltaT;
29 dL_top_y = L0_top_y * alpha_top * deltaT;
30
31 mismatch_x = abs(dL_bottom_x - dL_top_x);
32
33 %% 3. Plotting the Results
34 figure('Position', [150, 150, 800, 500], 'Color', 'w');
35
36 subplot(1,2,1);
37 set(gca, 'Color', 'w');
38 plot(T, dL_bottom_y, 'r-', 'LineWidth', 2); hold on;
39 plot(T, dL_top_y, 'b--', 'LineWidth', 2);
40 grid on;
41 title('Elongation vs. Temperature (Length / Y-Axis)');
42 xlabel('Temperature ( C )');
43 ylabel('Elongation \Delta L (mm)');
44 legend(sprintf('Bottom Plate (Al 5083, L_0=%dmm)', L0_bottom_y), ...
45         sprintf('Top Plate (Al 2045, L_0=%dmm)', L0_top_y), ...
46         'Location', 'northwest');
47
48 subplot(1,2,2);
49 set(gca, 'Color', 'w');
50 plot(T, dL_bottom_x, 'r-', 'LineWidth', 2); hold on;
51 plot(T, dL_top_x, 'b--', 'LineWidth', 2);
52 plot(T, mismatch_x, 'k:', 'LineWidth', 2);
53 grid on;
54 title('Elongation vs. Temperature (Width / X-Axis)');
55 xlabel('Temperature ( C )');
56 ylabel('Elongation \Delta L (mm)');
57 legend(sprintf('Bottom Plate (Al 5083, L_0=%dmm)', L0_bottom_x), ...
58         sprintf('Top Plate (Al 2045, L_0=%dmm)', L0_top_x), ...
59         'Shear Mismatch', 'Location', 'northwest');
60
61 set(gcf, 'InvertHardcopy', 'off');
62
63 %% 4. Print Results
64 fprintf('\n--- THERMAL ELONGATION AT %.0f C ---\n', T_target_max);
65 fprintf('Bottom Plate Length (290mm) expands by: %.3f mm\n', dL_bottom_y(
66     end));
67 fprintf('Top Plate Length (300mm) expands by: %.3f mm\n', dL_top_y(end));

```

```

67 fprintf('Bottom Plate Width (250mm) expands by: %.3f mm\n', dL_bottom_x(
    end));
68 fprintf('Top Plate Width (250mm) expands by: %.3f mm\n', dL_top_x(end));
69 fprintf('Maximum Width Mismatch (Shear displacement): %.3f mm\n',
    mismatch_x(end));

```

Appendix G: Arduino Source Code

The complete firmware controlling the casting cycle is listed below. The four-state machine, the velocity-locking logic, the hysteresis filter, and the pulse-generation routine described in Section 2.3 are all implemented here. The code is reproduced as flashed to the prototype; only the inline comments have been rendered in English for clarity.

```

1  /*
2   * Doctor Blade Coater -- Fully Automated System v4 (Parallel LCD)
3   * Speed: 0-45 mm/s | Return: Fixed 45 mm/s
4   * Microstep: 1600 | Lead = 8 mm, Gear Ratio = 1.33
5   * Delay constant for 800 microsteps is 6666, i.e. delay = 6666 / v.
6   * For 1600 microsteps the constant becomes 6666/2 = 3333, i.e. delay =
7     3333 / v.
8   */
9  #include <LiquidCrystal.h>
10
11 // -- LCD SETTINGS (parallel 4-bit interface) --
12 const int rs = 12, en = 11, d4 = 10, d5 = 6, d6 = A1, d7 = A2;
13 LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
14
15 // -- PIN DEFINITIONS --
16 const int stepPin = 9;
17 const int dirPin = 13;
18 const int enPin = 7;
19 const int potPin = A0;
20 const int limit1Pin = 3; // Home Switch
21 const int limit2Pin = 2; // End Switch
22 const int button1Pin = 4; // Start Button (Forward)
23 const int button2Pin = 5; // Reset Button (Backward)
24
25 // -- SYSTEM VARIABLES --
26 int systemState = 0;
27 unsigned long lastDisplayTime = 0;
28 const unsigned long DISPLAY_INTERVAL = 300; // ms
29 float lockedV = 0.0;
30 long lockedStepDelay = 0;
31 int lastPotValue = 0; // Added for the fluctuation (noise) filter
32
33 // -- SETUP --
34 void setup() {
35     lcd.begin(16, 2);
36     lcd.clear();
37     lcd.setCursor(0, 0);
38     lcd.print("BGMAD Coater ");
39     lcd.setCursor(0, 1);
40     lcd.print("Starting... ");
41     delay(1500);
42     lcd.clear();
43

```

```

44 pinMode(stepPin,      OUTPUT);
45 pinMode(dirPin,      OUTPUT);
46 pinMode(enPin,       OUTPUT);
47
48 pinMode(limit1Pin,   INPUT_PULLUP);
49 pinMode(limit2Pin,   INPUT_PULLUP);
50 pinMode(button1Pin,  INPUT_PULLUP);
51 pinMode(button2Pin,  INPUT_PULLUP);
52
53 digitalWrite(enPin, HIGH); // Disable the motor at start-up (silence it)
54 }
55
56 // -- MAIN LOOP --
57 void loop() {
58     bool btn1_pressed = (digitalRead(button1Pin) == LOW);
59     bool btn2_pressed = (digitalRead(button2Pin) == LOW);
60     bool lim1_pressed = (digitalRead(limit1Pin) == LOW);
61     bool lim2_pressed = (digitalRead(limit2Pin) == LOW);
62
63     switch (systemState) {
64
65         case 0: // Waiting for Start
66             {
67                 int potValue = analogRead(potPin);
68
69                 // Hysteresis (deadband) filter -- suppresses fluctuation
70                 if (abs(potValue - lastPotValue) > 3) {
71                     lastPotValue = potValue;
72                 }
73                 float currentV = lastPotValue * (45.0 / 1023.0);
74
75                 // Update LCD periodically to prevent garbage characters and lag
76                 if (millis() - lastDisplayTime > DISPLAY_INTERVAL) {
77                     lcd.setCursor(0, 0);
78                     lcd.print("Waiting Start...");
79                     lcd.setCursor(0, 1);
80                     lcd.print("Speed: ");
81                     if (currentV < 10.0) lcd.print(" ");
82                     lcd.print(currentV, 1);
83                     lcd.print(" mm/s ");
84                     lastDisplayTime = millis();
85                 }
86
87                 if (btn1_pressed && currentV > 0.1) {
88                     digitalWrite(enPin, LOW); // Enable the motor before moving
89                     lockedV = currentV;
90                     lockedStepDelay = 3333.0 / lockedV;
91                     digitalWrite(dirPin, HIGH);
92
93                     // Update LCD ONCE for State 1 to avoid motor stutter
94                     lcd.clear();
95                     lcd.setCursor(0, 0);
96                     lcd.print("Coating... ");
97                     lcd.setCursor(0, 1);
98                     lcd.print("Speed: ");
99                     if (lockedV < 10.0) lcd.print(" ");
100                    lcd.print(lockedV, 1);
101                    lcd.print(" mm/s ");
102

```

```

103         systemState = 1;
104         delay(300); // Debounce
105     }
106 }
107 break;
108
109 case 1: // Coating (Forward) -- LCD is NOT updated here to prevent
110         stutter
111     if (lim2_pressed) {
112         digitalWrite(enPin, HIGH); // Disable the motor at the end (
113             silence it)
114
115         // Update LCD ONCE for State 2
116         lcd.clear();
117         lcd.setCursor(0, 0);
118         lcd.print("Coating Done    ");
119         lcd.setCursor(0, 1);
120         lcd.print("Waiting Reset...");
121         systemState = 2;
122     } else {
123         moveStep(lockedStepDelay);
124     }
125     break;
126
127 case 2: // Waiting for Reset
128     if (btn2_pressed) {
129         digitalWrite(enPin, LOW); // Enable the motor before moving
130         digitalWrite(dirPin, LOW);
131
132         // Update LCD ONCE for State 3
133         lcd.clear();
134         lcd.setCursor(0, 0);
135         lcd.print("Returning...    ");
136         lcd.setCursor(0, 1);
137         lcd.print("Speed: 45.0 mm/s");
138
139         systemState = 3;
140         delay(300); // Debounce
141     }
142     break;
143
144 case 3: // Returning (Backward) -- LCD is NOT updated here to prevent
145         stutter
146     if (lim1_pressed) {
147         digitalWrite(enPin, HIGH); // Disable the motor once home is
148             reached
149         lcd.clear();
150         systemState = 0;
151     } else {
152         moveStep(3333 / 45); // 45 mm/s return speed
153     }
154     break;
155 }
156 }
157
158 // -- HELPER FUNCTIONS --
159 void moveStep(long delayTime) {
160     digitalWrite(stepPin, HIGH);
161     customDelay(delayTime);

```



```

158     digitalWrite(stepPin, LOW);
159     customDelay(delayTime);
160 }
161
162 void customDelay(long uSecs) {
163     if (uSecs > 16000) {
164         delay(uSecs / 1000);
165         delayMicroseconds(uSecs % 1000);
166     } else {
167         delayMicroseconds(uSecs);
168     }
169 }

```

Listing 3: Doctor Blade Coater firmware (Arduino Uno, parallel-LCD configuration).

Appendix H: Electrical Schematic and Wiring

The full electrical schematic of the system, including the AC protective cascade, the busbar distribution, the ATX DC rails, and the control-signal wiring between the Arduino, the TB6600 driver, and the PID/SSR thermal loop, is presented in Figure 63.

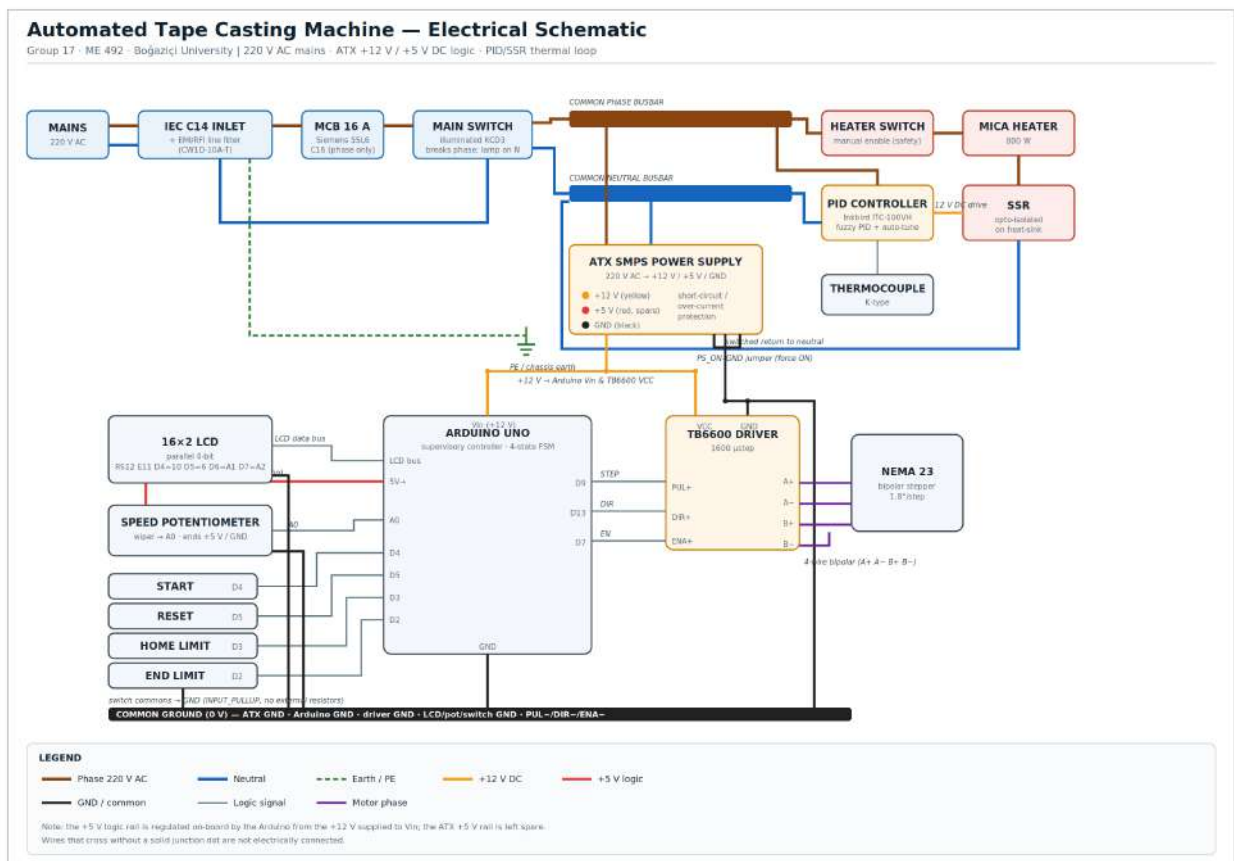


Figure 63: Complete electrical schematic of the automated tape casting machine.